

China Semiconductors

China Next Winners: Montage - riding on the Agentic AI boom



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This report is part of a series on China Next Winners in which we identify the next generation of industries and corporate winners in China. We highlight Montage as the key beneficiary to the accelerating Agentic AI adoption.

Memory interface chips offer amplified exposure to the CPU renaissance driven by agentic AI. We project its TAM to reach approximately USD20bn by 2030, growing at a 65% CAGR from 2025-30. Agentic AI expands the role of server CPUs beyond supporting accelerator workloads to orchestrating, retrieving data for and executing multistep tasks. Memory interface chips benefit through three compounding drivers: more server CPUs, more DIMMs per CPU and higher interface-chip value per DIMM. Together, these drivers allow the industry's revenue pool to grow substantially faster than CPU shipments alone.

Agentic AI strengthens the demand for DDR bandwidth and could accelerate the MRDIMM penetration. Agentic AI expands the AI memory opportunity beyond HBM into a broader hierarchy. Retrieval, embedding lookup, data preparation, CPU inference and host-side memory tiering create recurring traffic through system memory. While HBM remains the high-bandwidth tier closest to XPU, DDR increasingly serves as the high-capacity bridge between HBM and storage-based context repositories. Meanwhile, CPU per-core compute performance continues to increase faster than available DDR bandwidth. MRDIMM addresses this imbalance by doubling bandwidth while retaining DDR5 architecture.

MRDIMM offers the least disruptive path to a step-change in DDR bandwidth, and creates the largest change in interface-chip value per module. By aggregating two rank-level data streams onto a faster host interface, it approximately doubles bandwidth while retaining DDR5 DRAM technology, the 64-bit channel width and the established DIMM ecosystem. The performance gain comes primarily from module-level interface chips: MRDIMM uses one MRCD and ten MDBs, lifting our estimated interface chipset value from approximately USD6–7 per RDIMM to USD50–70 per MRDIMM. We project MRDIMM to reach approximately 25% of server DIMM shipments by 2030, making its adoption the largest driver of interface-chip value growth.

A concentrated supplier base positions incumbents to capture the opportunity and maintain lucrative margin. Montage, Renesas and Rambus collectively hold 90%+ of the memory interface chip market. Their positions rest on accumulated design expertise, long qualification cycles and close collaboration with CPU, DRAM and system vendors. Interface chips account for a small share of module cost but have a substantial impact on reliability and performance, limiting the incentive for customers to switch suppliers solely to reduce component prices.

Reiterate outperform rating on Montage, with our latest [company model](#) and [industry model](#) reflecting the CXL potential upside.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	23 Sep 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
6809.HK (Montage)	O	HKD	306.60	520.00	NA	CNY	1.97	3.44	5.69	132.9	76.2	46.0
688008.CH (Montage)	O	CNY	223.37	400.00	38.1%	CNY	1.97	3.44	5.69	113.4	65.0	39.3
ASIAx			1,993.86									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Montage Technology: We rate Montage Technology as **Outperform** and raise our **A-share TP to CNY 400**, based on **47x 2BF P/E**. We believe a 2BF framework better reflects the upcoming earnings inflection driven by the 2027–2028 product cycle, including the ramp-up of MRDIMM interface chips and CXL silicons, and the P/E is lower than the 5-year CAGR so should have more upside as well. For the **H shares**, we set a target price of **HKD 520**, implying a **15% premium** to our A-share TP (we use the CNY to HKD exchange rate at 1:1.13). The H share premium reflects that global investors favor Montage as a scarce China AI-exposed name without direct geopolitical risks, unlike many other Chinese semiconductor companies that face entity list restrictions or export control headwinds.

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DETAILS

TAM & GROWTH DRIVERS OF MEMORY INTERFACE CHIPS, A CORE BENEFICIARY OF THE AGENTIC AI ERA

AGENTIC AI CREATES A CPU RENAISSANCE, BENEFITING MEMORY INTERFACE CHIPS DIRECTLY

Agentic AI expands the amount of CPU work required to complete an AI task. The AI industry is entering a new phase. As workloads shift from training to inference to agentic applications, compute intensity migrates back toward the server CPU: GPUs handle massive parallel computation while CPUs sit largely idle in training, but in agentic workloads — chain-of-thought reasoning, tool invocation, multi-agent orchestration — the CPU becomes the essential orchestrator, scheduling thousands of small computational tasks per second, managing the model's working memory (KV cache), and coordinating external API calls. CPU-side processing can account for 50–90% of total task completion time in these workloads, and the CPU-to-GPU ratio is shifting dramatically, reaching approximately 1:1–2 in agentic architectures versus 1:12–16 in earlier training-era systems.

The CPU therefore participates in agentic AI through three broad roles. Host-node CPUs prepare and move data for accelerators and coordinate system resources. Sandbox CPUs execute tasks generated by agents. General-purpose CPUs run the retrieval, database, application and storage services that support the workflow. These roles expand the sources of CPU demand beyond the processors directly attached to GPU servers.

The economic incentive to provision adequate CPU resources is also strong. Accelerators represent a substantial investment, and insufficient CPU-side processing can leave them waiting for inputs or task coordination. Increasing CPU resources can improve the utilization of the wider system when CPU processing is the constraint. As agentic deployments scale, we expect customers to place greater emphasis on balancing accelerator capacity with the infrastructure required to keep it productive.

Memory interface chips benefit directly from this expansion. Server CPUs connect to multiple memory modules, and each DIMM requires interface silicon to support reliable communication between the memory controller and DRAM. **More CPU shipments therefore create an immediate volume opportunity. Increasing memory requirements per CPU and the migration to more sophisticated module architectures add further growth.**

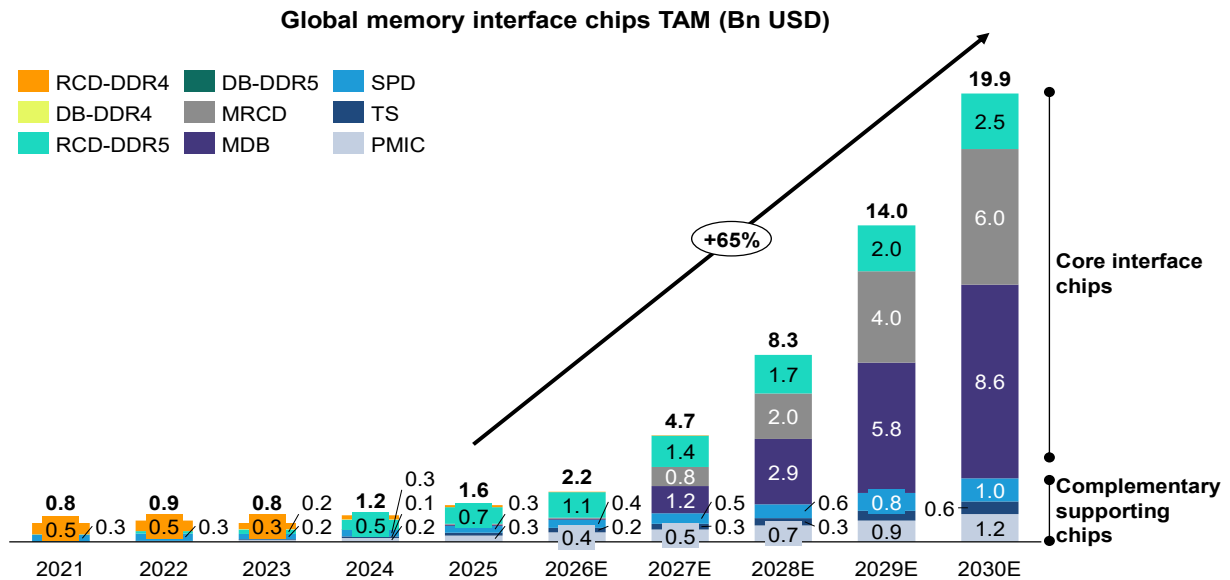
TAM AND THREE MULTIPLICATIVE DRIVERS

We project the global DDR module chipset TAM to reach USD 20Bn by 2030 (~65% CAGR), dramatically outpacing the ~24% growth in server CPU shipments, making the sector almost 3x more elastic on top of the CPU vol growth. Three growth drivers are multiplicative, not additive. The TAM is the product of CPU volumes, DRAM modules per CPU, and chipset value per module, nearly all of which are rising simultaneously. The relationship can be expressed simply:

Memory interface chip TAM = server CPU shipments × DIMMs per CPU × average interface-chip value per DIMM

By 2030, MRDIMM-related chips (MRCD and MDB) are projected to account for approximately 73% of the total memory interface chip TAM, a category that barely existed before 2025. This concentration underscores the transformative impact of the MRDIMM migration on the industry's revenue pool.

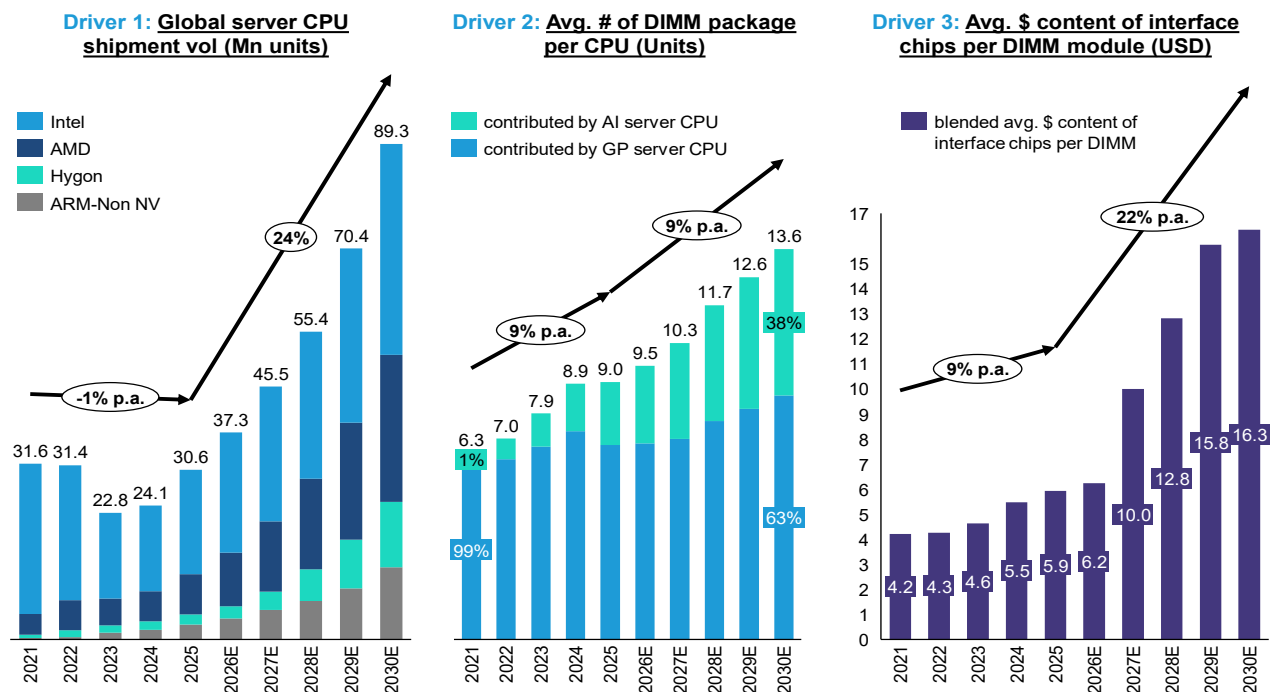
EXHIBIT 1: We raise our global memory interface TAM projection to USD 20Bn in 2030, with MRDIMM's MRCD and MDB contributing 73%, driven by significantly higher chipset value per module



Our TAM forecast does not include memory interface chips used in NVIDIA proprietary CPUs, such as voltage regulators and PMICs. We expect the impact to be only LSD, as NVIDIA CPUs account for a small portion of server CPU shipments and the related components are of relatively low value.

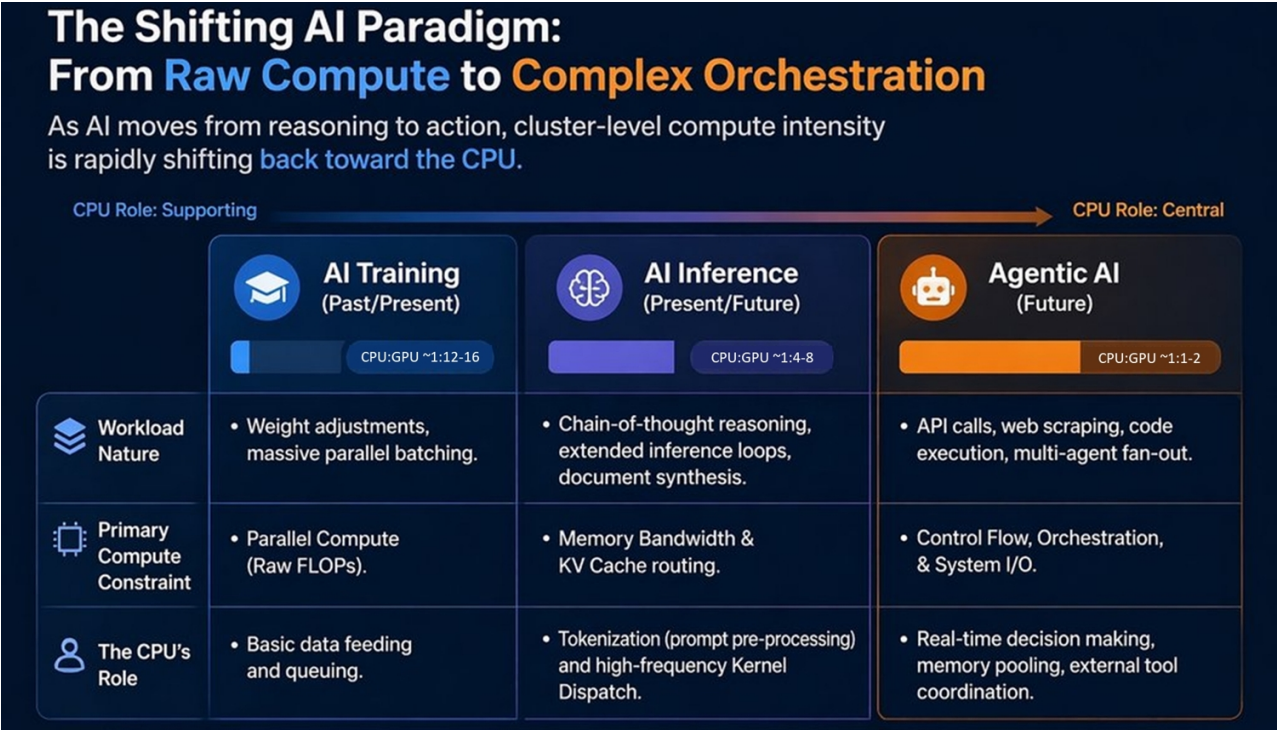
Source: Mercury, TrendForce, Gartner, Bernstein analysis and estimates

EXHIBIT 2: The expansion of the TAM is driven by three core factors, each of which is expected to accelerate in the second half of this decade relative to the past 5 years



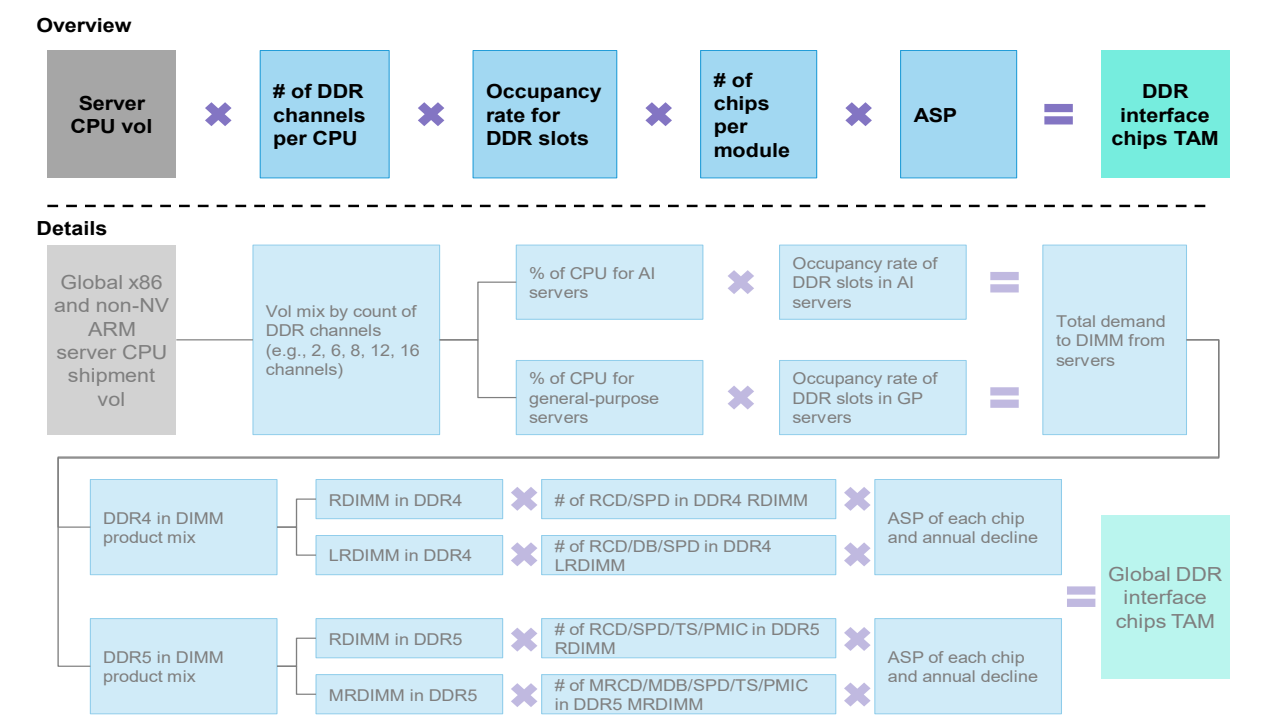
Source: Gartner, Mercury, Bernstein analysis and estimates

EXHIBIT 3: **As AI transitions from a training-heavy era (dominated by GPUs) to an inference-heavy era, the role of CPUs expands. Inference and agentic workflows requires the sequential and latency-sensitive capabilities of CPUs**



Source: public reports, Bernstein analysis and estimates

EXHIBIT 4: **Our TAM model follows a bottom-up approach, reflecting the impacts of three drivers on five multiplicative factors**



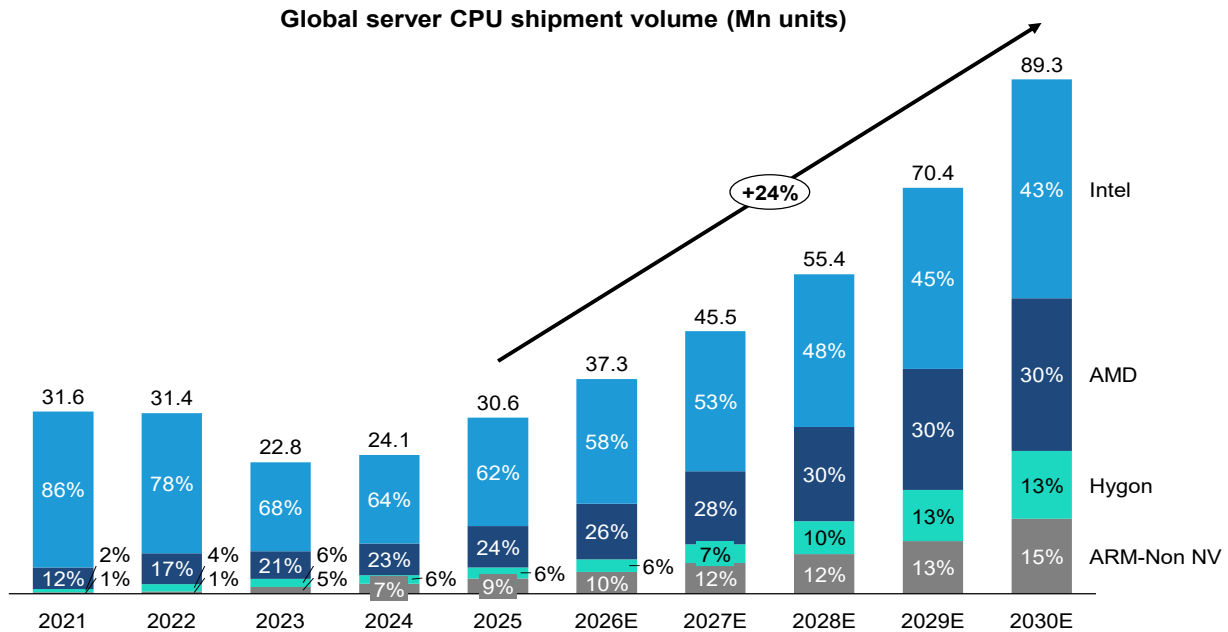
Source: Bernstein analysis

DRIVER 1 - SERVER CPU SHIPMENT VOLUME

We project global server CPU shipment volumes (excluding NVIDIA's proprietary CPUs) to grow from approximately 30.6 million units in 2025 to 89.3 million units by 2030, representing an 24% CAGR over the period. This growth is driven by the rapid expansion of AI server deployments, which require CPUs alongside GPU/accelerator arrays, as well as the continued build-out of cloud and enterprise infrastructure.

Within this total, AI server CPUs are growing significantly faster than general-purpose server CPUs. As NVIDIA shifts its proprietary CPUs (Grace) to LPDDR — which does not require the same DIMM chipset — DDR memory interface chip demand becomes primarily tied to x86 and non-NVIDIA ARM CPU shipment volumes.

EXHIBIT 5: Faster CPU volume growth should translate directly into stronger demand for memory interface chips

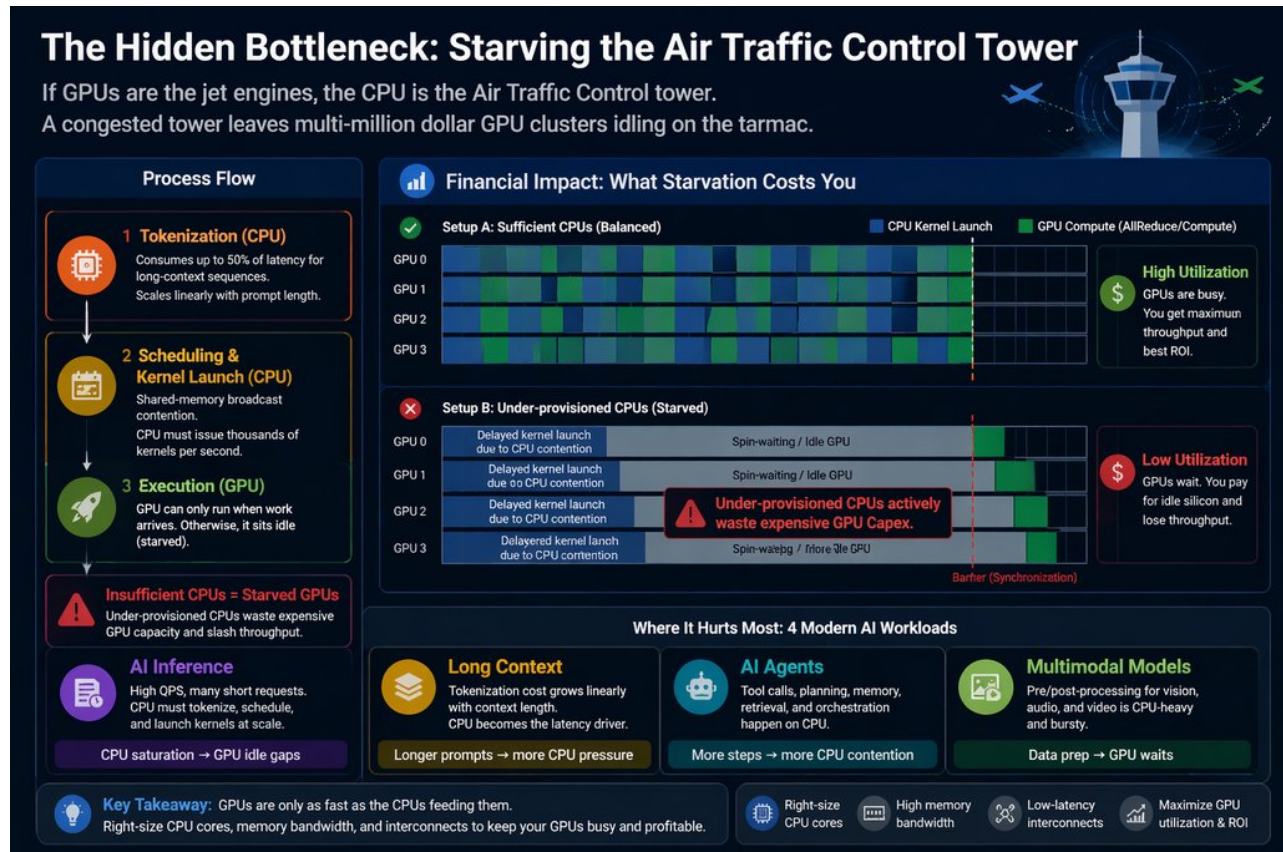


Source: Mercury Research, Bernstein analysis and estimates

Our global server CPU shipment volumes upward revision is driven by the convergence of AI agents, long-context processing, and multimodal models that push CPU utilization higher.

The economic implications are compelling. Research from Georgia Tech has demonstrated that CPU under-provisioning in multi-GPU LLM inference clusters can increase time-to-first-token by 1.4x to 5.4x, effectively leaving multi-million-dollar GPU clusters idling on the tarmac. Yet CPU cores are approximately 100–1,600x cheaper than equivalent GPU compute, meaning that increasing the CPU allocation typically adds only about 1.5% to total cluster cost while potentially unlocking dramatically higher GPU utilization. This creates a strong economic incentive for CSPs to aggressively increase CPU provisioning — a trend that directly drives DDR module chipset demand, since every additional CPU requires multiple DDR modules, each populated with module chipsets.

EXHIBIT 6: CPU under-provisioning leaves GPUs idle. Adding CPU cores is cost-effective to improve performance. Because CPU cores are approximately 100–1,600x cheaper than GPU compute, increasing the CPU allocation typically adds only about 1.5% to the total cost while potentially improving time-to-first-token by 1.36–5.40x



Source: Chung, Euijun, et al. "Characterizing CPU-Induced Slowdowns in Multi-GPU LLM Inference." Georgia Tech, 2026; Bernstein analysis

Beyond DDR, the CPU Renaissance is also structurally favorable for CXL-based memory expansion. AI agents require massive persistent memory for KV cache and context windows, and CPUs provisioned with large, disaggregated RAM pools via CXL memory pooling offer a more efficient architecture for accommodating the growing data volumes generated by long task chains — compared to HBM bound directly to GPUs. We view the CPU Renaissance as a material upside catalyst not yet reflected in either our estimates or Street consensus, and one that investors should monitor closely as agentic AI deployments scale through 2027–2030.

DRIVER 2 - RISING DIMM MODULES PER CPU

The second growth driver is the increasing number of DDR modules per CPU, which is a function of two sub-drivers: the migration to higher channel-count CPU architectures and the rising occupancy rate of DIMM slots.

Channel count migration: The number of DDR channels per CPU is migrating upward in response to AI workloads that are extremely bandwidth-hungry. The share of server CPUs with 12-channel DDR configurations will grow from 24% in 2025 to 45% in 2030, while 16-channel CPUs will rise from near zero to 30% over the same period. This transition has the potential to roughly double the memory interface content required per CPU.

The drivers behind this migration are clear. AI workloads demand massive memory bandwidth, since additional channels reduce contention and latency during parallel computing. CPU core counts are scaling faster than per-DIMM bandwidth, meaning that adding channels is necessary to sustain bandwidth per core and avoid memory starvation. Higher channel counts also enable more DIMM slots and greater total memory capacity per socket, supporting virtualization density, LLM inference, and large memory-resident workloads.

Competition between Intel and AMD to raise the bar on DDR channel count ensures that neither vendor can afford to fall behind on memory specifications, creating a secular ratchet effect. Intel's Sapphire Rapids and Emerald Rapids (2023–2024) operated on 8-channel DDR5 architectures. The step-change comes with Clearwater Forest (1H 2026), Intel's next-generation E-core platform, which doubles the channel count to 12-channel DDR5 with up to 288 cores. Looking further ahead, Diamond Rapids, Intel's next P-core platform (expected in mid-2027), is designed with 16-channel DDR5 support. On the AMD side, the channel count migration began earlier. AMD's Genoa (EPYC 9004, 2022) debuted with 12-channel DDR5. The upcoming Venice (EPYC 9006, 2026), built on Zen 6 architecture with TSMC 2nm and up to 256 cores, is expected to support up to 16 channels of DDR5 on the new SP7 socket.

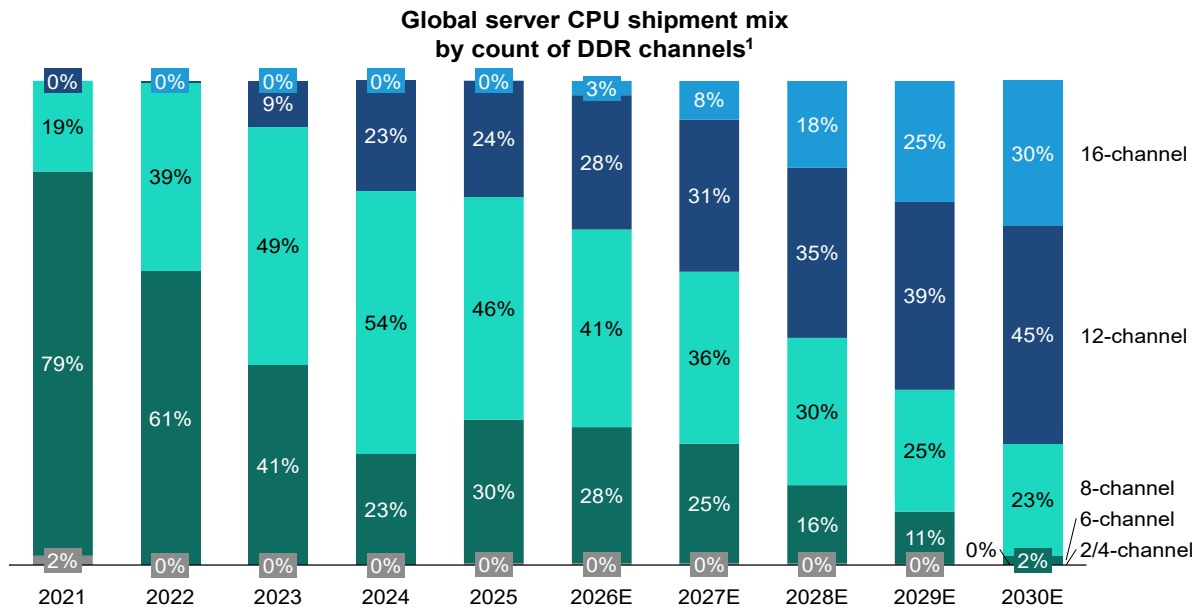
EXHIBIT 7: The competitive dynamic is self-reinforcing: neither Intel nor AMD can afford to lag on memory bandwidth specs. The migration to 12- and 16-channel architectures will proceed on schedule or even accelerate

Vendor	2022	2023	2024	2025	2026	2027	2028+
● Intel P-core	Sapphire Rapids 8-ch	Emerald Rapids 8-ch	Granite Rapids MCR RDIMM 8-ch			Diamond Rapids MRDIMM 16-ch	Coral Rapids TBD
● Intel E-core			Sierra Forest 144 cores 8-ch		Clearwater Forest 288 cores · 18A 12-ch		
● AMD EPYC	Genoa Zen 4 · 96c 12-ch	Bergamo Zen 4c · 128c 12-ch	Turin Zen 5 · 192c 12-ch		Venice Zen 6 · 256c · 2nm 12/16-ch	Verano Zen 7 · +LPDDR5X TBD	

8-channel DDR5
12-channel DDR5
16-channel DDR5
Announced, timeline TBD
Source: Intel, AMD, Bernstein analysis

Diamond Rapids reportedly delayed to mid-2027. Venice SP7 platform supports both 12-ch and 16-ch DDR5 configurations with RDIMM and MRDIMM. Verano adds LPDDR5X SOCAMM2 support alongside DDR5. Both Intel and AMD also support MRDIMM from their respective platforms.
Source: Intel, AMD, Bernstein analysis

EXHIBIT 8: The transition from 8-channel to 16-channel DDR architectures has potential to double memory interface content required per CPU



CPUs are categorized based on the higher DDR-channel count available among their versions (e.g., Clearwater Forest's 8-channel and 12-channel variants are grouped under 12-channel). In the explicit model, we instead use the weighted average DDR-channel count.

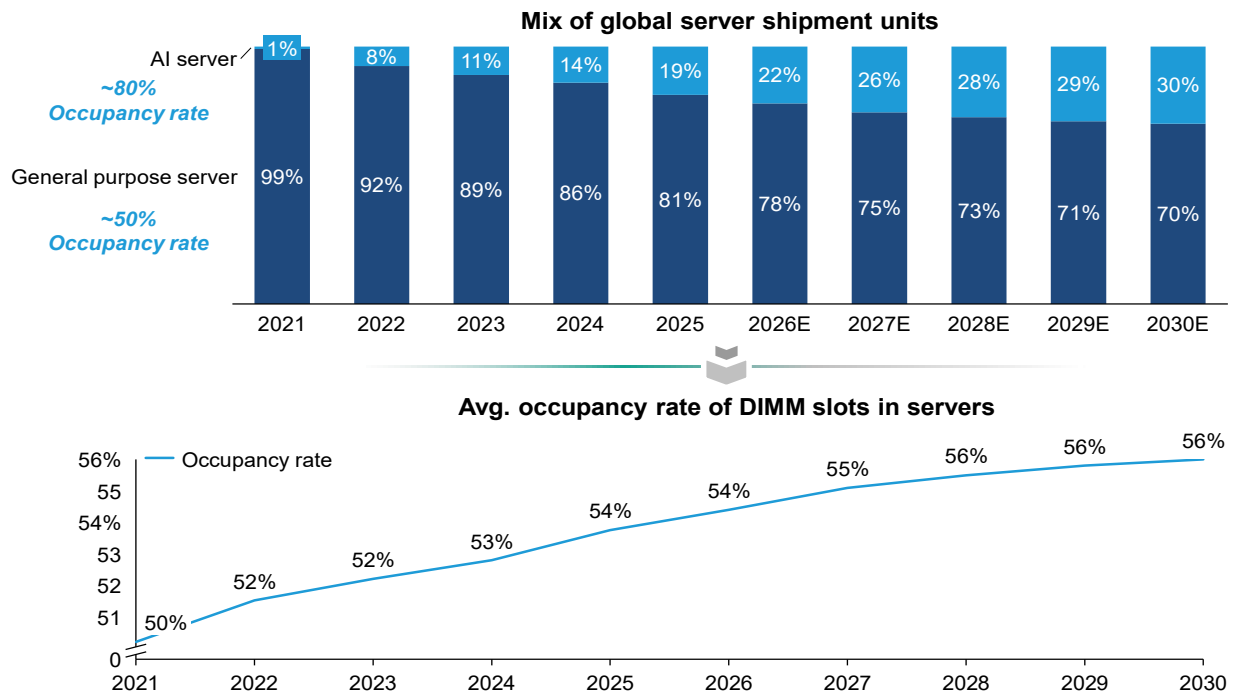
Source: Mercury Research, Bernstein analysis and estimates

Occupancy rate increase: AI servers fully populate DDR slots with approximately 70-80% occupancy to optimize accelerator utilization. The logic is straightforward: insufficient DDR capacity directly caps xPU utilization, and with accelerators representing the majority of the system BOM cost, maximizing xPU utilization far outweighs the incremental cost of populating every memory slot. Larger DDR capacity with manageable bandwidth trade-offs (losing 5–15%) is acceptable in AI's parallel compute profile, and symmetric DDR configurations are critical for scale-up AI systems.

By contrast, general-purpose servers leave roughly half of memory slots empty (~50% occupancy) to optimize fleet-level TCO and preserve headroom for future upgrades. DRAM accounts for approximately 50% of GP server CAPEX and is energy-intensive, making overprovisioning costly. GP servers expand via scale-out, with half-populated slots preserving flexibility to upgrade memory and add servers as needed.

As AI servers grow from approximately 14% of global server shipments in 2024 to an estimated 30% by 2030, the blended average occupancy rate of DIMM slots across the global server fleet rises from approximately 54% to 56% — a meaningful uplift in total DIMM demand per CPU shipped.

EXHIBIT 9: AI servers utilize nearly all DIMM slots to maximize accelerator performance, while traditional servers prioritize TCO and leave many slots unpopulated. The shift toward AI servers will drive higher average DIMM slot occupancy



Source: Gartner, Bernstein analysis and estimates

DRIVER 3 - HIGHER MEMORY INTERFACE CHIPS VALUE PER MODULE FROM MRDIMM MIGRATION

The migration from DDR4 to DDR5, and subsequently from DDR5 RDIMM to DDR5 MRDIMM, is a powerful driver of dollar content per package expansion for memory interface chips. DDR5 modules require additional chip types beyond the RCD and SPD that were only interface chips in DDR4 RDIMM; specifically, PMIC and TS are now integrated, lifting the total interface chip count per RDIMM module. More importantly, the transition to MRDIMM represents a step-change in silicon content: each MRDIMM module adopts a "1+10" architecture consisting of one MRCD chip and ten MDB chips, compared to one RCD and zero data buffers for a DDR5 RDIMM at 6400MT/s. This translates to a dramatic increase in memory interface silicon content per module, directly boosting Montage's dollar content per DIMM sold. We estimated that the memory interface chip total value could increase from about 7 USD in RDIMM to more than 70 USD in MRDIMM (with all interface chip combined).

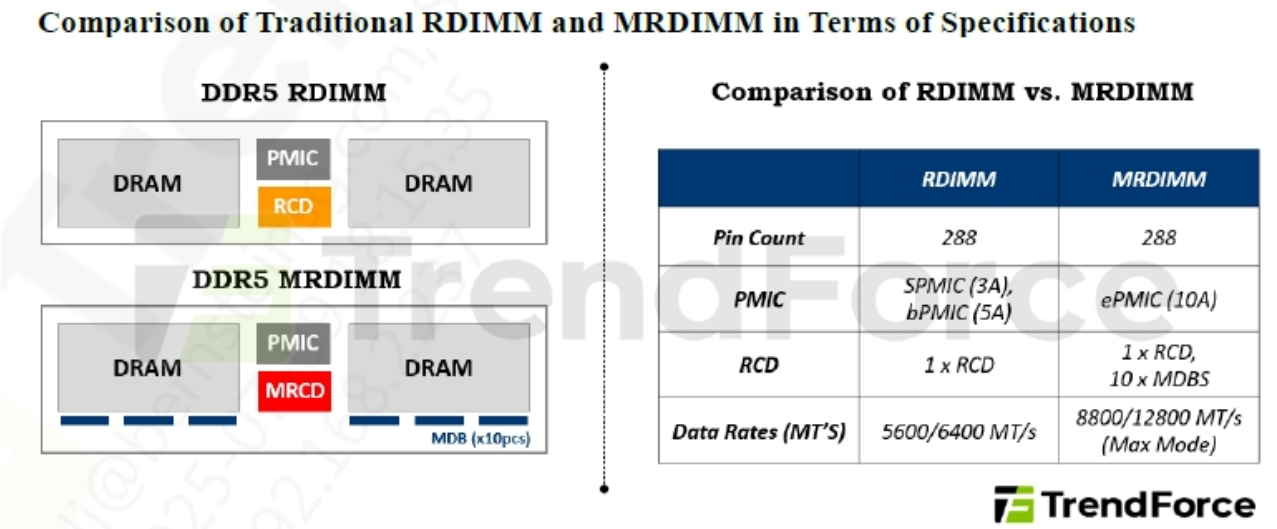
As a result, MRDIMM penetration is the single most sensitive variable in the TAM model. TrendForce estimates MRDIMM just accounts for 3% of global server DDR DIMM shipment units in 2026; and we project the rate will rise to 25% by 2030E.

Configuration comparison across DDR generations and DIMM formats

The chip count and composition per module varies significantly across DDR generations and DIMM formats, with each upgrade cycle introducing additional silicon content.

The key takeaway from the comparison across module formats in Exhibit 11 is the dramatic step-change in silicon content when moving from DDR5 RDIMM to DDR5 MRDIMM. MRDIMM adopts a "1+10" architecture (one MRCD plus ten MDB chips) versus just one RCD and zero data buffers for DDR5 RDIMM at 6400 MT/s. The total DDR chipset value amplifies by approximately 10 times for MRDIMM versus RDIMM: from roughly USD 7 in total interface chip value per RDIMM module to more than USD 70 per MRDIMM module (with all interface chips combined). This step-change is the single most important structural driver of TAM expansion.

EXHIBIT 10: The spec upgrade of MRDIMM versus RDIMM



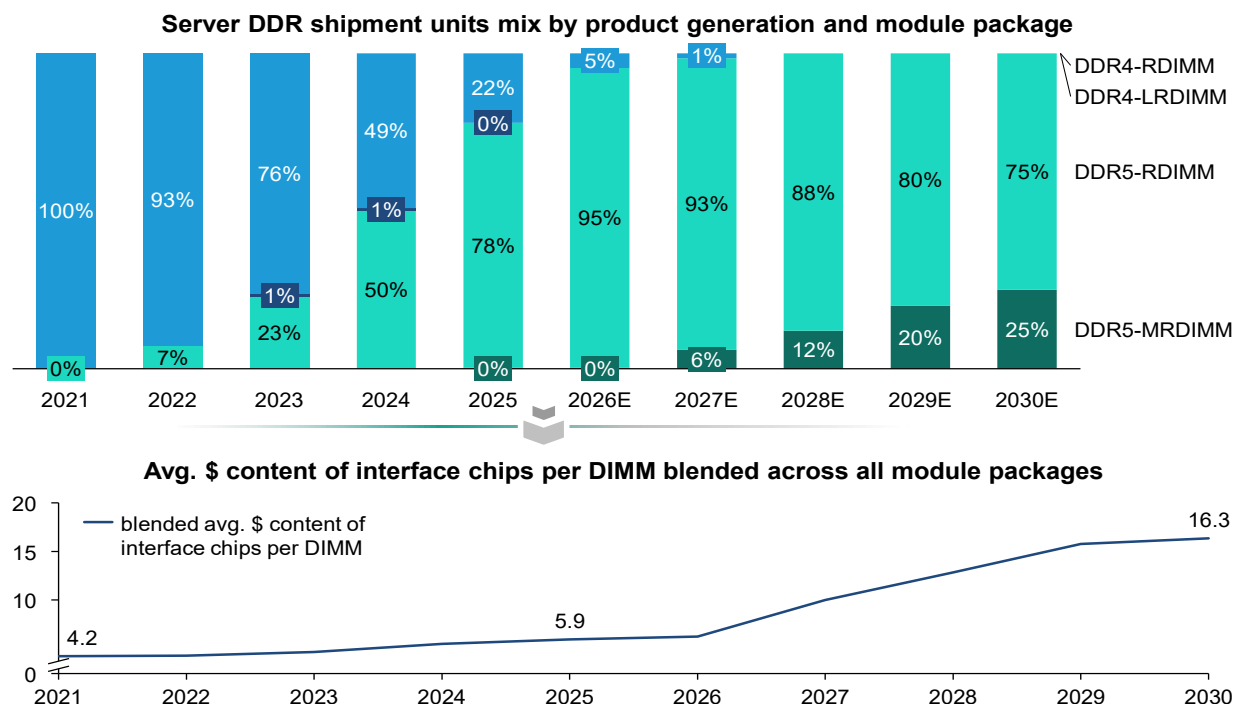
Source: TrendForce report in 2025 July

EXHIBIT 11: Memory interface chips configuration comparison across various formats of server DDR4 and DDR5

Unit of chips	DDR4 RDIMM	DDR4 LRDIMM	DDR5 RDIMM	DDR5 MRDIMM
RCD	1	1	1	
DB		9		
MRCD				1
MDB				10
SPD	0 or 1	0 or 1	1	1
TS			2	2
PMIC			1	1
ASP of the whole memory interface chipset	USD 2-3	USD 12-14	USD 6-7	USD 50-70

This chart shows our estimation of the chip price in 2026, we expect the price to further drop in the next few years.
Source: Montage reports, Bernstein analysis

EXHIBIT 12: MRDIMM adoption in the DDR5 era will nearly triple memory interface chips dollar content per module. We expect MRDIMM penetration to surpass 25% by 2030, a level that remains modest compared with the adoption trajectory of DDR5 RDIMMs



Source: TrendForce, Bernstein analysis and estimates

TAM SENSITIVITY ANALYSIS

EXHIBIT 13: 2030 TAM Sensitivity Analysis: Even in a bear-case scenario, the global memory interface chip TAM reaches our prior-version projection, reinforcing a well-protected downside for Montage at current share price

2030 TAM sensitivity analysis (Bn USD)		Server CPU shipment in 2030 (excl. NV, Mn units)						
MRDIMM penetration in 2030		60	70	80	89	100	110	120
	10%	8	9	10	12	13	14	16
	15%	10	11	13	14	16	18	19
	20%	12	13	15	17	19	21	23
	25%	13	16	18	20	22	24	27
	30%	15	18	20	23	25	28	30
	35%	17	20	23	25	28	31	34
	40%	19	22	25	28	32	35	38

Source: Bernstein analysis and estimates

WHY DDR BANDWIDTH DEMAND STRUCTURALLY RISES IN THE AGENTIC AI ERA

FROM COMPUTING SCALING TO CONTEXT SCALING: IMPORTANCE OF DDR ELEVATED IN THE MEMORY TIERING

The industry's focus in AI infrastructure is gradually shifting from pure compute scaling toward context delivery.

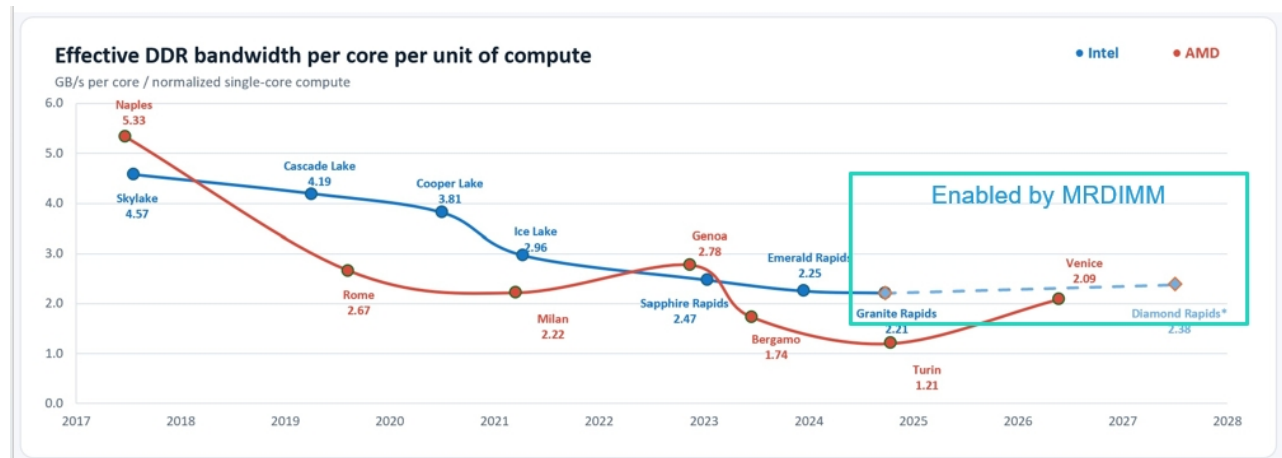
In traditional AI systems, memory was primarily used to store model weights and feed GPUs with sufficient data to maximize compute utilization. However, the rise of inference-centric and agentic AI workloads is creating a new bottleneck: the ability to efficiently store, retrieve, and manage large amounts of contextual information. Long-context reasoning, multi-step agent workflows, RAG retrieval, external tool invocation and persistent user memory all require AI systems to continuously access and update context beyond model parameters, making memory and storage performance increasingly critical to overall system efficiency.

As a result, AI infrastructure is evolving toward a more sophisticated memory hierarchy spanning HBM, DDR and storage-based context memory layers. And DDR plays a growing role as the bridge between HBM and storage-based context repositories. Rather than serving merely as a capacity expansion tier, DDR is becoming an active component of the context delivery pipeline, storing prompt histories, KV caches, retrieved knowledge and agent working memory. We believe this trend extends the role of DDR beyond serving traditional general-purpose servers and increases the strategic importance of DDR in AI systems. In this architecture, improving memory bandwidth and capacity is no longer just about supporting model execution, but increasingly about enabling low-latency access to context at scale, which is essential for sustaining high GPU utilization and efficient agentic reasoning.

The CPU memory wall reflects the widening gap between aggregate compute throughput and memory bandwidth.

Core counts per CPU continue to rise and per-core compute capability also improves, so each GB/s of DDR bandwidth has to support materially more total compute across successive CPU generations. Socket-level bandwidth increases as CPUs add channels and adopt faster DDR, but it does not scale at the same rate as the compound effect of core-count and per-core-performance uplift.

EXHIBIT 14: Effective DDR bandwidth for per unit of compute compresses across CPU generations, since the CPU core count and single-core compute scale faster than socket-level DDR bandwidth



Source: AMD and Intel companies reports, Bernstein analysis and estimates

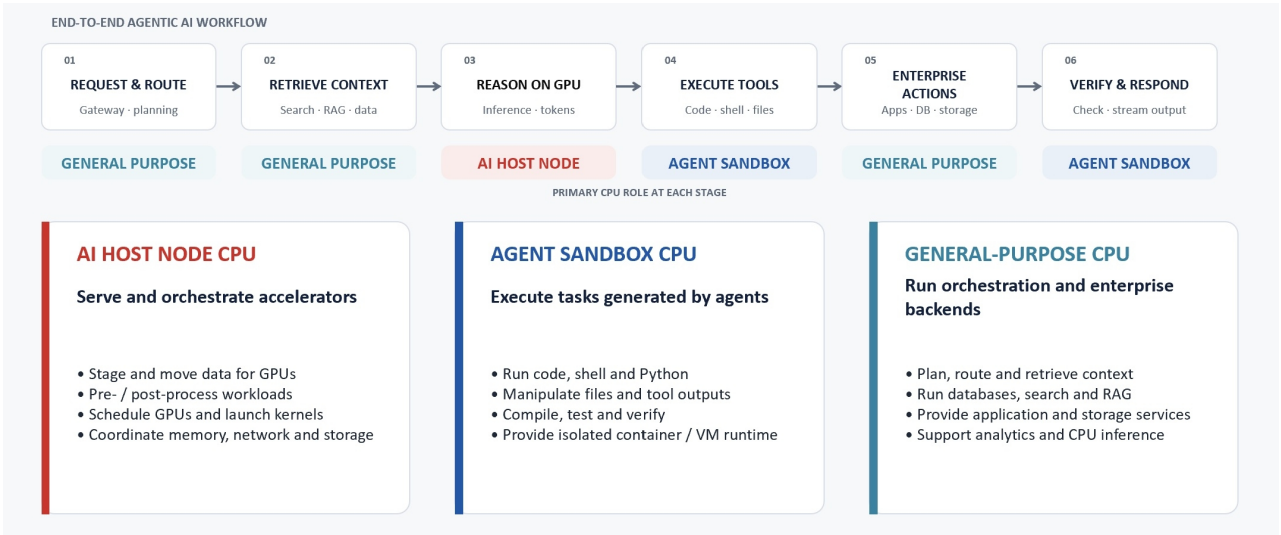
This compression does not make every compute workload memory-bound. It raises the probability that workloads with large active data sets, low arithmetic intensity, irregular access and high concurrency encounter DDR contention. MRDIMM becomes the most valuable, when the application already has sufficient CPU compute and I/O resources but cannot convert them into throughput because the memory subsystem saturates first.

AGENTIC AI ENGAGES THREE CPU ROLES WITH DIFFERENT DDR EXPOSURE

Agentic AI engages three types of CPU roles across the end-to-end workflow: **host-node** CPUs serve and orchestrate accelerators; **sandbox** CPUs execute tasks generated by agents; and **general-purpose** CPUs run retrieval, databases, enterprise applications, storage and CPU inference. CPU role determines what the processor does, while workload

characteristics determine whether DDR bandwidth becomes the bottleneck.

EXHIBIT 15: **AMD classifies three types of CPU engaged in the agentic AI workflow. Their sensitivity levels to DDR bandwidth are different**



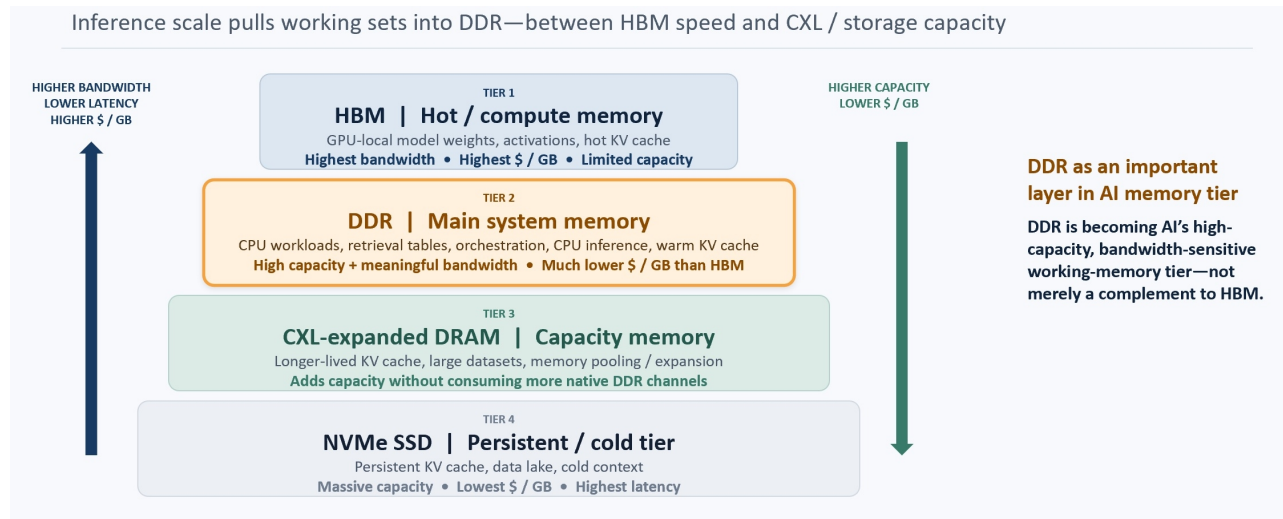
Source: AMD reports, Bernstein analysis

Host-node CPUs exhibit the highest DDR bandwidth demand

Host nodes sit on the system data path that stages and move data, launches accelerator kernels, and coordinates memory, networking and storage. DDR bandwidth can therefore become a system-level constraint: if the host node cannot prepare or move data fast enough, expensive GPUs wait. The economics value of relieving the bottleneck can exceed the direct CPU-performance benefit, because buyers prioritize accelerators utilization.

AI memory is also evolving from an HBM-centric view into a hierarchy. HBM remains the hot, compute-local tier; DDR provides high-capacity main-system memory and can hold retrieval tables, orchestration state, CPU-side inference data, intermediate buffers and warm working sets; CXL-expanded DRAM adds capacity; and NVMe remains the persistent cold tier. DDR complements HBM, and its relevance rise as context length, concurrency, and model dimensions expand the aggregate working set.

EXHIBIT 16: AI memory is evolving from an HBM-centric architecture into a multi-tier hierarchy, where DDR is becoming increasingly relevant for inference tasks



Source: Micron and SK hynix reports, Bernstein analysis

In a conventional PCIe-based host architecture, the CPU-GPU interconnect often becomes the bottleneck before DDR bandwidth is fully utilized, limiting the incremental benefit of faster memory. However, as next-generation CPU-GPU interconnects such as NVLink-C2C and UALink (UB) emerge and provide substantially higher bandwidth, the bottleneck is increasingly likely to shift to the DDR subsystem. Meanwhile, the opportunity cost of GPU underutilization continues to rise as GPU compute performance and pricing increase. This strengthens the incentive to provision sufficient memory bandwidth and avoid GPU stalls caused by data starvation. Consequently, we view host nodes as the most structurally attractive deployment opportunity for MRDIMM.

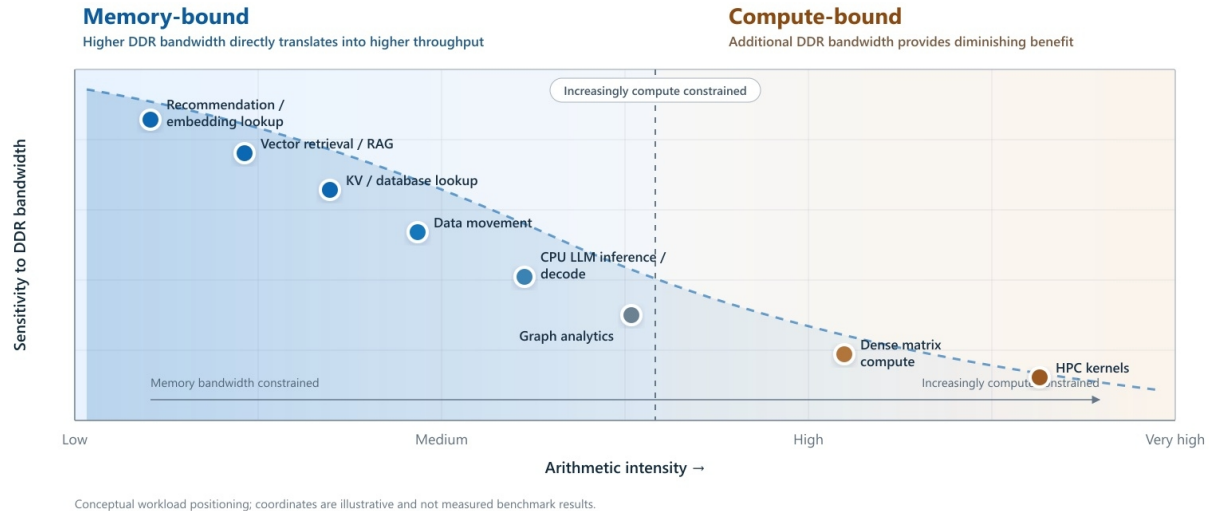
DDR bandwidth requirements for Sandbox and General-purpose CPUs vary, depending on workload mix

Sandbox CPUs execute heterogeneous tasks, including coding, compilation, file manipulation, browser and API calls, retrieval, analytics, and CPU inference. Many tasks are compute-, network-, storage- or latency-bound and receive limited benefit from additional DDR bandwidth. Others, particularly vector retrieval, embedding lookup, large data transforms, graph analytics, CPU LLM inference and memory movement, can saturate DDR channels. General-purpose CPUs show the same dispersion, because their workload universe spans web and application servers as well as recommendation, databases, search and in-memory analytics.

EXHIBIT 17: Sandbox and GP CPUs execute highly heterogeneous workloads. MRDIMM creates the most value in low-arithmetic-intensity workloads where data movement, not computation, constrains throughput

MRDIMM creates the most value for memory-bound workloads

Workloads with low arithmetic intensity move substantially more data per unit of computation, making system throughput more sensitive to DDR bandwidth. As arithmetic intensity rises, additional DDR bandwidth delivers diminishing performance benefit.

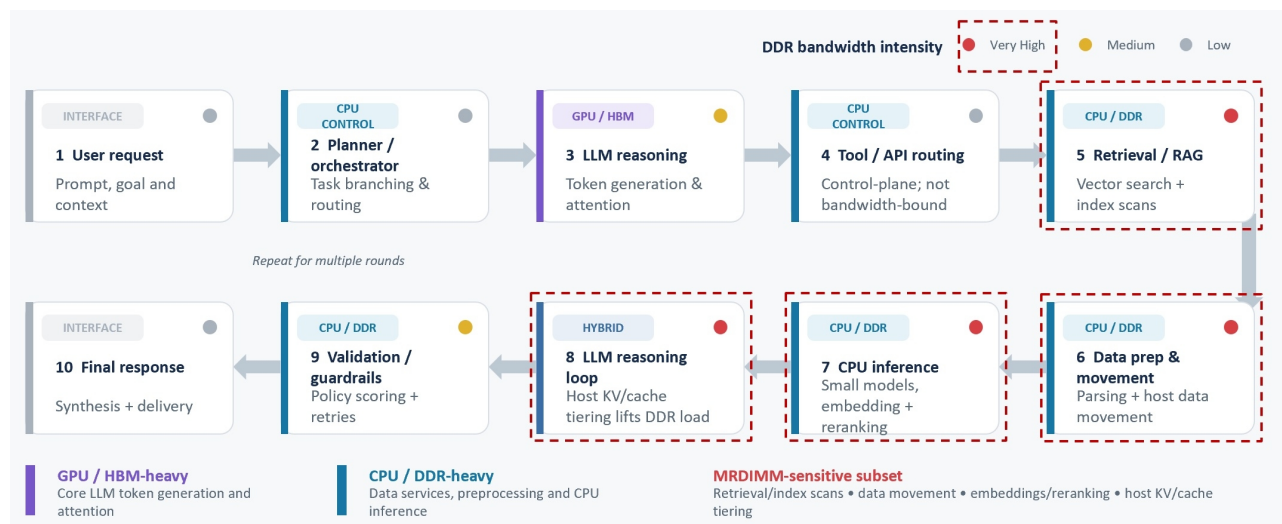


MRDIMM creates the most value where CPUs spend more time moving data than computing on it.

Source: company reports, Bernstein analysis

Within an agentic AI workflow, DDR bandwidth requirements can vary significantly across stages. Functions such as user routing and GPU-resident model inference typically exhibit limited direct sensitivity to DDR bandwidth, whereas **retrieval/RAG, data preparation and movement, CPU-based LLM inference, host-side KV cache tiering, and response validation can generate substantial memory traffic.** As a result, a single agentic workflow may encompass both MRDIMM-suitable and MRDIMM-insensitive stages. This suggests that MRDIMM deployment is more likely to be selective, with partial attachment to bandwidth-intensive nodes, rather than universal adoption across the entire system.

EXHIBIT 18: Several stages in the Agentic AI Workflow are DDR-BW-sensitive, creating ideal MRDIMM use cases within a heterogeneous task chain

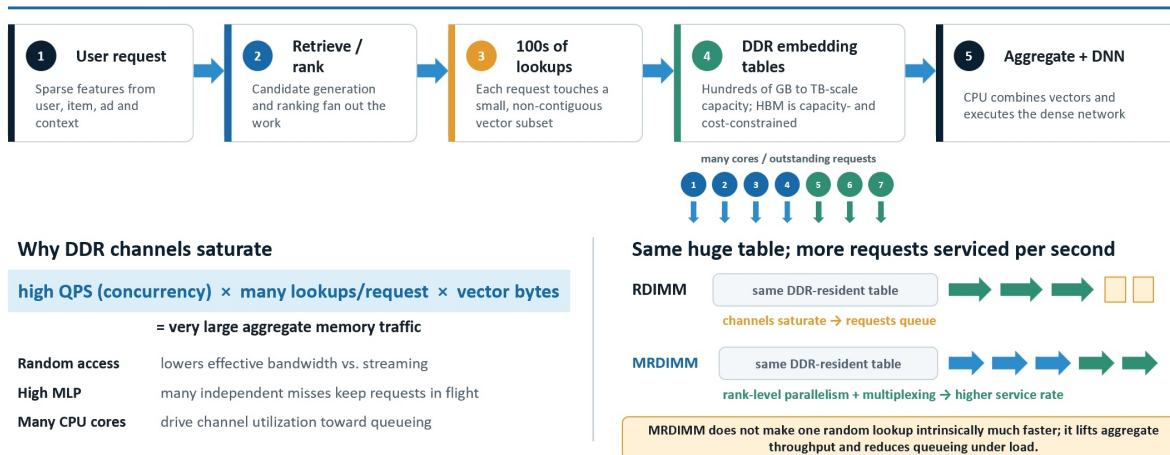


Source: company reports, Bernstein analysis

Recommendation and embedding represent the most ideal application. Their tables can exceed cache capacity by orders of magnitude, queries touch many sparse vectors with limited locality, and high QPS creates large numbers of independent memory requests. The bottleneck is aggregate random-access traffic under concurrency, not the bytes fetched by one request. MRDIMM does not make one random lookup intrinsically faster, but it can service more lookups in parallel and prevent the request queue from exploding under load.

EXHIBIT 19: Massive recommendation and embedding workloads combine random access, high concurrency, and DDR-restricted capacity, making them attractive MRDIMM workload profile

The bottleneck is aggregate random-access traffic under concurrency—not the bytes fetched by one query



INVESTMENT TAKEAWAY

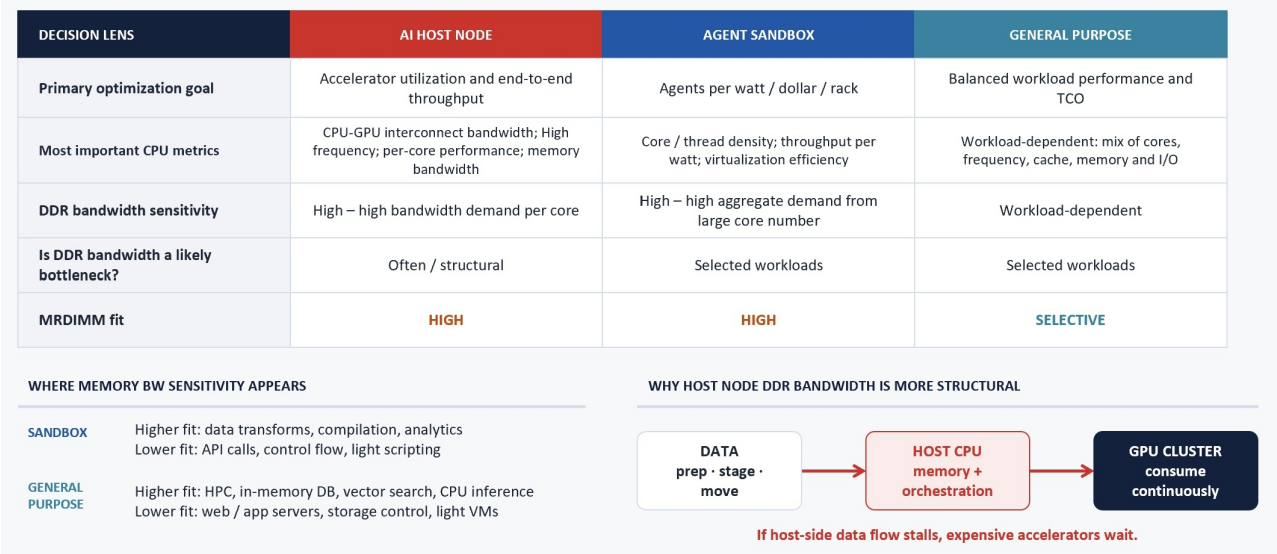
The MRDIMM fit is driven by high concurrency × many independent random lookups × massive DDR-resident capacity.

Source: Intel, Meta, Bernstein analysis

WE FORECAST MRDIMM SOCKET PENETRATION TO RISE TO 25%+ BY 2030

Host node CPUs represent the most structurally consistent MRDIMM opportunity, while sandbox and general-purpose CPUs broaden the addressable market in certain workloads where DDR bandwidth is the bottleneck to constrain throughput. Hence, we assume different MRDIMM attach rates across three CPU roles.

EXHIBIT 20: **Host node CPUs are structurally the most DDR-BW-sensitive; sandbox and general-purpose CPUs are workload-dependent, with recommendation, embedding, retrieval/RAG, and CPU LLM inference particularly attractive MRDIMM use cases**



Source: AMD reports, Bernstein analysis

We model overall MRDIMM socket penetration using two key variables for each CPU role: the global server CPU socket mix and the MRDIMM adoption rate within that specific CPU category. Role-level adoption is determined by several factors, including the share of DDR bandwidth-sensitive workloads, platform and memory-controller support, and customers' budget sensitivity.

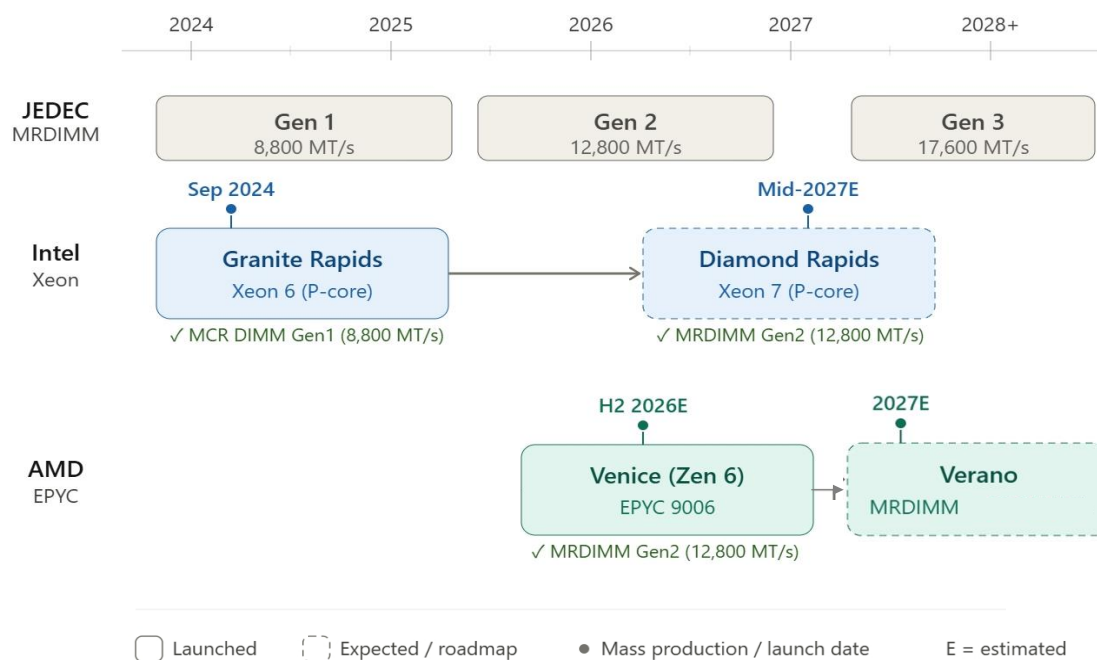
We expect penetration growth to be driven primarily by the increasing share of host-node and sandbox CPUs within the server installed base. While MRDIMM adoption should increase across all three CPU categories, the pace of penetration is likely to vary meaningfully by role. We anticipate the strongest and most structural adoption in host nodes, with more selective deployment in sandbox and general-purpose CPUs. Even under these relatively conservative assumptions, **we forecast overall MRDIMM penetration to increase roughly tenfold, from 2-3% in 2026 to 25%+ by 2030.**

CURRENT PROGRESS OF MRDIMM SUPPORT AND ADOPTION

CPU platform and DRAM roadmaps broaden the addressable pool of MRDIMM, while CSP workload mix determines its attach rate.

CPU support is a prerequisite. Intel initiated the migration since Grantie Rapids, supporting Gen1 MCR DIMM at 8,800 MT/s in Sep 2024. Diamond Rapids plans to support Gen2 MRDIMM in mid-2027. AMD enters directly at Gen2 with the latest-introduced Venice and plans to extend support in Verano in 2027E. This materially expands the set of high-end x86 sockets that can qualify MRDIMM.

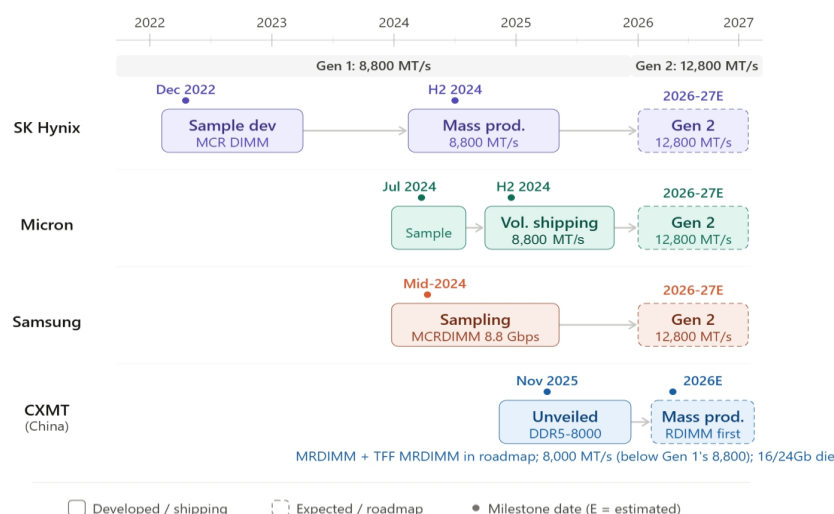
EXHIBIT 21: Intel led the migration to MRDIMM with Granite Rapids in 2H24, while AMD enters directly to MRDIMM Gen2 in 2026, with Venice as the first MRDIMM-capable platform



Source: Intel and AMD reports, Bernstein analysis

The memory supply chain is developing in parallel. Global leading vendors all have entered mass production stage. Meanwhile, CXMT also has introduced server-level DDR5 products and unveiled mass production plan for RDIMM as the first step. Multi-vendor MRDIMM support reduces supply concentration and gives CPU, OEM, and CSP customers greater confidence to invest in qualification.

EXHIBIT 22: DRAM suppliers have established solid MRDIMM product roadmaps; notably, CXMT is rapidly catching up with global leaders



Source: SK Hynix, Samsung, Micron, CXMT reports, Bernstein analysis

CSP potential adoption has showed early signals. Public deployment of Xeon 6 or Venice broadens the potential MRDIMM pool, though it does not prove that CSP customers deploy MRDIMM in every Xeon 6 or Venice supported server. Azure provides the most visible confirmation across both compatible CPU roadmaps, while AWS, Google Cloud, and Oracle have confirmed Xeon 6 adoption and memory-heavy workloads that support a high readiness assessment. Meta and ByteDance operate massive workloads of recommendation, ranking, and retrieval, theoretically having the highest incentive to adopt MRDIMM.

EXHIBIT 23: The adoption of Xeon 6 and AMD Venice by CSPs is a signal for the potential MRDIMM deployment, while workload mix determines actual attach rate. Azure is currently the most visible confirmed adopter of both MRDIMM-compatible CPU platforms

	CSP	Xeon 6 adoption	AMD Venice adoption	Memory-bound workload	MRDIMM current adoption probability
1	AWS	Confirmed – R8i / R8i-flex on custom Intel Xeon 6; X8i GA (Jan-26), marketed for very high memory bandwidth	Not publicly confirmed	Memory-optimized DB, in-memory cache, AI inference	High
2	Google Cloud	Confirmed – C4 GA on Xeon 6; X5 announced for large-memory SAP/HANA-style workloads	Not publicly confirmed	ML recommendation, large-memory DB / analytics	High
3	Microsoft Azure	Confirmed – Didsv7 / related Granite Rapids VMs	Confirmed – HDv2 and HXv2 on 6th Gen AMD EPYC ‘Venice’	AI / data processing, HPC, EDA, memory-sensitive VMs	Very High
4	Oracle OCI	Confirmed – X12 Standard Acceleron on Intel Xeon 6	Not publicly confirmed	DB, HPC, CPU inference, agentic AI	High
5	Meta	No public confirmation	No public confirmation	Recommendation / ranking, large-scale data serving	Medium-High
6	Alibaba Cloud	Platform readiness / no instance confirmation (Alibaba Cloud Linux 3 supports Xeon 6); no public instance confirmation	Not publicly confirmed	Cloud DB, analytics, AI services	Medium
7	Tencent Cloud	No public confirmation	Not publicly confirmed	Cloud / recommendation / data services	Medium
8	ByteDance	No public cloud-instance confirmation	Not publicly confirmed	Recommendation, embedding lookup, retrieval	Medium-High
9	Baidu AI Cloud	No public confirmation: public cloud materials still mostly reference earlier Xeon generations	Not publicly confirmed	Search / AI / retrieval / DB	Medium

MRDIMM readiness is based on our assessment

Source: companies reports, Bernstein analysis

COMPETITIVE DYNAMICS AND INDUSTRY MOAT

The core memory interface chip market is a textbook oligopoly with ~92% share held by three players. High barriers to entry protect incumbents, including JEDEC certification, multi-party validation, and advanced node fabrication. The complementary chip market is more fragmented, with many conventional analog or sensor suppliers.

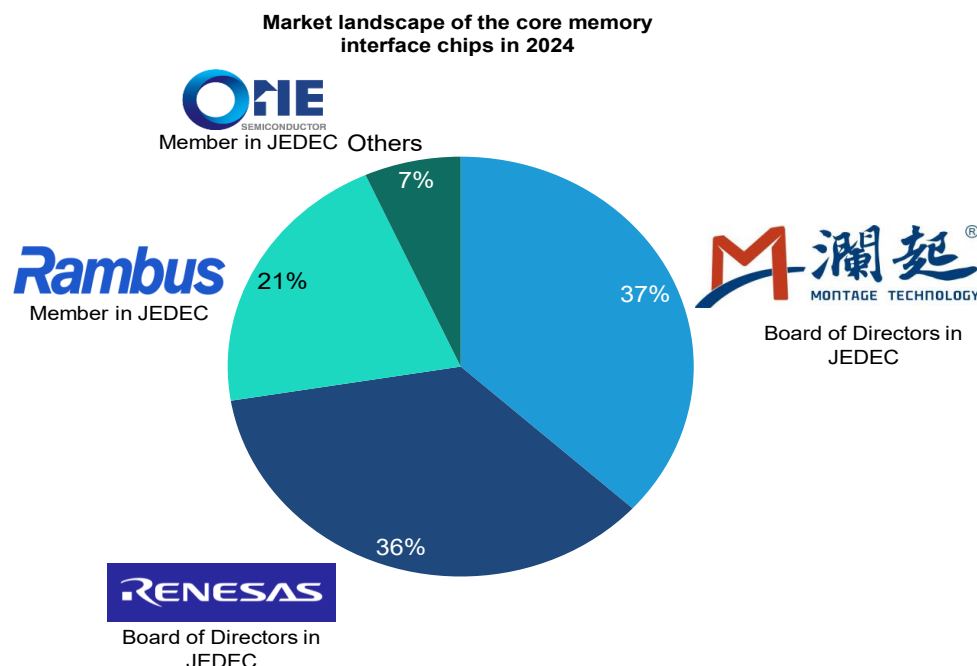
CORE MEMORY INTERFACE CHIPS MARKET: A CONSOLIDATED OLIGOPOLY

The core memory interface chip market exhibits a degree of concentration that is unusual even by semiconductor industry standards. Three players — Montage Technology (~37% market share), Renesas (~36%), and Rambus (~20%) — collectively controlled approximately 92% of global market share by revenue as of 2024. The only other meaningful participant is One Semiconductor, a Korean company that has emerged as a small player offering RCD and CKD products, manufacturing on Samsung Foundry's process platform. One Semiconductor was founded by the former Head of DRAM Development at Samsung Electronics and the former Chief Technology Officer of Samsung Electro-Mechanics. Despite its close ties to Samsung, the company has struggled to meaningfully scale its market share, underscoring the exceptionally high technological and ecosystem barriers in this sector.

These structural barriers have progressively eliminated competitors across successive DDR generations, resulting in the current level of industry concentration. Compliance with JEDEC standards demands deep expertise in high-speed serial interface design, analog/mixed-signal engineering, and memory controller protocols. In addition, the multi-party qualification and validation process—spanning foundries (primarily TSMC), CPU vendors (Intel, AMD), JEDEC committee participation, DRAM manufacturers (Samsung, SK Hynix, Micron, CXMT), module vendors (e.g., Kingston), and OEMs/CSPs—typically requires 18–24 months from initial silicon to volume production approval. During this period, any design flaw or interoperability failure can fully disqualify a supplier.

RCD and MRCD are the most technically complex components among the DIMM chipset family, fabricated at leading-edge CMOS nodes. DDR4 RCDs were manufactured on 40nm platforms, while DDR5 first-generation RCDs started at 28nm. Future RCD and MRCD generations for DDR5 will offer faster data rates and could employ even more aggressive technology nodes (down to 10nm below class), raising the technological bar for any potential new entrant.

EXHIBIT 24: The core memory interface chips market is a textbook oligopoly, with top 3 players consolidating 90+% of market share



Source: company reports, Bernstein analysis and estimates

COMPLEMENTARY SUPPORTING CHIPS: A MORE FRAGMENTED MARKET

In contrast to the oligopolistic structure of the memory interface chip market, the complementary supporting chip market (PMIC, SPD Hub, TS) is considerably more fragmented, with dozens of analog and discrete IC vendors participating. The technical barriers are relatively lower: these components do not require the same level of high-speed digital design expertise as RCDs, and they are fabricated on mature process nodes rather than leading-edge CMOS.

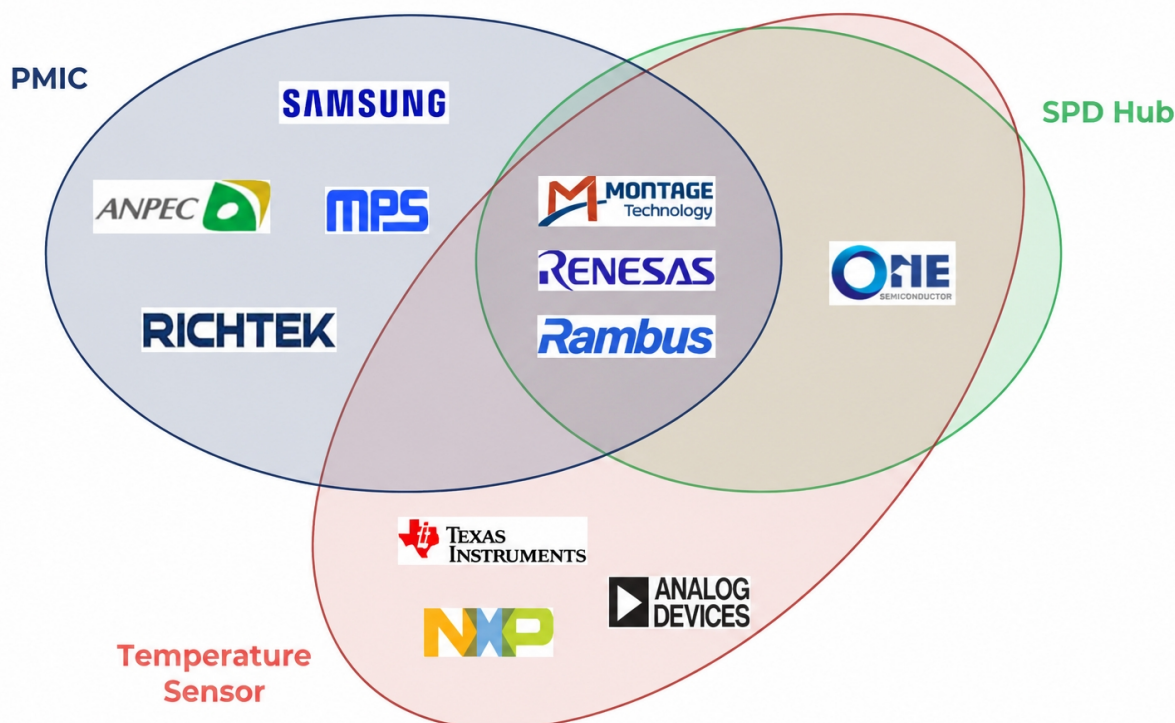
Renesas is the market leader in complementary chips with the largest market share of DDR5 PMIC/SPD/TS, leveraging the deep analog product expertise it acquired through its 2019 acquisition of Integrated Device Technology (IDT) for \$6.7 billion. IDT had been the market leader in DIMM chipsets prior to the acquisition, and Renesas used the deal to reinforce its position in advanced communications and automotive mixed-signal products.

Montage achieved portfolio completeness at DDR5 launch in October 2021, but partly through manufacturing partners: Giantec Semiconductor (Shanghai) for SPD Hub and Global Mixed-mode Technology (Taiwan) for PMIC. This preserved R&D focus on core RCD/DB competency. More recently, Montage appears to be building in-house PMIC and SPD design capability, with its US office now recruiting analog engineers for these products.

Rambus initially lacked complementary chips, a noted disadvantage; but it closed the gap through sequential in-house development: SPD Hub and TS in July 2022, server PMICs in April 2024, and the PMIC5030 for MRDIMM in October 2024.

Other notable participants in the complementary chip space include Samsung, Monolithic Power Systems, Richtek, Texas Instruments, NXP, and Analog Devices. The lower barriers and higher number of participants mean that pricing power in complementary chips is significantly weaker than in memory interface chips.

EXHIBIT 25: **Besides the three leading players with full chipsets, a range of analog and discrete suppliers also provide complementary supporting chips**



Source: Yole, Bernstein analysis

EVALUATION OF THREE CORE MEMORY INTERFACE CHIP PLAYERS

Montage, Renesas, and Rambus collectively controlled over 90% of global core memory interface chips market share in 2024. Barriers to entry are formidable: chips must pass 18–24 month qualification cycles with DRAM vendors, CPU platform providers, and CSPs, while incumbents shape the JEDEC specifications that new entrants must meet. Suppliers who fall behind a DDR generation have historically been eliminated. We evaluate the three players across product portfolio breadth, DDR5 RDIMM roadmap, and MRDIMM positioning.

Product portfolio breadth: all three now offer complete DDR5 chipsets, but through different paths

DDR5's expanded module architecture introduced PMIC, SPD Hub, and Temperature Sensor alongside the traditional RCD and DB, making a complete chipset offering a competitive imperative for module manufacturers seeking one-stop sourcing.

Renesas entered DDR5 with the broadest in-house portfolio from the outset, inheriting deep analog and power management expertise through its 2019 acquisition of IDT (USD 6.7Bn), which had launched the industry's first DDR5 PMIC in 2017 and held the largest PMIC share among the three incumbents.

Montage achieved portfolio completeness through manufacturing partnership at the beginning, but is gradually building up in-house design capabilities for complementary supporting chips.

Rambus initially lacked complementary chips, but has closed the gap through sequential in-house development since 2022. With this build-out complete, Rambus now offers a fully in-house chipset for both RDIMM and MRDIMM.

In short, portfolio breadth is no longer a differentiator. The remaining distinction lies in the development model: Renesas leads on in-house analog heritage, Rambus has caught up through organic investment, and Montage bridges the gap through a hybrid

of partner-sourced and increasingly in-house capabilities.

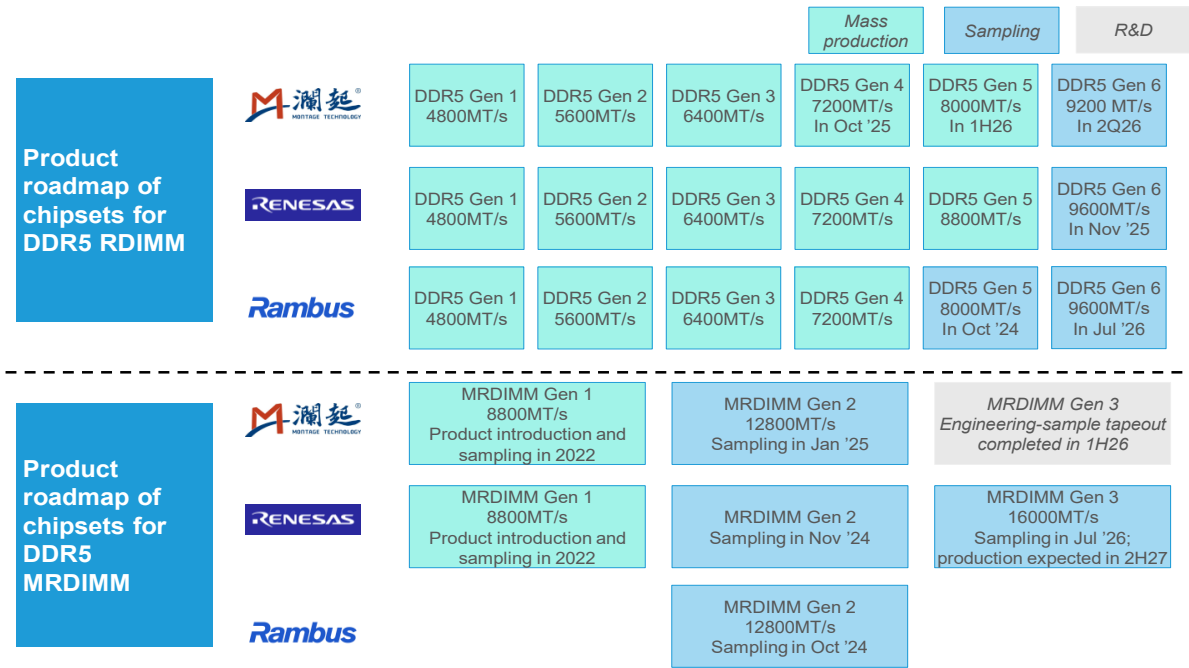
DDR5 RDIMM roadmap: Renesas maintains a narrow lead, while Montage and Rambus continue to narrow the gap

Renesas remains marginally ahead in DDR5 RDIMM chipsets, having started sampling Gen 6 devices (9600 MT/s) in November 2025 after achieving mass production of Gen 5 (8800 MT/s). Montage reached mass production of Gen 4 (7200 MT/s) in October 2025 and Gen 5 (8000 MT/s) in 1H26, while Gen 6 (9600 MT/s) remains under sampling. Rambus started customer sampling of Gen 5 (8000 MT/s) in October 2024 and Gen 6 (9600 MT/s) in July 2026. Overall, the three players remain closely matched technologically, with practical competitive implications moderated by server platform qualification cycles that tend to compress vendor timing advantages within each CPU generation.

DDR5 MRDIMM chipsets: Montage and Renesas remain leading innovators, with Renesas currently ahead on Gen 3 development

For MRDIMM Gen 2 (12800 MT/s), all three vendors are now actively competing, with Rambus beginning sampling in October 2024, Renesas in November 2024, and Montage in January 2025. Looking beyond Gen 2, Renesas currently appears to hold a modest lead in MRDIMM Gen 3 (16000 MT/s), having commenced customer sampling in July 2026. Montage has completed engineering-sample tapeout, while Rambus has yet to disclose a Gen 3 roadmap. While Renesas currently leads the Gen 3 roadmap, Montage retains a strong competitive position given its co-leadership in Gen 1 and Gen 2 MRDIMM platforms and its deep engagement with the emerging MRDIMM ecosystem. Ultimately, market-share outcomes are likely to be driven more by customer qualification, ecosystem support and manufacturing execution than by a several-quarter development lead.

EXHIBIT 26: **Three industry leaders remain closely matched in DDR5 RDIMM and MRDIMM offerings. However, Montage and Renesas hold a first-mover advantage in MRDIMM memory interface chips**



Source: companies reports, Bernstein analysis

NETWORK EFFECT AND HIGH SWITCHING COST PROTECT INCUMBENTS

The memory interface chip market possesses deep structural moats, built on reinforcing pillars.

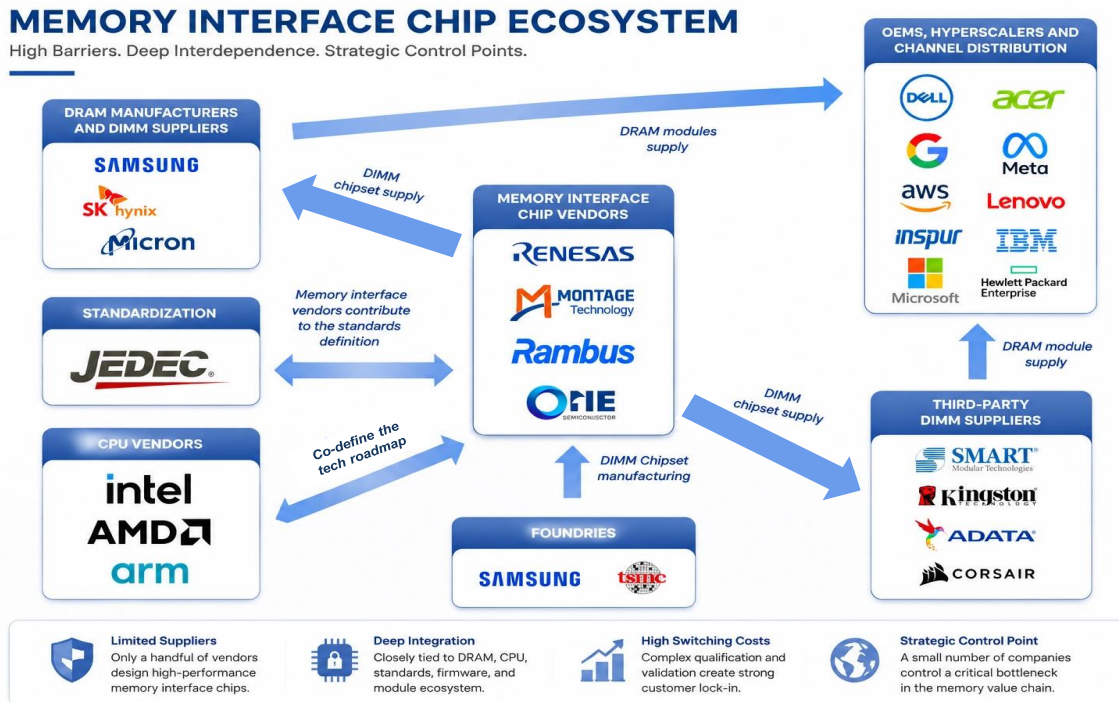
Standards definition through JEDEC participation: The three incumbent suppliers are all active members of JEDEC, the industry body responsible for defining DDR standards. Notably, Montage and Renesas sit on the JEDEC board of directors, granting them early visibility into upcoming specifications and the ability to influence standard setting. This creates a “roadmap definition” dynamic in which incumbents help shape the technical requirements that new entrants must ultimately meet—

effectively defining the competitive landscape on their own terms.

Multi-party validation ecosystem and strong network effects: Bringing a new memory interface chip to market requires rigorous interoperability validation across a complex ecosystem, including foundries (primarily TSMC for advanced nodes), CPU vendors (Intel, AMD), JEDEC committees, DRAM manufacturers (Samsung, SK hynix, Micron, CXMT), module vendors (e.g., Kingston), and end customers such as OEMs and cloud service providers. This qualification process typically spans 18–24 months and must be repeated for each new product generation. For new entrants, this implies a minimum of 3–4 years from initial R&D investment to first meaningful revenue, assuming no qualification setbacks. The deeply interconnected and established ecosystem reinforces both network effects and high switching costs, creating strong structural protection for incumbent players.

Industry consolidation across technology generations. Historical industry dynamics point to increasing consolidation over successive DDR generations. The number of viable suppliers has declined rather than expanded: Intel exited after DDR2 due to insufficient returns, and Texas Instruments followed after DDR3. The remaining players have each invested billions of dollars over multiple DDR cycles in IP, engineering capabilities, and foundry partnerships. For potential new entrants, the combination of deep technical expertise requirements, lengthy qualification cycles, and the risk of missing critical generational transitions, which has historically proven fatal, results in prohibitively high barriers to entry.

EXHIBIT 27: Well-established ecosystem ties and close collaboration requirements have formed high barriers that safeguard incumbents in the core memory interface chips sector



Source: Yole, Bernstein analysis

CUSTOMER ECONOMICS SUPPORT SUPPLIER PROFITABILITY

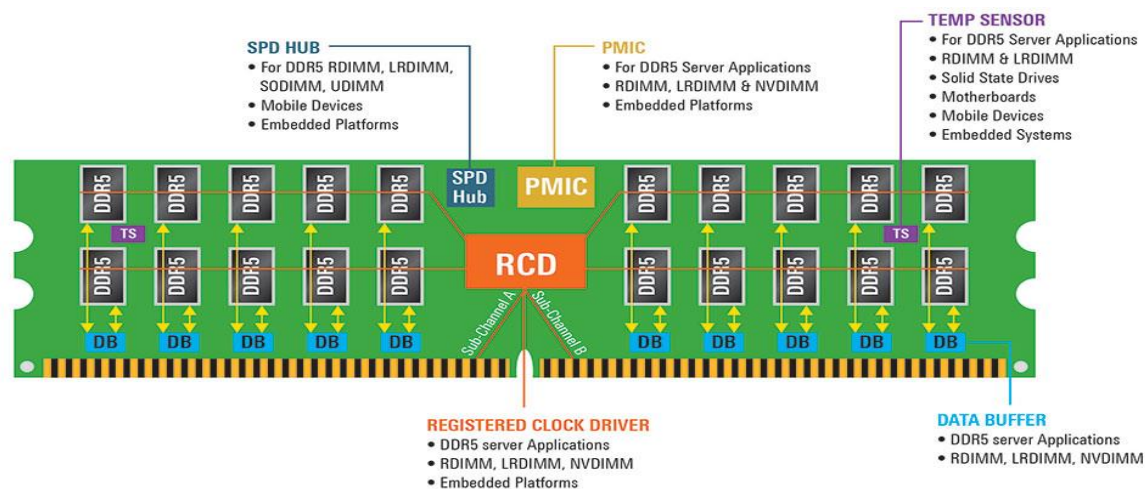
Interface chips have a greater impact on system performance than their share of module cost suggests. A malfunction or compatibility problem can affect memory stability and the operation of the server. **The savings from a lower-priced component must therefore be weighed against qualification costs and operational risk.**

For a customer purchasing a high-value memory module, a small reduction in interface-chip price offers limited benefit, if it requires substantial validation work or introduces uncertainty. Suppliers with a proven record can consequently compete on reliability, availability and roadmap execution as well as price.

These economics support incumbent profitability, although they do not eliminate competition or normal price declines. Sustained returns still require suppliers to maintain technical competitiveness and secure positions on the next generation of products.

DDR module chipsets are a family of specialized ICs mounted on server DIMM modules. They divide into two tiers — high-value memory interface chips (RCD, DB, MRCD, MDB) and complementary supporting chips (SPD, TS, PMIC) — with silicon content per module increasing dramatically across DDR generations and DIMM formats. Memory interface chips are the core high-value silicon, and Complementary supporting chips have grown in relevance since the DDR5 generation.

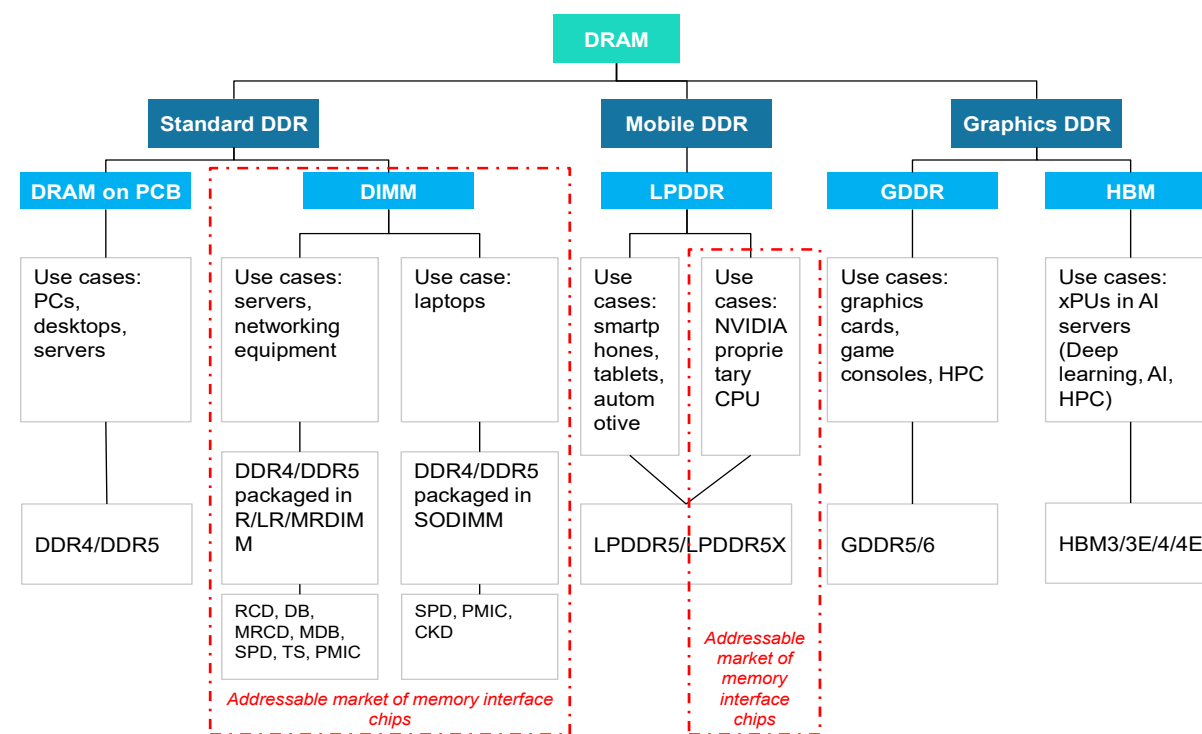
Memory interface chips	Core interface chips	• RCD: registered clock driver, compatible with RDIMM • DB: data buffer, compatible with RDIMM • MRCD: multiplexed rank registering clock driver, compatible with MRDIMM • MDB: multiplexed data buffer, compatible with MRDIMM
	Complementary supporting chips	• SPD: serial presence detect • PMIC: power management IC • TS: temperature sensor • CKD: clock driver, compatible with SODIMM/ UDIMM in PC



Source: Renesas, Bernstein analysis

Not all DRAM packages require memory interface chips. In PCs and low-end servers, where memory capacity and bandwidth requirements are modest, the CPU's integrated memory controller can communicate directly with DRAM chips without intermediary buffering. However, in high-end servers, particularly AI servers that demand maximum memory bandwidth and capacity, the CPU cannot talk to too many DRAMs simultaneously without signal degradation. This is where DDR module chipsets become essential.

EXHIBIT 29: **Among all DRAM variants, DDR packaged in DIMMs requires RCD/MRCD and DB/MDB to interface with server CPUs, while LPDDR used by NVIDIA Vera CPUs also needs some simplified memory interface chips**



Source: Semiconductor Engineering, Bernstein analysis

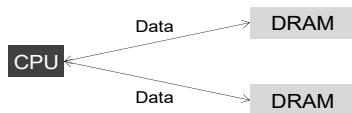
CORE INTERFACE CHIPS

Core interface chips are the most technically sophisticated and commercially valuable components in the DDR module chipset. They sit physically between the CPU memory controller and the DRAM dies on the module, buffering and conditioning signals to enable higher memory capacity and bandwidth than would otherwise be possible. A useful analogy: if the CPU is the "CEO" of a company, the RCD/MRCD is the "manager" that directs instructions, and the DB/MDB is the "delivery person" that physically moves data to and from the DRAM workforce.

EXHIBIT 30: Memory interface chip: core networking chip to connect CPU and DRAM in servers to support more memory capacity/bandwidth

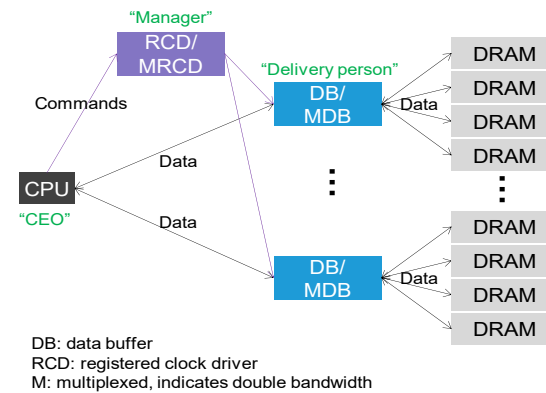
DRAM for PC and low-end servers: CPU connect to DRAM directly

- Limited requirement on the DRAM capacity/bandwidth
- CPU can talk to DRAM directly for data read/write



DRAM for high-end servers: CPU connect to DRAM through interface chips

- More DRAM capacity/bandwidth needed, especially for AI servers, but CPU as the “CEO” cannot talk to too many DRAMs at the same time
- **DB/MDB** as the additional “delivery person” to help CPU distribute and deliver the data to more DRAM
- **RCD/MRCD** as the additional “manager” to control the DB/MDB when and which to send data to CPU



Source: company reports, Bernstein analysis

RCD (Registered Clock Driver): The RCD is the bridge between the memory controller and the DRAM chips on the module. It provides command, address, and clock signals to achieve synchronous data transfer. RCDs are employed on RDIMM and LRDIMM server modules to enable higher capacity and better reliability than PC modules (UDIMM and SODIMM). One RCD is mounted per RDIMM module. The main functional blocks on a typical RCD chip include a logic microcontroller (usually ARM-based) for managing I/O ports, an I2C/I3C interface for inter-chip communication, and a clocking block for handling reference signals.

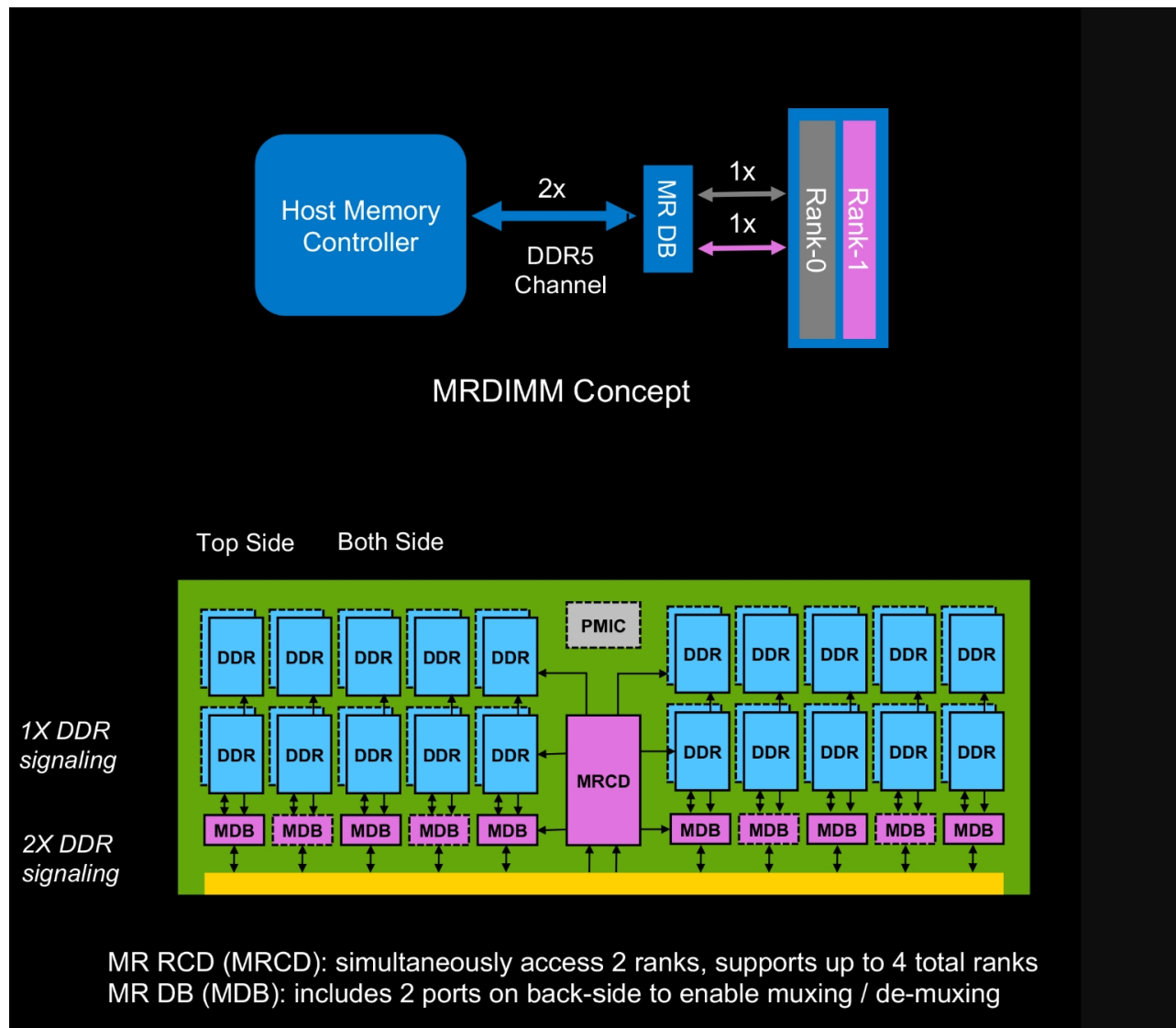
MRCD (Multiplexed Rank RCD): The MRCD is the next-generation registering clock driver designed specifically for DDR5 MRDIMM modules. Unlike conventional RCDs that passively buffer signals, the MRCD actively reshapes and re-clocks signals, maintaining signal integrity at the higher data rates that MRDIMM targets (8800–12800 MT/s). One MRCD is mounted per MRDIMM module.

RCDs/ MRCDs are the most complex and advanced ICs among the DIMM chipset. They were first introduced with DDR3 and have gained relevance with each subsequent generation. The leading vendors typically manufacture their RCDs using TSMC’s CMOS processes, with DDR4 RCDs on 40nm platforms and DDR5 first-generation RCDs starting at 28nm. Future generations will migrate toward even more aggressive nodes (down to 10nm class) to support higher data rates.

DB (Data Buffer): Data buffers store copies of data blocks being transferred between the CPU and the DRAM dies. Positioned spatially closer to the DRAM chips than RCDs, they reduce I/O trace lengths, leading to less pronounced signal reflection and lower data load on the module’s data bus. Consequently, modules with data buffers (LRDIMMs) can support higher memory capacities than RDIMMs. Data buffers were first implemented on DDR4 LRDIMMs, with 9 per module. Since JEDEC defines a specific number of DBs per module, they are sold in bundles to memory module vendors.

MDB (Multiplexed Rank Data Buffer): The MDB is the next-generation data buffer for DDR5 MRDIMM. Like the MRCD, it actively manages data signals through multiplexing, enabling concurrent access to two memory arrays on the module, effectively doubling per-module bandwidth versus conventional RDIMM. 10 MDBs are mounted per MRDIMM module, representing a dramatic increase in silicon content compared to DDR5 RDIMM, where no data buffers are required at 6400 MT/s.

EXHIBIT 31: The mechanism of MRCD and MDB in MRDIMM: MRCD allows simultaneous access to both ranks, and MDB enables muxing and demuxing. Hence, two DDR ranks run in a multiplexed mode to double the speed



Source: Micron, Bernstein analysis

COMPLEMENTARY CHIPS

The DDR5 specification introduced several new on-module components that were absent in DDR4, expanding the total chipset content per module and creating additional revenue opportunities for chipset suppliers.

SPD Hub (Serial Presence Detect): The SPD Hub stores module configuration information — including size, speed, voltage, and timing parameters — that is read by the system BIOS during boot. It contains an embedded temperature sensor and manages the standalone temperature sensors mounted on the DIMM. A 1KB EEPROM device is also included to store DIMM topology configuration and DRAM information. One SPD Hub is present on every server DDR module across all formats and generations.

TS (Temperature Sensor): DDR5 temperature sensors must achieve approximately 0.25°C accuracy to trigger effective thermal management actions, including lowering IC speed or increasing cooling fan activity. Two standalone temperature sensors are mounted on each DDR5 server module (RDIMM and MRDIMM), compared to zero standardized sensors on DDR4 RDIMM modules. The sensors are managed by the SPD Hub through the I3C serial bus.

PMIC (Power Management IC): The transition to DDR5 moved power regulation from the motherboard to the DIMM module itself. DDR5 PMICs include step-down switching regulators and low-dropout regulators to manage voltage conversion and distribution. This on-module power management improves power efficiency while reducing vulnerability to power noise on the motherboard — a critical advantage for high-density server deployments where multiple DIMMs operate in close proximity. One PMIC is present on each DDR5 module.

DECODING THE TECH BEHIND MRDIMM

FOUR FUNDAMENTAL PATHS TO BOOST MEMORY BANDWIDTH

The peak bandwidth between CPU and DRAM is measured by how many data can go through the CPU host, measure by **gigabyte per second (GB/s)**. If we break down the bandwidth calculation into different elements, it follows a simple equation:
CPU Bandwidth (GB/s) = native DRAM data rate (GT/s) x Rank count (#) x interface width (bits) x channel count (#) / 8

These four parameters capture different aspect of data transfer:

1. **Native DRAM data rate:** how fast data package moves out of DRAM. This is measured by giga transfers per second (**GT/s**). A transfer does not specify how many payload bits or bytes it carries, which will be defined by interface width.
2. **Rank count:** A **rank** is a group of DRAM devices selected together to supply data across the relevant interface. Think about this as the smallest package of data that can be transferred at a time coming out of the interface. RDIMM package can only run one rank at a time, while MRDIMM package can run two ranks at a time.
3. **Interface width** measures the number of payload bits transferred simultaneously. For server DRAM this is generally 64 bits. For HBM, the interface width will be much bigger, for example HBM3 has 1,024 bits, while HBM4 has 2,048 bits.
4. **Channel count** measures the number of independent memory paths whose bandwidth adds across the CPU.

Dividing by 8 converts bits into bytes, as 1 byte = 8 bits.

There four parts essentially capture how fast data package moves (1), how much data can be moved in one package (2,3), and how many packages can be delivered in parallel (4). Using cargo truck shipment in freight system as analogy Exhibit 32:

1. DRAM data rate is how fast a truck can move;
2. Rank count is how many stacks of cargo container per truck (package delivery in parallel);
3. Interface width is how long the cargo container is (smallest package);
4. Channel count is how many independent routes trucks can move in parallel.

AMD's EPYC 9006 (Venice) server CPU provides an illustrative example. Its top memory configuration supports 16 DDR5 channels and Multiplexed Rank Dual Inline Memory Module (MRDIMM) modules at 12,800 MT/s (6,400 MT/s with 2 ranks in parallel), equivalent to 12.8 GT/s. Each channel carries 64 bits per offloading operation. It therefore reaches 1.6TB/s of peak bandwidth per CPU, with the calculation process below:

Venice peak BW = 6,400 MT/s x 2 ranks x 64 bits x 16 channels / 8 = 1.6 TB/s

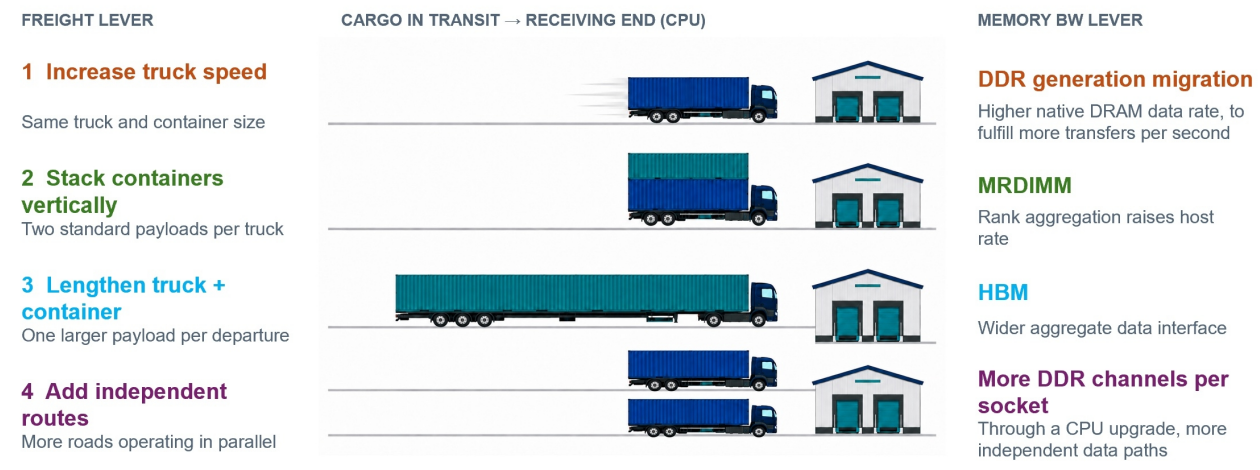
For MRDIMM (Multiplexed Rank Dual In-line Memory Module), effective data rate measures transfers per second on each data pin. The effective data rate can be decomposed further to two sub factors:

Effective host data rate = native DRAM data rate x rank-stream aggregation factor.

Conventional RDIMMs transfer data from one rank at a time. MRDIMM combines two concurrent rank streams through its data buffers, allowing the host interface to operate at approximately twice the data rate of either rank. Two rank-side streams at 6,400 MT/s can therefore feed a host interface at 12,800 MT/s, which is the key technology innovation to deliver higher bandwidth. As a result, MRDIMM use the same memory die (the DRAM data rate is the same as RDIMM), but delivers 2X effective host data rate.

EXHIBIT 32: **The freight system offers the simplest analogy. Higher bandwidth equals to more cargo received per second. Fundamentally, there are four routes to deliver more cargo**

$$\text{BW (GB/s)} = \text{native DRAM data rate (GT/s)} \times \text{Rank count(\#)} \times \text{Interface width (bits)} \times \text{Channel count} \div 8$$



conceptual analogy; MRDIMM serializes two rank streams onto a faster host link of unchanged width, and it does not widen the interface or physically stack DRAM
 Source: Bernstein analysis

As a result, there are four distinct hardware routes to expand memory bandwidth: increase native DDR data rate, increase rank-stream aggregation, widen the interface, or add more channels. MRDIMM doubles the host interface speed by exploiting rank-level parallelism, offering the lowest-disruption and cost-efficient path

Route 1: increase native DRAM data rate; deliver trucks more frequently. Faster DRAM generations shorten the interval between transfers and raise the data rate of the DDR cells. This route depends on DRAM process upgrade, I/O signaling and signal integrity advances. The scaling cadence is therefore tied to DRAM generations, while power and timing margins become progressively harder at higher speeds. DDR5 speed improvement is approaching the ceiling.

Route 2: exploit rank-level parallelism; stack two containers vertically on each truck. Two depots supply separate containers concurrently, and MRDIMM stack these two containers vertically on one truck. Each container retains its original size. The doubled cargo is converted into a faster sequence of transfers, increasing effective host data rate. The MRCD coordinates rank access, and MDBs multiplex the two rank-side streams onto the faster host interface. The channel count and 64-bit width remain unchanged, and DRAM cells do not need to run at the 2x host rate. MRDIMM instead double the trucks traffic in one time slot on the existing roads, resulting in higher loaded throughput.

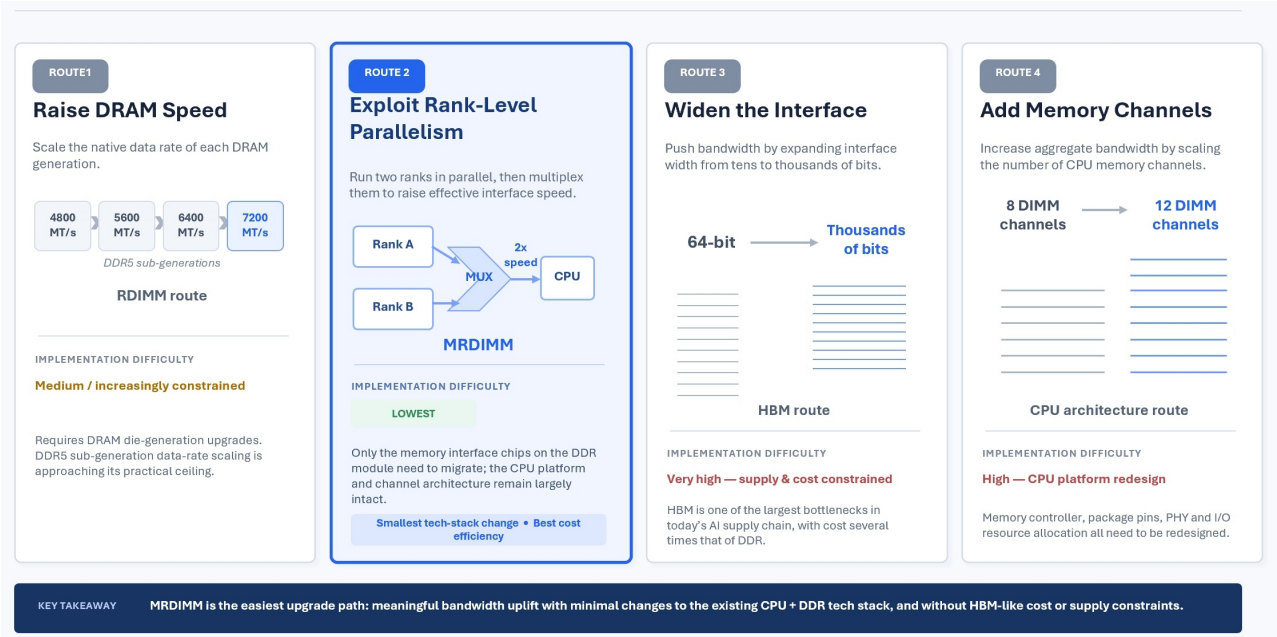
Route 3: widen the interface; lengthen the truck and its container. Each delivery carries one larger payload, which the warehouse accepts in a single unloading operation through an expanded receiving interface. Unlike Route 2's two vertically-stacked containers, this Route operates one longer container to transfer more bits simultaneously. It expands interface width, while allowing transfer frequency to remain unchanged. Enabled by dense I/O connections and close physical placement, HBM illustrates this approach through thousands of bits wide versus 64 bits for a DDR channel. The result is very high bandwidth, but it requires close memory placement, advanced packaging and a different memory supply and cost structure. Capacity and serviceability are also trade-offs.

Route 4: add memory channels; open more delivery lanes. Each lane carries an independent stream of trucks, so adding lanes increases aggregate cargo received. However, every channel requires another memory-controller interface, PHY resources, CPU package pins, socket connectivity, and motherboard redesign. The bandwidth uplift is direct, but the change sits at the CPU-platform level and consumes scarce package and board area.

The location of the architectural change determines disruption. Route 1 waits for the next generation of DRAM process node. Route 3 redesigns the CPU package, socket and board. Route 4 changes the physical memory architecture and packaging ecosystem. MRDIMM takes the Route 2 and concentrates most incremental complexity on interface chips on the DIMM module. The CPU memory controllers must still support the MRDIMM protocol and the system must qualify higher module power and thermals, but channel count, bus width and the broader DDR5 module ecosystem remain familiar. Hence, **MRDIMM is the**

narrowest architectural intervention that can deliver a step-change in host-side bandwidth.

EXHIBIT 33: Among four tech routes to increase memory bandwidth, MRDIMM doubles the host interface speed by exploiting rank-level parallelism, offering the lowest-disruption and cost-efficient path

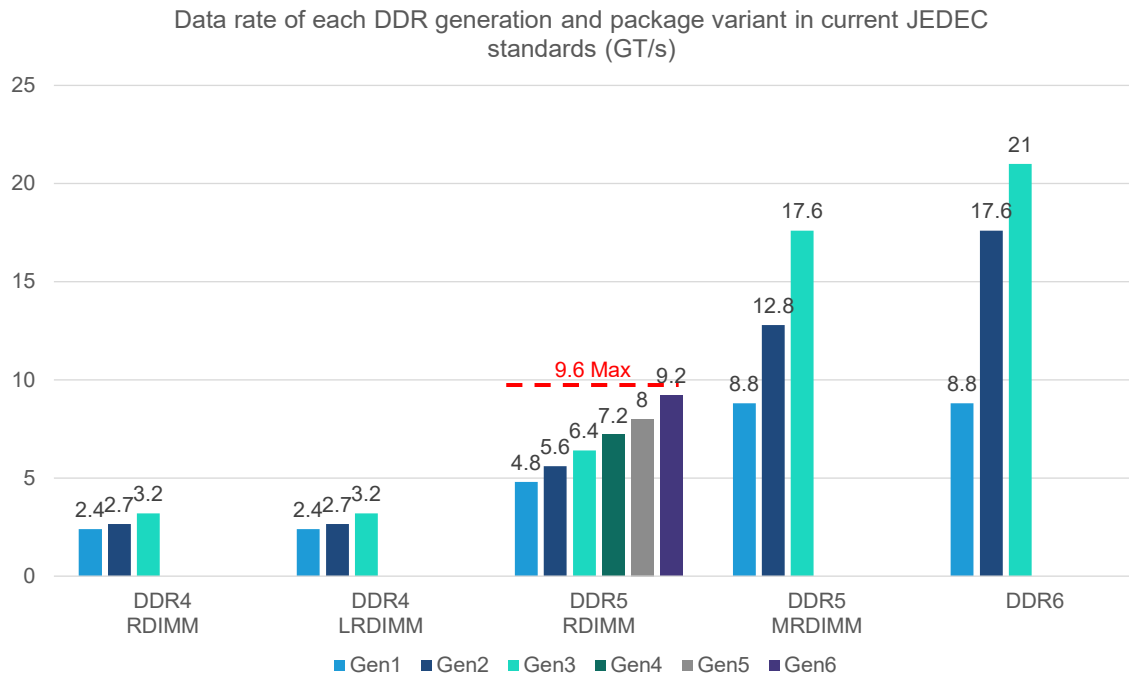


Source: company reports, Bernstein analysis

MRDIMM DOUBLES HOST-INTERFACE SPEED BY EXPLOITING RANK PARALLELISM

In JEDEC’s current roadmap, DDR5 RDIMM progresses from 4,800 MT/s in Gen1 to 9,200 MT/s in Gen6, while MRDIMM standards advance from 8,800 MT/s to 17,600 MT/s across three generations. MRDIMM therefore reaches DDR6-class data rates, while continues to use DDR5 DRAM technology and a DIMM-based physical architecture.

EXHIBIT 34: DDR5 MRDIMM doubles RDIMM bandwidth and achieves DDR6-class data rates, while leveraging the existing DDR5 die and physical architecture

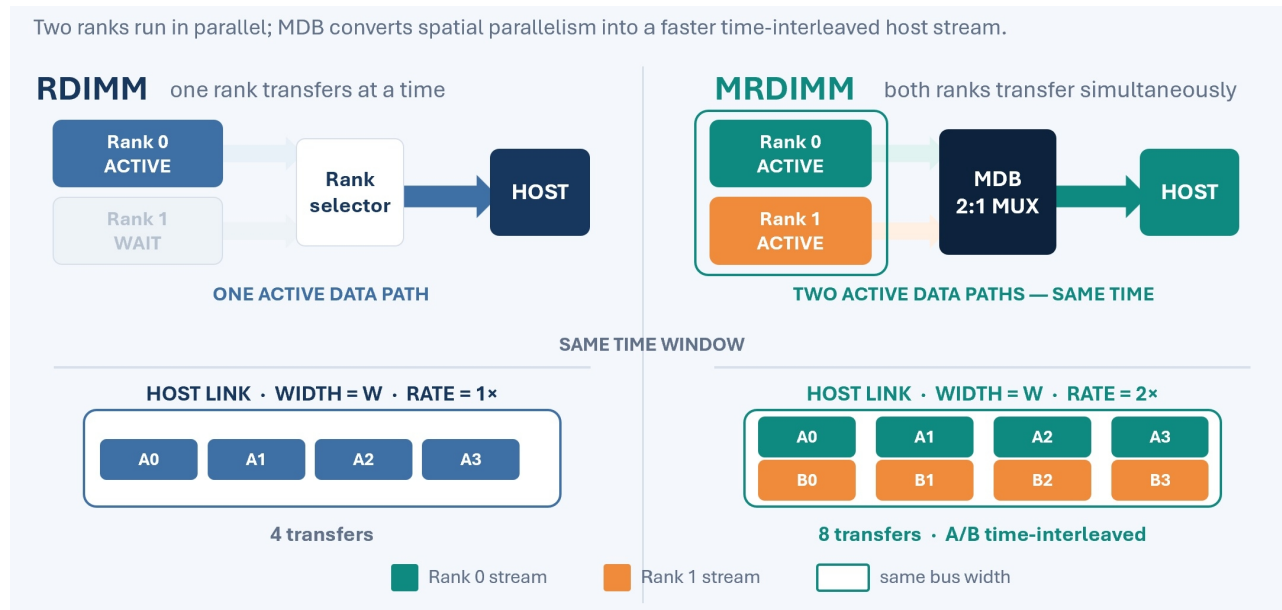


JEDEC's earlier long-term roadmap targeted up to 17.6GT/s for MRDIMM Gen3, while current Gen3 memory interface chipset by Renesas and Montage support 16GT/s

Source: JEDEC, Bernstein analysis

The mechanism for MRDIMM is rank-level parallelism. A conventional RDIMM selects one rank at a time, leaving the second rank idle during the same transfer window. MRDIMM's MRCD can address two ranks concurrently, while the MDBs multiplex and time-interleave the two data streams before transmitting them to the host. The channel remains 64 bits wide, but it carries twice as many transfers in the same period.

EXHIBIT 35: **MRCD allows simultaneous access to two DDR ranks, and MDB performs muxing and demuxing. Together, two DDR ranks run in a multiplexed mode to double the speed and correspondingly the bandwidth, without broadening the host interface channel width (still in 64bit)**



Source: JEDEC, Bernstein analysis

MRDIMM OPTIMIZES TASK-LEVEL LATENCY BY REDUCING THE QUEUING DELAY

Although MRDIMM introduces a small amount of additional intrinsic latency due to the active interface chips in its data path, its higher interface bandwidth can significantly reduce task-level latency that is often dominated by queuing effects. Under light traffic, RDIMMs may retain a modest latency advantage because MRDIMMs add multiplexing, buffering, and retiming overhead. However, the benefits of MRDIMM become increasingly apparent as memory utilization rises. Near saturation, queuing delay in RDIMMs grows nonlinearly, whereas MRDIMM's higher service bandwidth helps maintain lower utilization and shorter memory-request queues.

High-concurrency workloads, such as recommendation systems, retrieval applications, CPU inference, and data-movement-intensive tasks, can benefit substantially because both aggregate throughput and tail latency are strongly influenced by queue depth. Consequently, MRDIMM's primary value proposition is the ability to deliver higher throughput while maintaining lower latency under high utilization.

EXHIBIT 36: With sufficient bandwidth, MRDIMM can substantially reduce queuing delay, resulting in system-level latency improvement for high-load tasks

MRDIMM shifts the memory queueing wall to the right

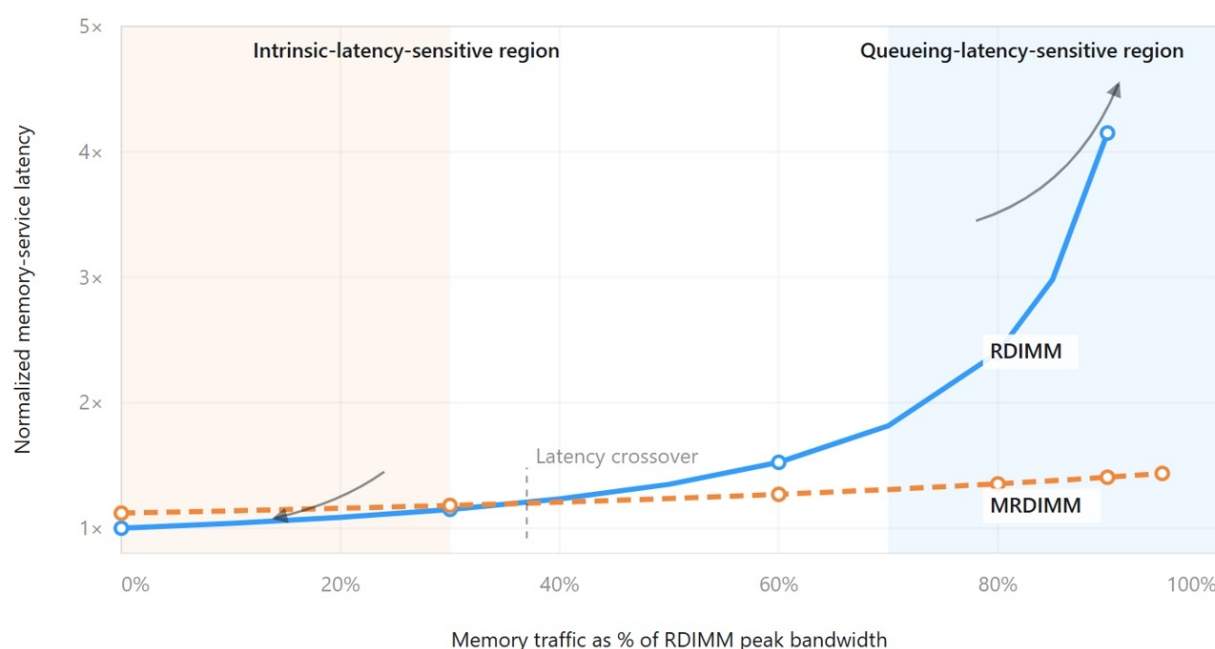
Higher service bandwidth trades modest unloaded latency for materially lower loaded latency.

Low load: intrinsic latency matters

With little queueing, RDIMM retains a small latency advantage; MRDIMM carries added mux and buffer-path overhead.

High load: queueing latency dominates

RDIMM latency rises non-linearly near saturation, while MRDIMM's higher bandwidth keeps utilization—and queueing—lower.



Illustrative normalized model, not measured benchmark data. MRDIMM effective service bandwidth assumed at ~2x RDIMM.

The line chart is a illustrative normalized model, not measured benchmark data. MRDIMM effective bandwidth is assumed at 2x RDIMM
Source: company reports, Bernstein analysis

POWER, THERMAL DENSITY, AND PHYSICAL SPACE ARE THE PRIMARY TRADE-OFFS

MRDIMM buys bandwidth through active electronics and higher I/O activity. The MRCD, ten MDBs, and simultaneous rank operation **increase module power and thermal density versus RDIMM, making cooling design the principal architectural constraint**. Tall-form-factor (TFF) MRDIMM expands PCB and cooling surface area to relieve local thermal challenges, but consumes more system space and can complicate airflow or liquid-cooling layouts.

However, module watts alone do not determine the system-level efficiency outcome. If higher DDR bandwidth raises CPU utilization, reduces request queuing or protects the utilization of much more expensive accelerators, the system can still deliver attractive workload throughput per watt. Nevertheless, we do not assume this benefit for every workload; power-constrained and low-bandwidth systems can rationally remain on RDIMM or select a more compact architecture.

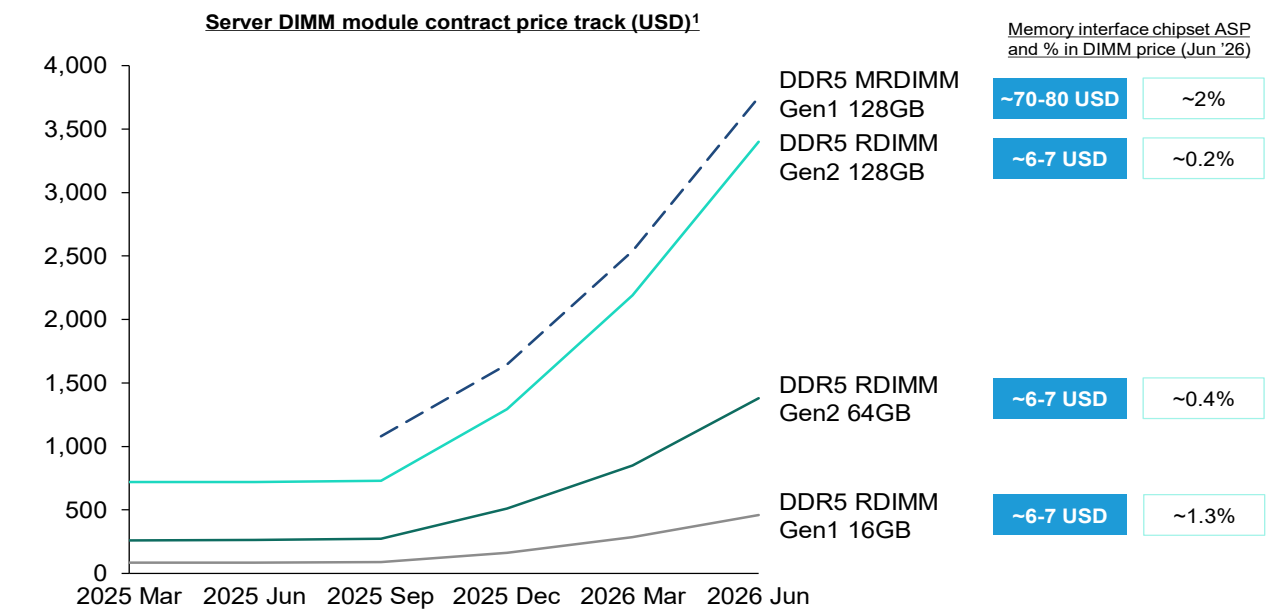
MRDIMM REQUIRES MODEST INCREMENTAL MODULE BOM

The additional BOM derived from interface chips is only LSD in the DRAM module ASP. The MRDIMM interface chipset sold in USD70-80 represents only 2% in the module price of a 128GB DDR5 MRDIMM at present. The extra silicon is

substantial for interface-chip suppliers, but remains limited for the server buyers.

The percentage premium also falls mechanically, when DRAM prices hike. Because the interface-cost increment is relatively fixed, while DRAM dies dominate total module value. We estimate MRDIMM currently carries a USD300-350 premium over RDIMM at the same capacity, equivalent to roughly 10% at current 128GB module price. This narrows the adoption barrier from economic perspective, as the bandwidth problem becomes more acute.

EXHIBIT 37: **Memory interface chips account for only an LSD portion of module costs. Rising memory prices are also reducing MRDIMM's premium over RDIMM to around 10%, reducing adoption barriers of MRDIMM**



MRDIMM remains a niche product without an established contract pricing track record. The MRDIMM pricing shown in the chart is an estimate based on public information; in general, we estimate MRDIMM enjoys a \$300-350 markup over RDIMM at the same capacity
Source: TrendForce, Bernstein analysis and estimates

MRDIMM VS. SOCAMM2

TECH COMPARISON OF TWO DDR PACKAGE FORMATS FOR SERVERS

MRDIMM and SOCAMM2 should not be framed as a simple winner-takes-all performance contest. MRDIMM prioritizes absolute CPU memory bandwidth, capacity and compatibility with the mature DIMM ecosystem. SOCAMM2 packages LPDDR5X in a compact, detachable module and prioritizes bandwidth per watt and physical density—an attractive objective for NVIDIA's vertically integrated, GPU-centric rack design. At the CPU level, a 16-channel Gen2 MRDIMM configuration delivers roughly 1,638 GB/s versus c.1,229 GB/s for eight SOCAMM2 modules on NVIDIA Vera. The capacity gap is larger: high-density MRDIMM configurations can exceed 16TB per CPU, compared with only 0.75TB SOCAMM2 used by Vera. SOCAMM2's clearest advantage is power efficiency. NVIDIA states that LPDDR5x delivers roughly 2x the bandwidth at the half the power of conventional DDR5 as baseline, implying 4x bandwidth per watt. The size of a SOCAMM2 module is roughly 30% of that of an RDIMM module. The smaller package can free board and airflow space.

In shorthand, SOCAMM2 optimizes GB/s per watt and physical compactness; MRDIMM optimizes total GB/s and GB per socket. But migration from DIMM to SOCAMM2 requires more than substituting one module for another.

EXHIBIT 38: configuration comparison between SOCAMM2 and DDR5 RDIMM



- ~133mm x ~31mm x ~2.5mm
- 288 pins on edge connector
- Vertical, insertion socket



- ~86mm x ~14mm x ~1mm
- 694 pins on back
- Horizontal, screw mounted

Source: AMD website, Bernstein analysis

EXHIBIT 39: MRDIMM maximizes absolute bandwidth and capacity; SOCAMM2 optimizes power efficiency and compactness

DERIVED METRIC	Aggregated bandwidth	÷	Power	=	Bandwidth per W	Superior	Moderate	Inferior
	DDR5 RDIMM (Gen 3)		DDR5 MRDIMM (Gen 2)		LPDDR5x SOCAMM2 for NV Vera (original design)		LPDDR5x SOCAMM2 for NV Vera (adjusted design)	
Data rate	6,400 MT/s (Gen 3)		12,800 MT/s (Gen 2)		9,600 MT/s			
Aggregated bandwidth for CPU	819.2 GB/s for 16-channel CPU (6400*64 bit*16-ch/8)		1,638.4 GB/s for 16-channel CPU (12800*64bit*16-ch/8)		1,228.8 GB/s for NV Vera (9600*128bit*8 modules/8)			
Memory capacity per module	16/32/64/96/128/256 GB		128/256/512 GB		192 GB		96 GB	
Total capacity per CPU	Up to 8TB+ (16-ch * 2 DIMM/Ch * 256GB)		Up to 16TB+ (16-CH * 2 DIMM/Ch * 512GB)		1.5TB (8 SOCAMM2 * 192GB)		0.75TB (8 SOCAMM2 * 96GB)	
Power efficiency	Moderate		Slightly higher than RDIMM, due to additional interface chips		Best-in-class: deliver 2x bandwidth at half the power of traditional DDR			
Bandwidth per W	0.25x (GB/s/W)		0.24x (GB/s/W)		Assuming in x (GB/s/W)			
Compactness	288 pins, requiring vertical slot space		Similar to RDIMM		Highly compact low-profile compression module, smaller footprint			
Price per GB	\$24.3		\$27.7 (small sampling price)		\$29 (small sampling price)			
Per module price	\$3,200+ (128GB)		\$3,550 (128GB)		\$5,600 (196GB)		\$2,800 (96GB)	

Source: NVIDIA reports, Bernstein analysis and estimates

HIGH SWITCHING COST AS THE MOAT OF DIMM ECOSYSTEM

The relevant switching cost sits at the CPU-platform level. A vendor must redesign the memory controller and PHY, package and I/O allocation, board layout, connector, power delivery, firmware, cooling, validation and service model. The redesign requires requalifying the supply chain and forfeiting backward compatibility. NVIDIA can justify this greenfield investment because it controls the GPU, CPU, interconnect, power and rack architecture. Intel, AMD and other ARM vendors already operate inside a mature DDR ecosystem and can obtain higher bandwidth through MRDIMM without absorbing the same system-level disruption.

SOCAMM2 therefore remains a credible NVIDIA-specific solution rather than a broad substitute for MRDIMM. It also adds modest interface-chip content, but only a few dollars per module versus roughly USD70–80 for MRDIMM. We expect its impact on the global memory-interface-chip TAM to remain limited as long as deployment stays concentrated in NVIDIA's proprietary CPU platforms.

CXL - A NEW GROWTH ENGINE FROM DDR4 REUSE

A SHORT PRIMER

The history of CXL

Compute Express Link (CXL) has emerged in less than a decade as one of the most consequential interconnect technologies in modern computing. Born from the industry's growing frustration with memory bottlenecks and stranded resources, CXL has evolved from a niche proposal into a universal standard that is fundamentally redesigning data center architecture. Its history traces a deliberate journey from point-to-point memory expansion to a full fabric for rack-scale composable infrastructure.

Origins: The Memory Wall and the Pre-CXL Landscape

By the late 2010s, the computing industry faced an intensifying "memory wall." CPU core counts were scaling rapidly, driven by Moore's Law and the rise of data-intensive workloads such as AI, in-memory databases, and cloud-native microservices. Yet memory capacity per server was constrained by physical DIMM slots and channel limits, and memory resources were often stranded—underutilized on one server while another nearby starved for capacity. Traditional PCIe, the dominant I/O interconnect, was not designed for cache-coherent memory sharing between processors and devices.

Several proprietary and semi-open approaches attempted to fill this gap, including IBM's OpenCAPI, Gen-Z, and Intel's Omni-Path. Each offered pieces of a solution but suffered from limited vendor buy-in, software incompatibility, or narrow use cases. The industry needed a single, open standard built on the ubiquitous PCIe physical layer that could deliver cache coherence, memory expansion, and pooling across heterogeneous compute environments.

Birth of the CXL Consortium and Version 1.0 (2019)

In March 2019, Intel formally announced Compute Express Link, and the CXL Consortium was incorporated shortly thereafter with founding members including Alibaba, Cisco, Dell EMC, Meta (then Facebook), Google, HPE, Huawei, Microsoft, and Intel. Crucially, rather than inventing an entirely new physical layer, CXL was built atop PCIe 5.0 electrical and physical infrastructure, leveraging the massive existing PCIe ecosystem and enabling reuse of existing PCB designs, connectors, and signal integrity knowledge.

CXL 1.0 and the follow-on 1.1 specification (released in June 2019) introduced three foundational protocols:

- **CXL.io:** Based on PCIe, for traditional I/O operations.
- **CXL.cache:** Allows accelerators to coherently cache host CPU memory.
- **CXL.mem:** Allows the host CPU to coherently access memory attached to a device.

This triad was revolutionary. For the first time, an industry-standard interconnect enabled bidirectional cache coherence between a host CPU and an endpoint device at near-native latencies.

The 2.0 Revolution: Pooling and Switching (2020)

The release of CXL 2.0 in November 2020 marked the technology's maturation from a point-to-point interconnect to a true fabric technology. The most significant addition was **CXL switching**, which enabled memory pooling across multiple hosts. Instead of each server owning its own dedicated memory pool, memory resources could be disaggregated into a shared pool and dynamically allocated to servers on demand.

CXL 2.0 also introduced support for persistent memory, enhanced RAS (Reliability, Availability, and Serviceability) features, and formalized the concept of memory expander controllers—later branded as MXC (Memory eXpander Controller) chips. This version arrived at a fortuitous moment: the AI boom was beginning, and hyperscalers were grappling with skyrocketing memory costs and supply constraints. DDR4 reuse via CXL memory expanders quickly emerged as a killer use case, allowing data centers to reuse millions of legacy DDR4 DIMMs from retiring servers into new DDR5-based AI clusters.

CXL 3.0 and Beyond: Fabric Composability (2022–Present)

Released in August 2022, CXL 3.0 doubled the per-lane bandwidth to 32 GT/s (matching PCIe 6.0) and introduced advanced fabric capabilities including multi-level switching, enhanced memory sharing across many hosts, and support for global shared memory spaces. The specification added flexible port configurations and improved Quality of Service (QoS), laying the groundwork for fully composable disaggregated infrastructure where compute, memory, and storage are independent resources allocated over a CXL fabric.

CXL 3.1, announced in 2024, further refined the standard with focus on AI-optimized features, lower latency modes, and improved management interfaces. By this point, the CXL Consortium had grown to over 300 member companies spanning semiconductors, systems, hyperscalers, and software vendors.

Industry Adoption and the Road Ahead

What began as an Intel-led initiative has become a truly cross-industry standard. AMD, Arm, and all major CPU vendors now support CXL on their server platforms. Marvell, Astera Labs, Montage, and others have launched commercial memory expander controller chips. Hyperscalers including Meta, Google, and Microsoft have publicly disclosed CXL deployments at scale—[Meta's Vistara ASIC](#), for instance, has recycled millions of DDR4 modules across its production fleet.

Looking forward, CXL is positioned to become the backbone of the AI-era data center. As agentic AI workloads drive insatiable demand for memory capacity and bandwidth, CXL's ability to disaggregate, pool, and share memory resources across racks will transition from a cost-saving optimization to an architectural necessity. In barely seven years, CXL has traveled from a technical specification to a transformative force—proof that when the industry unites behind an open standard, it can rewrite the rules of computing architecture.

DDR4 reuse could serve as the first killer application that moves CXL from ecosystem build-out into meaningful deployment, helping improve CSP familiarity, software maturity and hardware availability. Beyond simple expansion, memory pooling can reduce stranded capacity and improve utilization across CPU- and GPU-centric systems, while memory sharing could ultimately reduce data movement and improve system-wide utilization, giving CXL a much longer growth trajectory than DDR4 reuse alone.

Three key use cases with CXL

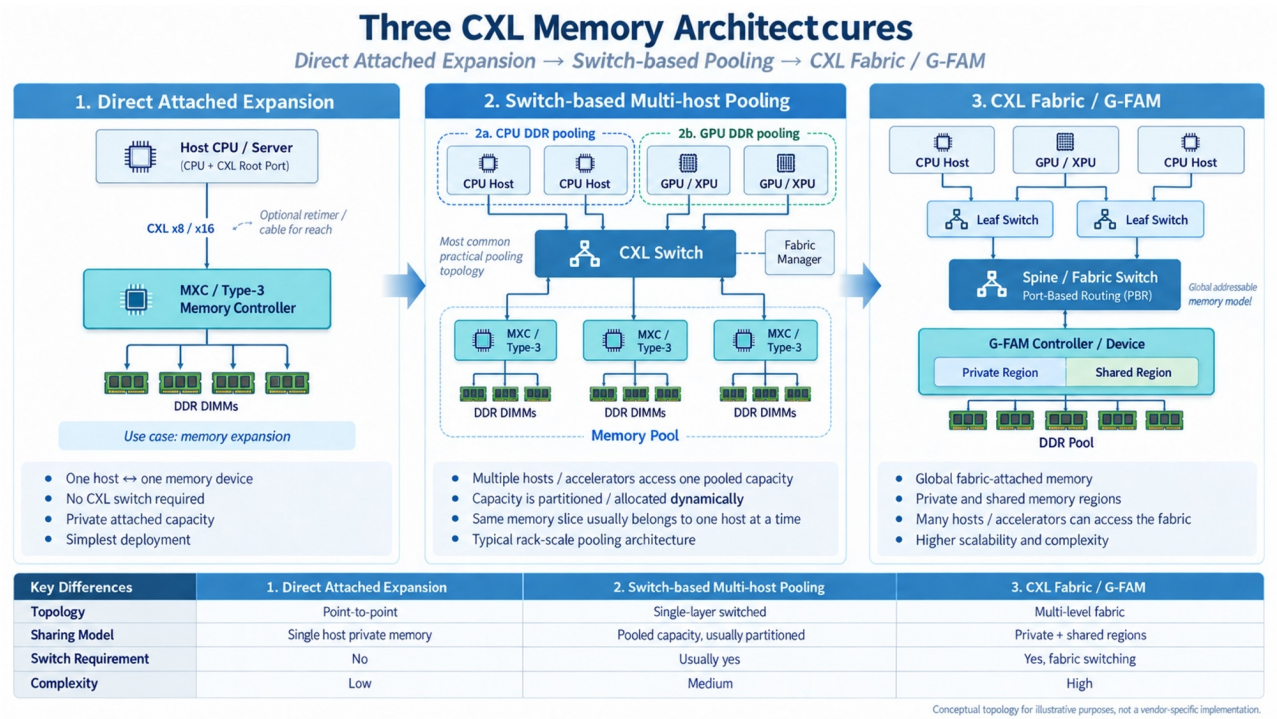
CXL's three memory architectures address three distinct needs: expanding server memory capacity, improving memory utilization across servers, and enabling data sharing across compute nodes. Direct-attached expansion adds dedicated memory to an individual server; switch-based pooling allocates a common pool of DRAM capacity across multiple hosts; and CXL Fabric/G-FAM extends connectivity and supports shared memory across nodes. Hardware requirements increase accordingly, from a memory expansion controller to CXL switches and, ultimately, a multi-level fabric with support for cross-host memory coherency. The choice depends on workload requirements and the trade-offs between capacity, access performance, and system cost.

Memory expansion: Direct-attached expansion addresses the memory capacity constraints imposed by a server's CPU memory channels and DIMM slots. For memory-intensive workloads such as in-memory databases and data analytics, insufficient memory can force customers to add servers or upgrade platforms even when existing compute resources remain underutilized. In this architecture, a CXL-capable CPU root port connects over an x8/x16 PCIe lanes to a MXC, which in turn connects to DDR memory. The controller translates CXL.mem requests into accesses to the attached DDR, making the additional capacity addressable by the host. CXL switch is not mandatory, and the attached memory remains dedicated to that host. This provides a relatively simple expansion path, although application performance depends on link bandwidth, controller characteristics, and memory access patterns.

Memory pooling: Switch-based multi-host pooling addresses stranded memory capacity caused by uneven demand across servers. Provisioning each server for its own peak requirement can leave some nodes short of memory, while capacity sits idle elsewhere. A CXL switch connects multiple CPUs or GPUs to memory devices and their underlying DDR capacity, forming a common memory pool. A Fabric Manager assigns resources to hosts, with host software supporting capacity changes. This allows memory allocation to follow workload demand and can reduce excess provisioning. In typical CXL 2.0 pooling, each physical memory region belongs to only one host at a time: the benefit comes from reallocating capacity across hosts. The GPU/XPU pooling example similarly represents access to external DDR capacity by compatible accelerator systems; implementation requires appropriate accelerator and platform support and should not be interpreted as pooling GPU-local memory.

Memory sharing: CXL Fabric/G-FAM extends the architecture to larger systems and shared data access, with the potential to reduce duplicate memory copies and data movement across nodes. When multiple compute nodes use the same dataset, maintaining separate copies consumes additional capacity and creates synchronization traffic. The illustrated architecture connects CPUs, compatible GPUs/XPUs, and Global Fabric Attached Memory (G-FAM) devices through multiple layers of CXL switches, using Port-Based Routing (PBR) to support a larger fabric. The G-FAM controller manages the underlying DDR pool and exposes private and shared memory regions. With the required coherency support, multiple nodes can access the same shared region, supporting shared-data analytics and heterogeneous computing. However, commercial maturity remains a constraint. While vendors have introduced early memory-sharing solutions, the complete hardware and software ecosystem for large-scale, hardware-coherent sharing across a CXL fabric remains immature. Broader deployment requires coordinated support across host processors, switches, memory controllers, system software, and applications. We therefore expect this architecture to require a longer commercialization timeline than direct-attached expansion and conventional memory pooling.

EXHIBIT 40: **CXL's three memory architectures address three distinct needs: expanding server memory capacity, improving memory utilization across servers, and enabling data sharing across compute nodes**



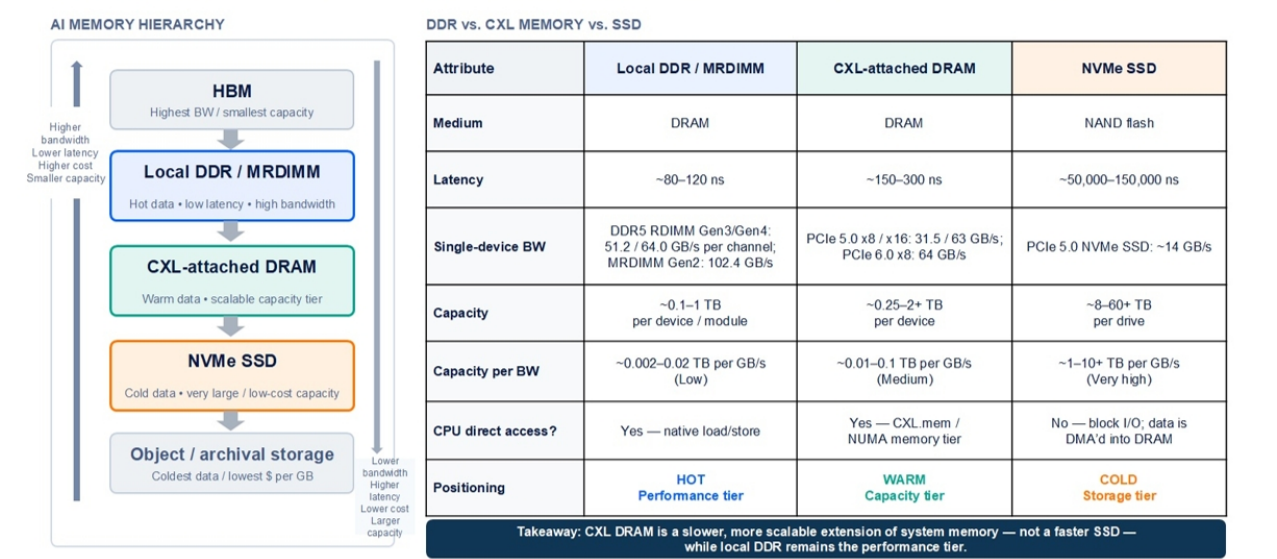
Source: CXL institution reports, Bernstein analysis

Reshaping the datacenter memory hierarchy

Agentic AI strengthens the case for memory tiering, and CXL memory becomes an increasingly important layer.

Workloads place increasingly different demands on bandwidth, capacity, and latency, requiring more diverse memory segments. Model inference, retrieval, tool execution, and the retention of long-running agent context do not require every byte of data to reside in the fastest memory. With HBM capacity constrained by packaging and cost, and local DDR expansion limited by CPU memory channels and DIMM slots, a more granular hierarchy allocates scarce resources according to access frequency and performance requirements. HBM serves bandwidth-intensive accelerator workloads, local DDR holds frequently accessed CPU data, and NVMe SSDs and archival storage accommodate progressively colder data. CXL-attached DRAM adds a warm memory tier between local DDR and NVMe, delivering memory-level access latency far below NVMe while enabling more flexible capacity expansion beyond local DDR limits. CXL memory latency ranges at approximately 150–300ns, far faster than 50–150µs of NVMe SSDs, with roughly 0.25–2+TB per CXL device. PCIe 5.0 x8/x16 links provide theoretical one-way bandwidth ceilings of approximately 31.5/63GB/s before CXL protocol overhead. This positions CXL for less frequently accessed context, caches, and other capacity-intensive data that still benefit from memory-speed access, while software placement policies preserve faster tiers for performance-critical data.

EXHIBIT 41: **CXL fills the gap between local DDR and NVMe by providing memory-speed access with flexible capacity expansion**

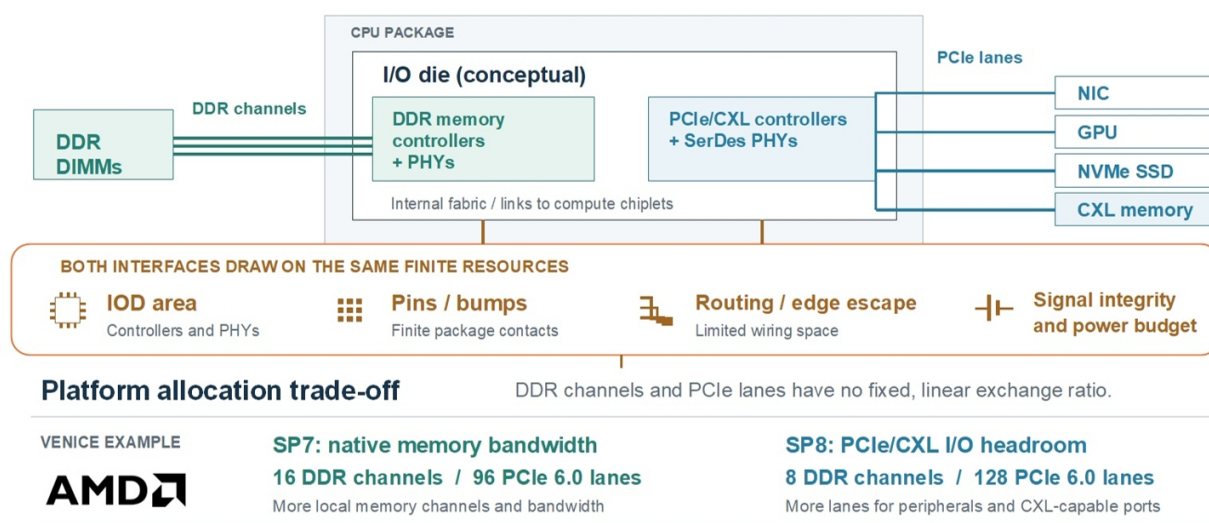


latency/ bandwidth ranges are implementation-dependent
Source: Meta Vistara report, companies reports, Bernstein analysis

CXL memory expansion is particularly relevant for CPU platforms with relatively few DDR channels and available CXL-capable I/O capacity, but its scalability remains constrained by finite platform resources. DDR controllers and PHYs compete with PCIe/CXL controllers and SerDes for silicon area, package pins and bumps, routing space, and power budgets. Within a given platform design budget, a higher DDR channel count can therefore leave less headroom for PCIe/CXL connectivity. CXL consequently cannot provide unlimited memory expansion: its capacity and bandwidth scale within the limits of available CXL-capable ports, lane allocation, and platform support. For CPUs with fewer DDR channels, connecting a Type-3 memory controller and its attached DRAM through available PCIe/CXL ports provides a practical way to increase addressable memory without adding native DDR channels. The trade-off is higher access latency and consumption of I/O lanes that could otherwise connect GPUs, NICs, or NVMe SSDs. We therefore view CXL's expansion value as platform-dependent: CPUs configured with more DDR channels provide greater native memory capability but may offer less PCIe/CXL headroom, while platforms with fewer DDR channels and more available CXL connectivity can use tiering to accommodate larger working sets.

EXHIBIT 42: CXL offers a practical expansion path for CPUs with fewer DDR channels, but its scalability remains constrained by I/O resources. Shared I/O resources create a trade-off between PCIe/CXL and DDR

Both interfaces compete for finite silicon, package connectivity and electrical budgets.



Platform differences illustrate allocation choices, not a measured causal exchange of DDR channels for PCIe lanes
Source: AMD reports, Bernstein analysis

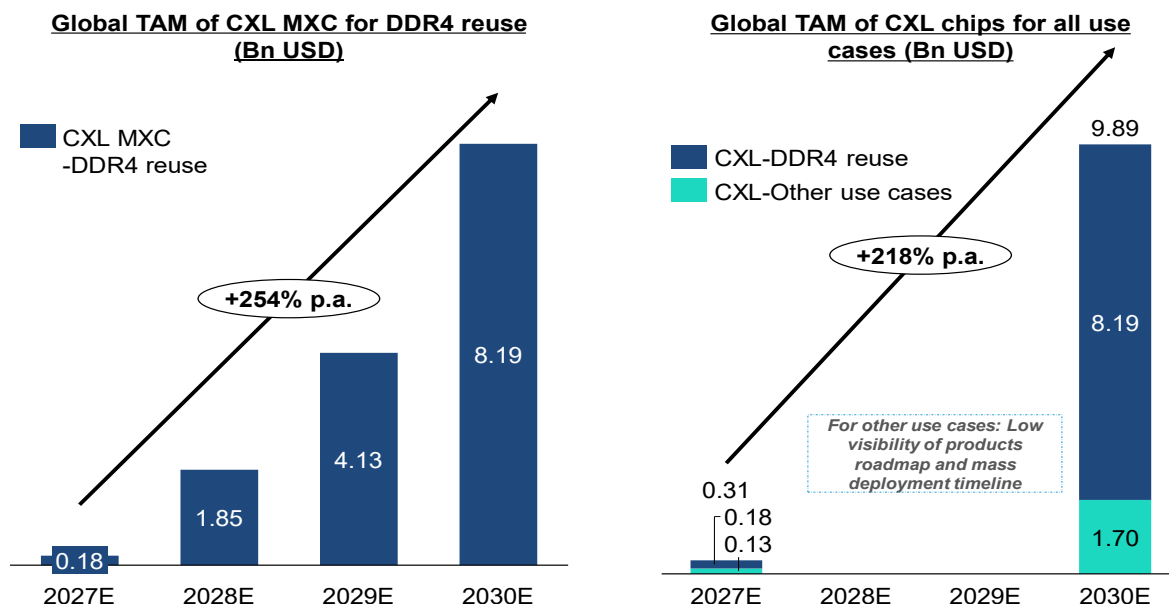
ECONOMICS DEEP DIVE

\$8bn incremental CXL TAM by 2030 from DDR4 reuse

We believe DDR4 reuse could become the first large-scale commercial application of CXL memory expansion, creating a global MXC TAM of roughly US\$8.2bn by 2030E. Assuming Montage captures 50% of the China market and 10% of the non-China market, this could translate into roughly 10% potential revenue upside, making CXL a meaningful new growth driver beyond its core memory-interface business.

We estimate the TAM for CXL MXC used in DDR4 reuse through a bottom-up approach anchored in the server DDR supply-demand gap. We first estimate the shortfall in server DDR capacity and assume that CXL-enabled DDR4 reuse addresses 50% of this gap in 2028–2030. We then derive CXL MXC dollar content per GB of expanded memory capacity from existing controller solutions, based on controller pricing and the DDR4 capacity supported by each configuration. Multiplying the capacity shortfall addressed through DDR4 reuse by the implied MXC dollar content per GB yields our annual controller TAM.

EXHIBIT 43: **We project DDR4 reuse will be the first killer application of CXL and its global TAM to reach 8.2Bn USD by 2030, driving CXL chips all use cases TAM to 9.9Bn USD in 2030**

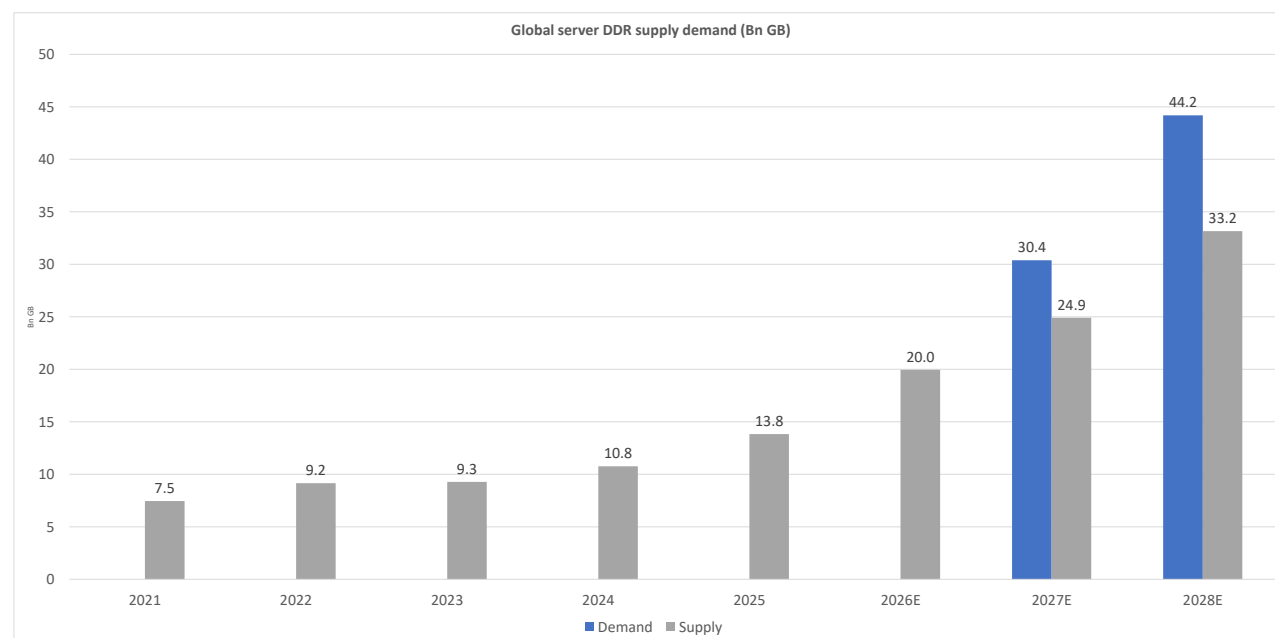


Source: Bernstein analysis and estimates

Global shortage on server DRAM

A structural server-memory shortage creates the need for DDR4 reuse, as server DRAM demand is growing much faster than the supply. We estimate the global server DDR shortfall will reach approximately 5.5bn GB in 2027E and 11bn GB in 2028E. These gaps establish the need for alternative capacity sources.

Our server DDR demand forecast implies a 47.3% CAGR over 2025–28E. Server DDR demand benefits from two reinforcing drivers: more CPUs and more memory attached to each CPU. We forecast server CPU shipments to grow at a 24% CAGR over 2025–30E, supported by the expansion of agentic AI. Over the same period, we forecast DDR capacity per CPU to grow at an 18% CAGR.

EXHIBIT 44: **We project the global server DDR shortage in 2029 and 2030 to be 5.5Bn GB and 11Bn GB, respectively**

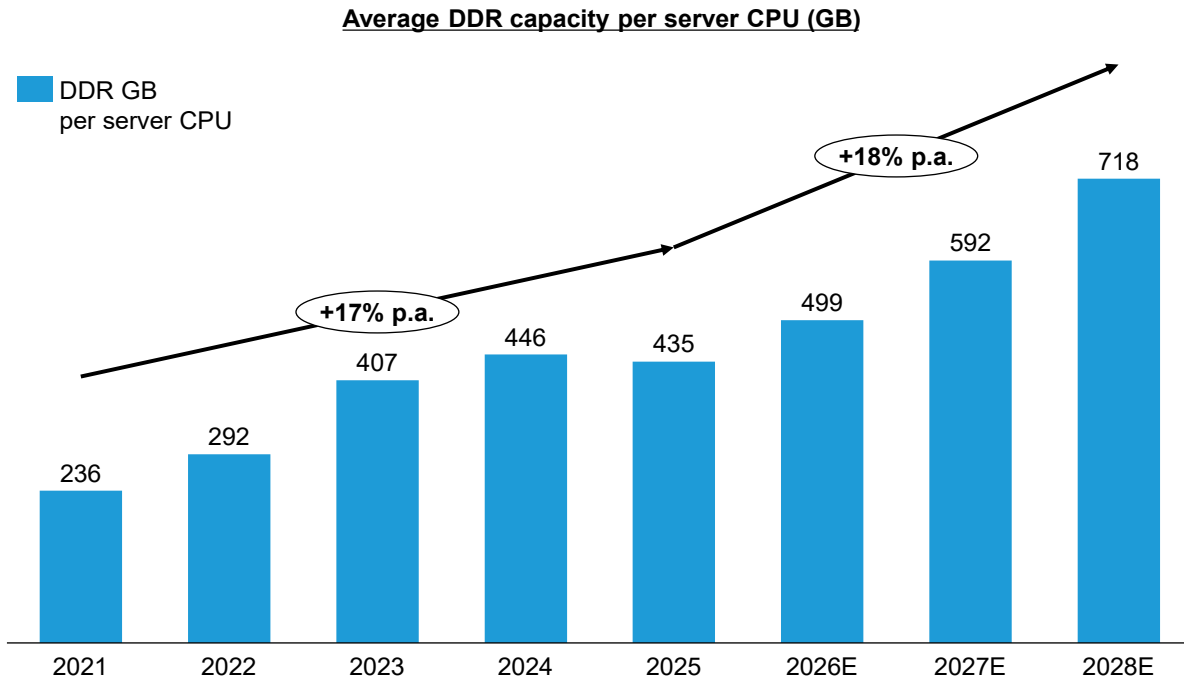
Source: TrendForce, Mercury, Bernstein analysis and estimates

Agentic AI extends infrastructure demand beyond model inference. Agents require environments in which to run browsers, execute code, retrieve information, call tools and maintain state across successive actions. These activities create CPU and memory requirements alongside accelerator demand. As more agents operate concurrently, the supporting infrastructure must accommodate more execution environments and larger aggregate application footprints.

Memory capacity per CPU grew at 17% CAGR during 2021-2025, and we expect the growth of the number will accelerate to 18% CAGR in the next two years. GB per CPU becomes increasingly important as operators consolidate workloads onto higher-core-count processors. Adding cores creates limited economic value if insufficient memory prevents those cores from hosting additional services. Larger working datasets, more concurrent sessions and retained application state increase the capacity needed to sustain CPU utilization. We therefore expect CPU DDR demand to grow faster than server CPU shipments alone.

Bandwidth requirements increase alongside capacity, but the two do not rise uniformly across all data. Frequently accessed working sets require high bandwidth and low latency, while retained state and less active pages may consume substantial capacity with limited sustained traffic. This distinction creates room for a second memory tier: local DDR serves the hottest data, while a lower-cost tier accommodates a portion of the expanding footprint.

EXHIBIT 45: Memory capacity per CPU growth will accelerate, as more concurrent sessions and retained application state increase the capacity needed to sustain CPU and XPU utilization



Source: TrendForce, Mercury, Bernstein analysis and estimates

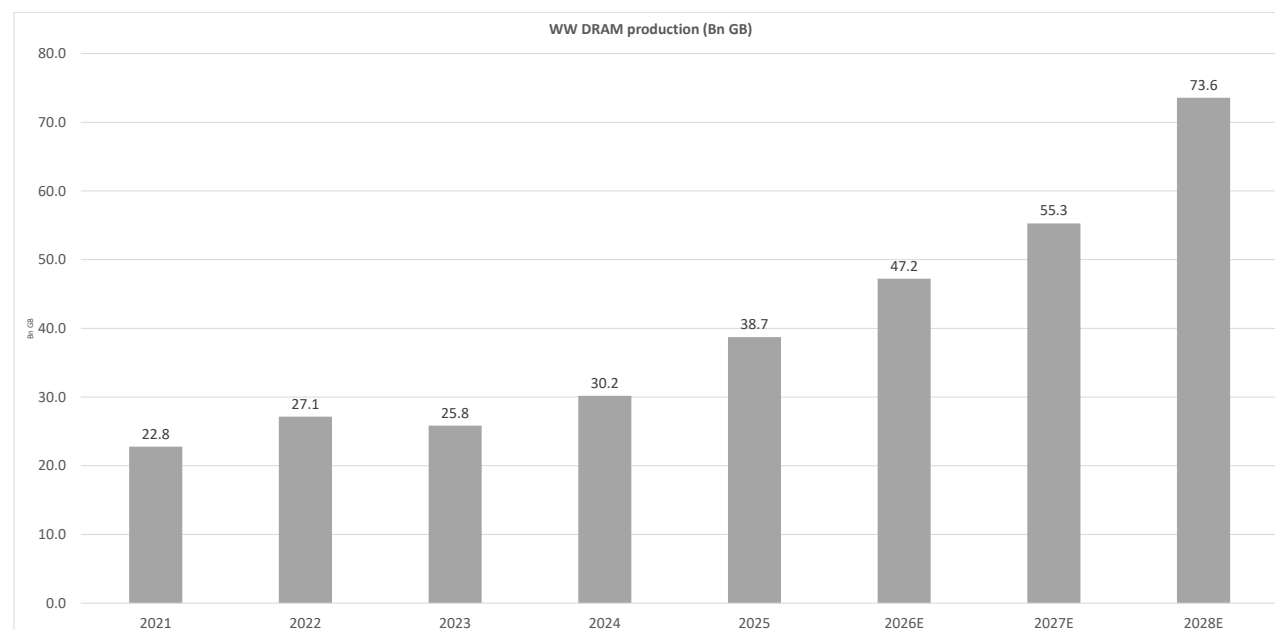
Disciplined capacity expansion and HBM competition constrain server DDR supply. We expect DRAM supply growth to remain insufficient to meet the increase in server-memory requirements. We anticipate approximately 22% CAGR in global DRAM bit capacity over 2026–28E, reflecting disciplined expansion and the time required to bring new capacity into qualified production. This is an aggregate DRAM measure; the availability of server DDR also depends on suppliers' allocation across products and end markets.

Announced investment does not immediately translate into usable memory supply. Suppliers must complete facilities, install equipment, ramp processes and qualify products before customers can deploy the resulting output.

HBM adds another constraint because it consumes manufacturing resources disproportionately to the number of bits produced. Micron has disclosed that HBM3E requires approximately three times the wafer supply of DDR5 to produce an equivalent number of bits on the same process node. As suppliers allocate more resources to HBM, growth in total investment need not generate a corresponding increase in conventional server DDR availability.

We also expect Chinese suppliers to increase output, but our forecast assumes that remaining technology and yield gaps versus global DRAM leaders limit their ability to eliminate the shortage. Wafer capacity alone is insufficient to assess the impact: qualified output, bit density, yields and the product mix determine how much additional supply can serve server demand. We therefore retain a supply-constrained outlook despite domestic expansion.

EXHIBIT 46: **We anticipate the worldwide DRAM production will reach 73.6Bn GB by 2028, growing at the CAGR of 23.8% in 25-28E**



Source: Bernstein analysis and estimates

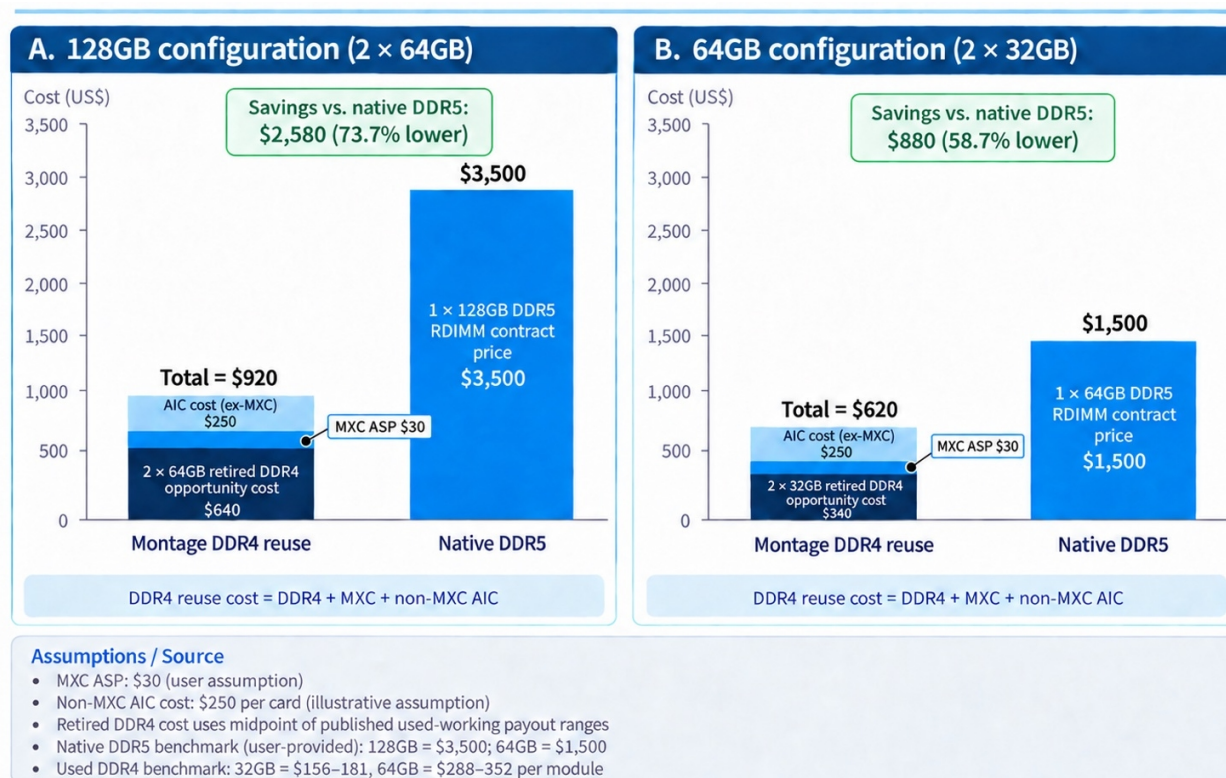
Elevated memory price makes the ROI attractive

Server retirement can release usable DDR4 well before the memory reaches the end of its service life. In its July 2026 Vistara presentation, Meta cited a 5–7-year server retirement cycle versus a 10–14-year DRAM lifetime, illustrating the scope to reuse memory across successive server generations. These figures support the underlying economic case: platform retirement need not coincide with memory retirement. **We therefore see qualified DDR4 recovered from retired servers as a potential source of incremental capacity**, subject to module health, remaining service life and compatibility with the receiving platform.

CXL allows a DDR5-generation server CPU to access retired DDR4 through a memory-expander controller, effectively turning otherwise stranded memory into a usable warm-memory tier.

Based on current DDR5 contract pricing, we estimate that a CXL DDR4 reuse configuration can be 58–73% cheaper than same-capacity native DDR5; a 128GB configuration can save more than US\$2,580 even after including MXC and AIC costs. The DDR5–retired-DDR4 price spread can absorb the controller and card costs while leaving a substantial saving. Our comparison uses August 2026 DDR5 contract-price benchmarks, per TrendForce, of US\$1,500 for a 64GB DIMM and US\$3,500 for a 128GB DIMM. We assume US\$30 per MXC and approximately US\$250 of non-MXC add-in-card costs. For a 128GB configuration, two reused 64GB DDR4 DIMMs carry an assumed aggregate opportunity cost of US\$640, the midpoint of used DDR4 price listed in SellUsedDRAM. Adding the controller and card brings the acquisition cost to US\$920, compared with US\$3,500 for equivalent native DDR5 capacity. This implies a saving of US\$2,580, or 74%. For 64GB, two reused 32GB DIMMs cost US\$340 on the same opportunity-cost basis; the US\$620 total reuse cost saves US\$880, or 59%, against native DDR5.

EXHIBIT 47: With Montage's solution as an illustrative example, DDR4 reuse via CXL can save more than half of cost, compared to native DDR5



used DDR4 costs use the midpoints of indicative buyback ranges of USD 156-181 per 32GB module and USD288-352 per 64GB. All memory prices reflect market condition in Aug 2026

Source: SellUsedDRAM, TrendForce, Bernstein analysis and estimates

Current available solution on CXL MXC

The CXL MXC market is currently concentrated among a small group of suppliers, Montage, Astera Labs, Marvell and Meta's internal solution. The relevant competitive distinction is the architecture and economics of a qualified deployment. Their solutions cover smaller capacity increments, multi-channel expansion and a hyperscaler-specific design. Meta is a captive implementation rather than a merchant competitor.

Astera's newer Leo 2 E- and P-Series support PCIe 6.0, four DDR channels and configurations with twelve DDR4 DIMMs; Astera announced the expanded family in September 2026 and described it as sampling with hyperscaler customers. Its older CXL 2.0 Leo products primarily address DDR5 expansion.

Marvell's Structera X 2404 targets DDR4 expansion with 4 channels and up to 12 DIMMs, while the 2504 addresses DDR5. Using twelve 128GB DDR4 DIMMs provides 1,536GB of capacity. DIMM population also affects operating speed, as discussed in the bandwidth section.

Meta provides direct evidence that DDR4 reuse can become an operational memory tier. Its July 2026 Vistara presentation describes a production platform with 768GB of local memory and 256GB of CXL memory, using two custom ASICs and Linux memory tiering. Each ASIC has two DDR4 channels with 2DPC; the production configuration uses four 32GB DIMMs per ASIC and x8 host connectivity. This validates the complete reuse approach.

EXHIBIT 48: Spec of current product offerings of CXL MXC for DDR4/DDR5 expansion

   				
1 DDR4-reuse-oriented MXC products			Targeting near-term cost efficiency by reusing existing DDR4 infrastructure	
Product	M88MXS891	Leo 2 / A2000 (DDR4 configuration)	Structera X 2404	Vistara (custom DDR4 reuse ASIC)
Host interface (PCIe / CXL, lanes, upstream BW)	PCIe 5.0 x8 / CXL 2.0 ~31.5 GB/s per direction	PCIe 6.0 x16 / CXL 3.2 128 GB/s raw per direction (Gen5 host: ~63 GB/s ceiling)	PCIe 5.0 x16 or 2 x x8 / CXL 2.0 ~63 GB/s per direction (x16)	PCIe 5.0 x16 / CXL 2.0 x8 in production: ~31.5 GB/s (per direction)
Memory topology (# channels, max DIMMs)	1 DDR4 channel 2 DPC, max 2 DIMMs	4 DDR4 channels 3 DPC, max 12 DIMMs	4 DDR4 channels Up to 3 DPC, max 12 DIMMs	2 DDR4 channels, 2 DPC 4 DIMMs per ASIC
Memory speed / downstream BW	DDR4-3200 25.6 GB/s (aggregate)	DDR4-3200 (up to 3 DPC) 4 x 25.6 = 102.4 GB/s (aggregate)	DDR4-3200, 2 DPC: 102.4 GB/s DDR4-1600, 3 DPC: 51.2 GB/s (aggregate downstream BW)	Production: DDR4-2400 2 x 19.2 = 38.4 GB/s (theoretical downstream BW / ASIC)
Total capacity	256GB configured capacity (using 2 x 128GB DIMMs)	1,536GB configured capacity (12 x 128GB DIMMs)	1,536GB configured capacity (12 x 128GB DIMMs)	Production: 128GB / ASIC (4 x 32GB; 256GB with 2 ASICs) Internal design; not merchant silicon
2 DDR5-expansion-oriented MXC products			Enabling higher capacity and bandwidth with DDR5 memory expansion	
Product	M88MXS851	Leo A1000 / first-gen Leo	Structera X 2504	No public merchant DDR5-expansion MXC
Host interface (PCIe / CXL, lanes, upstream BW)	PCIe 5.0 x8 / CXL 2.0 ~31.5 GB/s per direction	PCIe 5.0 x16 / CXL 2.0 ~63 GB/s per direction	PCIe 5.0 x16 or 2 x x8 / CXL 2.0 ~63 GB/s per direction (x16)	—
Memory topology (# channels, max DIMMs)	Not publicly disclosed	2 DDR5 channels 4 DDR5 RDIMM slots total	4 DDR5 channels 2 DPC, max 8 DIMMs	—
Memory speed / downstream BW	Up to DDR5-5600 (channel count / aggregate BW not publicly disclosed)	DDR5-5600 2 x 44.8 = 89.6 GB/s (aggregate)	DDR5-6400 4 x 51.2 = 204.8 GB/s (aggregate)	—
Total capacity	Not publicly disclosed	Up to 2TB	>6TB-class (DDR5 support)	—

configured capacity is not a controller maximum, but is estimated on mainstream DDR4 DIMMs capacity

Source: companies reports, Bernstein analysis

Montage offers a simple, low-cost architecture well suited to DDR4 reuse. Montage's M88MX5891 uses a relatively simple PCIe 5.0 x8 + single DDR4 channel + 2DPC architecture, making it particularly suitable for lower-cost DDR4 reuse and positioning Montage as the most direct domestic beneficiary of adoption in China. Montage's opportunity is to make a useful capacity increment economical without requiring a large expansion subsystem. This provides a smaller deployment unit than the 12-DIMM designs by ALAB and Marvell, and can fit workloads that require incremental warm capacity rather than terabytes of additional memory.

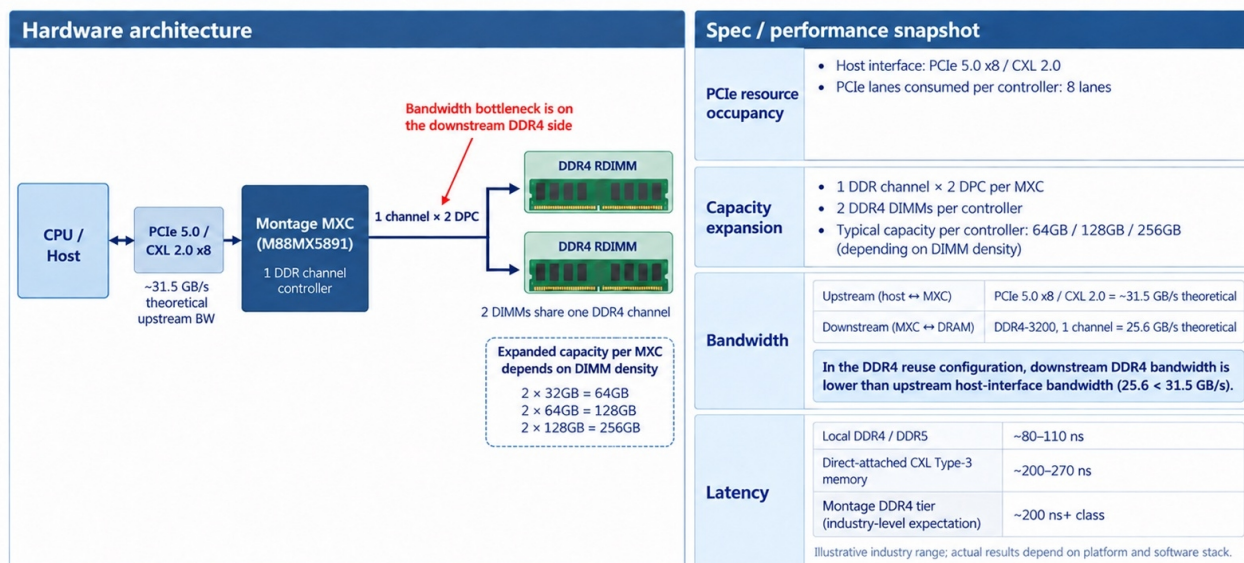
Montage's existing memory-interface business provides relevant relationships across the server-memory supply chain. We believe this can support CXL MXC evaluation and integration, particularly in China. However, CXL qualification introduces additional requirements around host interoperability, firmware, error handling and software placement, requiring more time for large-scale deployment.

EXHIBIT 49: **Montage's current CXL MXC for DDR4 reuse, M88MX5891, uses a relatively simple PCIe 5.0 x8 + single DDR4 channel + 2DPC architecture, making it particularly suitable for lower-cost DDR4 reuse**

Montage DDR4 Reuse via CXL: Architecture and Performance

Illustrative M88MX5891-based design for repurposing retired DDR4 as a warm-memory tier

EXTEND
CAPACITY.
UNLOCK VALUE.



Source: company reports, Bernstein analysis

TECH FEASIBILITY DEEP DIVE

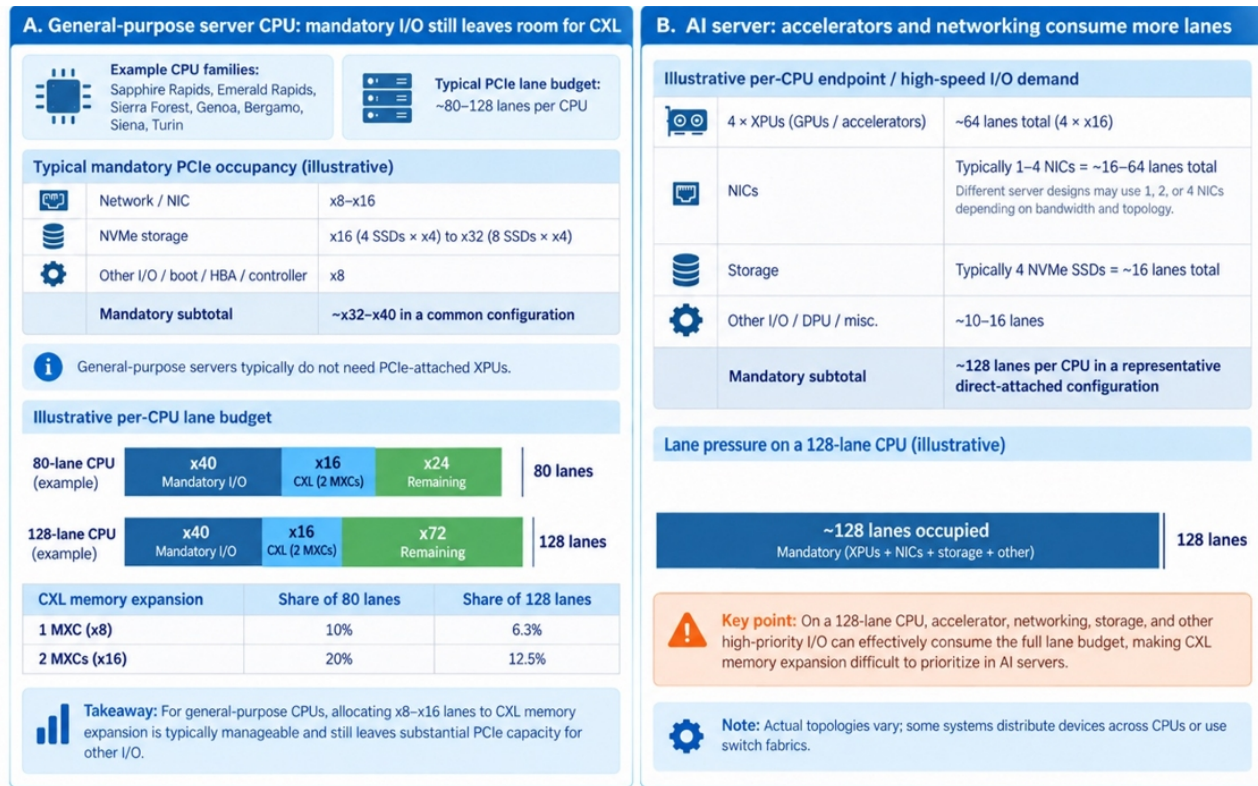
Sufficient PCIe channels to support CXL MXC

Our analysis shows that PCIe lane availability is generally sufficient on general-purpose CPUs. Unlike AI servers where accelerators and high-speed networking consume most of the available PCIe budget, allocating x8–x16 lanes to one or two MXCs is generally manageable for general-purpose server CPUs with roughly 80–128 PCIe lanes.

General-purpose servers provide a more accessible initial deployment environment because their PCIe resources face less competition from accelerators. Representative platforms include Intel Sapphire Rapids, with 80 PCIe 5.0 lanes per CPU, and AMD EPYC 9004 in a single-socket configuration, with 128 lanes, of which up to 64 can support CXL connectivity. These examples establish a useful range for evaluating the incremental resources required by one or two MXCs. In our illustrative general-purpose configuration, networking, NVMe storage and other required I/O consume 40 lanes. An 80-lane CPU retains 40 lanes before CXL, while a 128-lane CPU retains 88. Adding two x8 lanes MXCs consumes 16 lanes, leaving 24 and 72 lanes, respectively. The precise allocation varies by server, but these examples show why modest CXL expansion can fit within a conventional I/O budget. Available lanes must also connect to usable CXL-capable ports and slots. Motherboard routing, supported device counts, firmware and platform qualification determine whether nominal headroom translates into deployable memory. We therefore assess the complete server configuration rather than assuming every unallocated PCIe lane is interchangeable.

Accelerator-heavy servers face a higher opportunity cost. GPUs, high-speed NICs and storage compete for connectivity and bandwidth, with PCIe switches often shaping the final topology. CXL expansion may still be technically possible, but an additional MXC competes with resources that directly support accelerator utilization.

EXHIBIT 50: General-purpose servers typically have spare PCIe lanes for 1-2 CXL MXCs, while AI servers allocate much more high-speed I/O to XPU and NICs



Illustrative lane budgets for conceptual comparison. Exact lane usage depends on server topology, CPU generation, NIC count, storage design, and XPU attachment scheme

Source: Intel and AMD products spec, Bernstein analysis and estimates

The solution comes with limitations on bandwidth and latency

Bandwidth is comparable to one native DDR channel, not the full local-memory subsystem

CXL expansion bandwidth depends on the complete memory path, not simply the number of DIMMs attached. A PCIe 5.0 x8 link provides approximately 31.5GB/s per direction before transaction-level overhead, while x16 provides approximately 63GB/s. A single DDR4-3200 channel provides 25.6GB/s of theoretical data bandwidth. Sustainable application bandwidth is lower and depends on access patterns, controller efficiency and protocol overhead.

Two bottleneck scenarios follow:

- In a multi-channel configuration whose combined DDR bandwidth exceeds the host link's capacity, the upstream CXL connection limits throughput. Additional DIMMs can expand capacity substantially without increasing host-visible bandwidth proportionately.
- In a single-channel DDR4 configuration such as the Montage design used here, the downstream 25.6GB/s DDR4 interface provides the lower theoretical ceiling relative to the x8 host link.
- DIMM population can change this comparison. Marvell's Structera X2404 supports DDR4-3200 at 2DPC, but its product brief specifies DDR4-1600 at 3DPC. Across four channels, these configurations imply theoretical downstream bandwidth of 102.4GB/s with eight DIMMs and 51.2GB/s with twelve DIMMs, respectively. The twelve-DIMM configuration therefore cannot automatically be treated as host-link-limited solely because it has four DDR channels.

The appropriate comparison for a single Montage DDR4 expander is approximately one native DDR4-3200 channel. A sufficiently provisioned PCIe 5.0 x16 expander has a host-link ceiling in the same broad range as a fast native DDR5

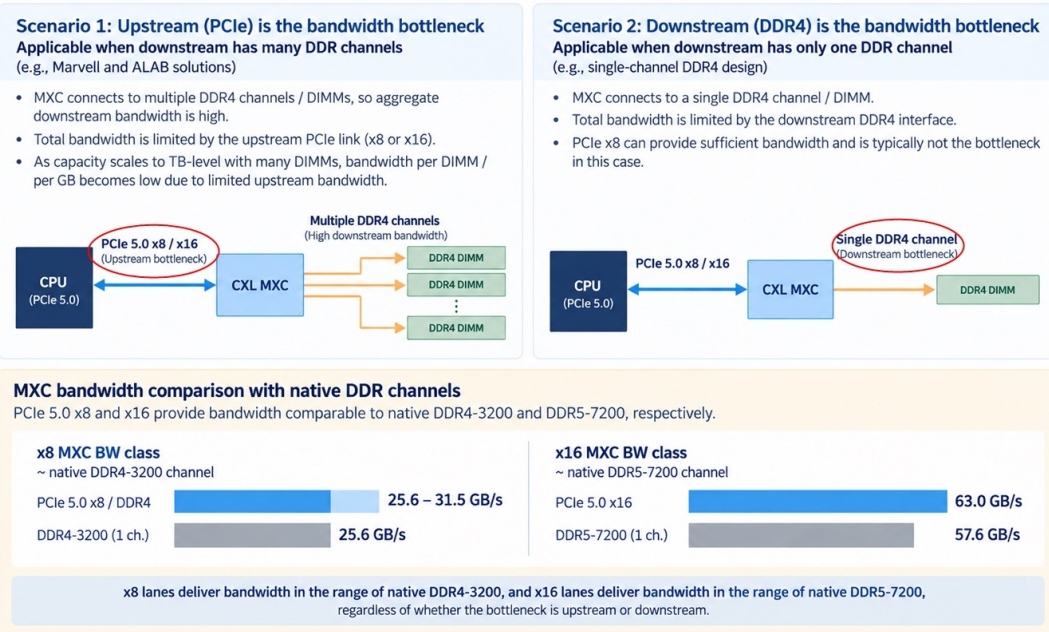
channel, although actual throughput remains configuration-dependent. Neither comparison equates an individual expander with the CPU's complete multi-channel local-memory subsystem.

CXL can still add aggregate system bandwidth when it supplements existing local memory. The performance trade-off arises when data that would otherwise receive local-memory service moves to the slower tier, or when many workloads concentrate requests on one expander. Deployment should therefore preserve sufficient local capacity and bandwidth for the active working set.

EXHIBIT 51: Total bandwidth is limited either the upstream PCIe link or the downstream DDR4 interface, depending on the design

CXL MXC DDR4 reuse: bandwidth performance penalty

Total bandwidth is limited by either the upstream PCIe link or the downstream DDR4 interface, depending on the design.



Source: Companies products brief, Bernstein analysis and estimates

Latency is the more meaningful performance penalty

Latency is the more consequential constraint for workloads that repeatedly access data on the critical path. We use approximately 200–270ns as an illustrative low-load range for selected direct-attached CXL ASIC implementations, **broadly around 2–2.5x local DRAM in comparisons.** Published characterization reports ASIC-device latencies of 214ns, 239ns and 271ns, while also showing substantial variation across platforms and implementations. These measurements are reference points rather than a guaranteed Montage specification. CXL performance characterization.

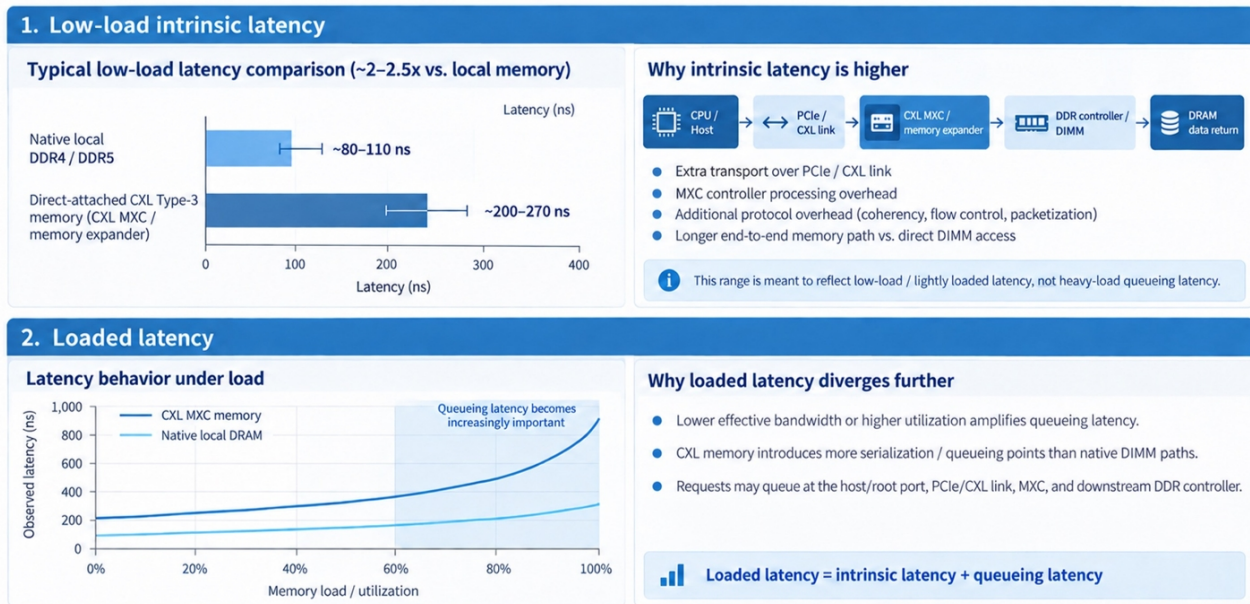
The extra delay reflects the longer access path. A request traverses the host interface, CXL transport and expander before reaching the DDR controller and DIMM. Controller design and physical implementation affect the result, so the protocol alone does not determine a universal latency penalty.

Under load, queuing adds to this baseline. Requests can accumulate when traffic approaches the service capacity of the host link or downstream DDR interface. Shared links, controller queues and concurrent access streams influence how quickly latency rises.

The economic implication is that capacity utilization and bandwidth utilization should be assessed separately; hence, we do not expect full unsatisfied server DDR demand will be penetrated by the CXL DDR4 reuse solution. A card can be valuable while storing a large amount of infrequently accessed data and carrying modest traffic. Conversely, a smaller dataset can perform poorly, if it generates intense random accesses or sustained transfers. Customers must validate throughput and tail

latency at the intended concurrency, including migration traffic between tiers.

EXHIBIT 52: CXL MXC memory expansion faces latency performance penalty, with intrinsic latency in low-load scenario and additional queueing latency in heavy-loaded scenario



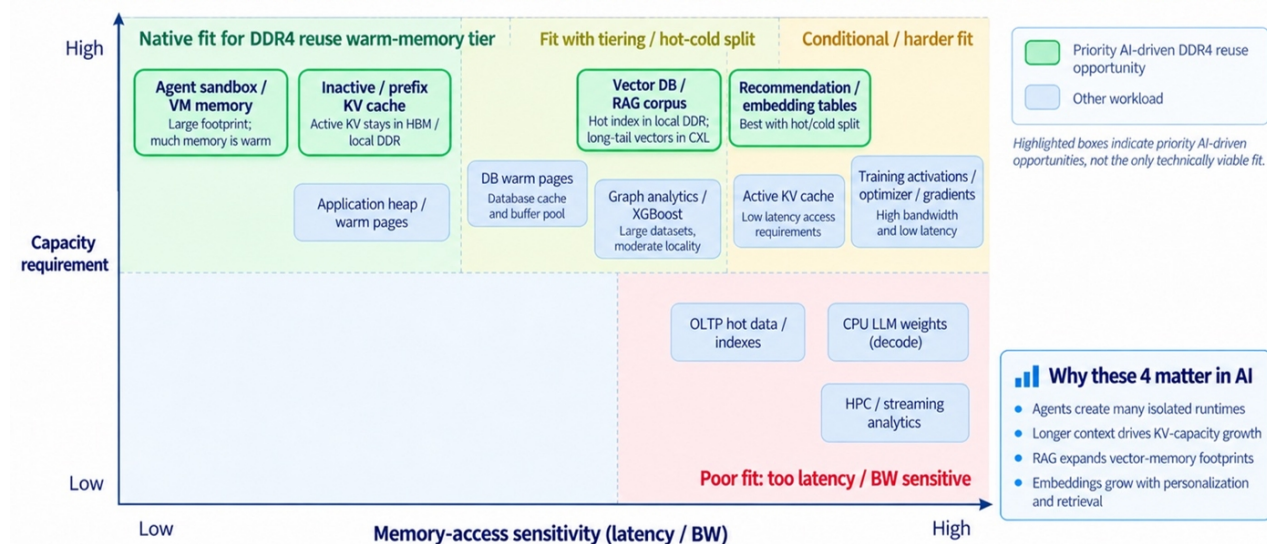
illustrative industry-level ranges and conceptual load curves; exact latency depends on platform, controller design, topology, utilization, and workload
Source: Microsoft report, Bernstein analysis

AI IS EXPANDING THE WORKLOADS THAT FIT CXL DDR4 REUSE

The best-fit workloads require large capacity but only a limited hot working set. The right way to assess CXL DDR4 reuse is not simply whether a workload is memory-intensive, but whether it needs large capacity without requiring all data to be accessed at native-DRAM latency and bandwidth. The strongest reuse candidates require substantial resident capacity while concentrating most accesses on a smaller hot working set. Capacity determines the potential procurement saving; the traffic directed to the slower tier determines the performance cost. We therefore assess workloads along two dimensions: memory-capacity requirements and sensitivity to memory access. The distinction applies within applications as well as across them. A database may have latency-critical indexes alongside less active buffer pages. An inference service may hold active KV blocks alongside reusable state from inactive sessions.

Production offloading experience supports this approach. Meta's Transparent Memory Offloading work reported 20–32% memory savings across its deployment by identifying colder pages and moving them to lower-cost resources. That work supports the existence of tiering opportunities. For DDR4 reuse, the useful middle ground is data that benefits from remaining memory-addressable but does not need continuous local-DRAM performance. Data with virtually no reuse may fit better on storage, while frequently accessed data should retain the fastest appropriate tier. The size of this middle ground determines the useful capacity available to CXL.

EXHIBIT 53: We assess workloads by two dimensions: memory-capacity requirements and sensitivity to memory access. The best-fit workloads for CXL DDR4 reuse require large capacity but limited hot working set



conceptual positioning

Source: company reports, Bernstein analysis

Agent sandbox and VM memory offer a compelling use case of CXL DDR4 reuse

We see agent sandbox and VM memory as a particularly compelling incremental opportunity because concurrency can expand allocated capacity faster than the active working set. Agents may require isolated browsers, Python environments, application processes and retained session state. Many such environments can remain resident while awaiting model responses, external tools or user input, creating periods during which portions of their memory receive limited traffic.

This creates a potential separation between the memory required to preserve an execution environment and the memory required to advance its current task. Warm pages can remain in CXL DDR4 while actively used code and data stay in local DDR. If the platform can preserve application responsiveness, operators may support more concurrent environments without purchasing equivalent additional native DDR5 capacity. We expect the strongest adoption case where environments have a substantial warm footprint, activity is staggered and performance targets allow measured tiering. This places agent infrastructure naturally within the general-purpose server opportunity discussed above.

RAG/vector DB and embedding tables provide additional opportunities

Inactive or reusable KV cache creates a capacity opportunity when retaining state costs less than recomputing or retrieving it. Longer contexts and concurrent sessions increase the volume of potentially reusable KV data. Blocks associated with paused sessions or cached prefixes may remain valuable without being accessed continuously. CXL DDR4 reuse focuses on inactive or reusable blocks that can be promoted or transferred before reuse. The economic benefit depends on whether retention avoids sufficient recomputation and whether transfer time fits the application's response target.

RAG and vector databases offer a related opportunity in their less frequently accessed data. A growing knowledge base increases the number of vectors and associated records, while query popularity can concentrate accesses on a subset. Hot search structures and frequently accessed vectors can remain local, with an appropriate long tail assigned to CXL. Astera's published RAG discussion describes this separation between hot vectors in local DRAM and warmer data in CXL memory. The qualification is that retrieval itself can be latency-sensitive. Graph traversal, pointer chasing and concurrent queries may expose slower accesses directly to users. A large corpus creates a capacity opportunity, but its suitability for DDR4 reuse depends on the index design, access distribution and achieved tail latency. We therefore treat RAG as a tiering opportunity rather than assuming the entire vector database belongs in CXL.

Recommendation and embedding tables are attractive when accesses are sufficiently concentrated. Large tables can consume substantial memory while a smaller set of entries accounts for much of the traffic. Keeping frequently accessed embeddings local and assigning colder entries to a capacity tier can reduce native-memory requirements. However, random

lookups remain performance-sensitive, so the cold fraction must be large enough and sufficiently stable to justify placement. Published AMD–Asteria work demonstrates CXL's relevance to recommendation models, though its configuration uses DDR5 expansion.

EXHIBIT 54: **Four AI workloads offer different paths to monetize a warm-memory tier, fit with CXL DDR4 reuse**

AI workload	Why AI demand grows	Hot-set ratio	Acceptable latency	Typical memory footprint	Required BW	DDR4 reuse fit	Implementation note
Agent sandbox / VM memory	More concurrent agents need isolated runtimes, tools, file systems, and working environments	Low; much of allocated memory is idle or warm	Medium-high for warm pages	Tens to hundreds of GB per host; fleet-level TBs	Low to medium; bursty	Very high	Keep active runtime state in local DDR; place warm pages in CXL DDR4 tier
Inactive / prefix KV cache	Longer context windows and more concurrent sessions expand KV-cache capacity needs	Low-medium; only active KV is truly hot	Medium for inactive or reusable KV; low for active KV	Hundreds of GB to multi-TB per cluster	Low to medium for inactive tier	High	HBM/local DDR for active KV; CXL DDR4 for inactive, reusable, or prefetched KV blocks
Vector DB / RAG corpus	Enterprise RAG and retrieval systems drive very large embedding corpora	Medium-low; hot index small, long tail much larger	Medium; retrieval path is sensitive but tail data can be slower	Hundreds of GB to TB+	Medium; random access	Medium-high with tiering	Keep hot index and top-hit vectors in local DDR; place long-tail corpus in CXL DDR4
Recommendation / embedding tables	Personalization and retrieval use ever-larger embedding tables	Medium-low due to skew; small hot set, large long tail	Medium-low; sensitive for hot lookups	Hundreds of GB to multi-TB	Medium-high; random lookups	Medium with hot/cold split	Retain hot embeddings in local DDR; use CXL DDR4 for long-tail tables

 Key takeaway

- Best fit = large memory footprint, low hot-data ratio
- CXL DDR4 is a warm-memory tier, not a DDR5 replacement
- AI creates new demand in agents, KV cache, RAG, and embeddings

illustrative workload characteristics; actual thresholds also depend on software stack and hot-set management

Source: Bernstein analysis

FUTURE TECHNOLOGY ROADMAP

DDR6 AND ITS IMPLICATIONS FOR MEMORY INTERFACE CHIPS

JEDEC completed the initial DDR6 specification draft at the end of 2024, with final ratification expected within 2026. Compared with DDR5, DDR6 introduces two fundamental architectural changes. First, the peak data rate roughly doubles for the basic format: DDR6's JEDEC spec defines a base rate of 8,800 MT/s scaling up to 17,600 MT/s, versus DDR5 RDIMM's range of 4,800–8,400 MT/s. Second, DDR6 replaces DDR5's dual 32-bit sub-channel architecture with a quad 24-bit sub-channel design. By splitting data transfer across four narrower channels, each bit line is shorter and experiences less capacitance and crosstalk, enabling the data rate doubling without proportional increases in power consumption. DDR6 is also expected to adopt the CAMM2 connector standard, replacing decades-old DIMM and SO-DIMM form factors, which could change the physical design and BOM for memory modules, including the role of on-module interface chips.

All three major DRAM manufacturers — Samsung, SK hynix, and Micron — have kicked off DDR6 development, completed prototype chip designs, and are now working with memory controllers and CPU platform players on interface testing. However, commercialization timelines remain uncertain and diverge significantly across the industry: SK hynix's public roadmap, disclosed at its AI Summit in November 2025, positions DDR6 in the 2029–2031 timeframe, reflecting its strategic priority of extending DDR5 through MRDIMM and HBM4/5 for AI workloads. Next-generation CPU platforms from Intel and AMD are reportedly undergoing DDR6 interface validation, though neither has disclosed a clear product roadmap with confirmed DDR6 support. On the memory interface chip side, none of the three major vendors has yet disclosed concrete DDR6 interface chip products. Montage claims it is conducting DDR6-related R&D programs. Historically, interface chip development lags the DRAM standard by one to two years, as vendors require a stable specification before committing silicon design resources.

Critically, DDR6's base data rate of 8,800 MT/s is equivalent to the effective throughput of DDR5 MRDIMM Gen 1, while DDR5 MRDIMM Gen 3 is expected to match DDR6's high-end spec at 17,600 MT/s. It means the full DDR6 data rate range is already achievable within the DDR5 generation through advanced interface chip architectures; hence, this substantial overlap weakens the near-term urgency for server OEMs and CSPs to migrate to DDR6. This dynamic reinforces our projection that DDR5 MRDIMM penetration will continue to rise through the decade, before DDR6 becomes a compelling alternative. Looking further ahead, the outlook for DDR6 is structurally positive for the memory interface chip industry. For DDR6 to offer a meaningful bandwidth advantage over DDR5 MRDIMM, it will need to push data rates well beyond the current 17,600 MT/s ceiling of its basic specification, achieving that will require a new generation of advanced memory interface chips for active retiming, signal conditioning, and multiplexed-rank architectures analogous to what MRDIMM brought to DDR5. In other words, DDR6 in its basic module format does not offer a compelling upgrade path over DDR5 MRDIMM; only with higher interface chip content can DDR6 unlock data rates that justify the generational transition. This ensures that each successive push for higher memory bandwidth demands greater interface chip intelligence and dollar content per module, a secular trend that is positive for Montage, Renesas, and Rambus regardless of which DDR generation dominates.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Montage Technology**

We set our A-share TP at CNY 400, based on 51x 2BF P/E. For the H shares, we set a target price of HKD 520, implying a 15% premium to our A-share TP (we use the CNY to HKD exchange rate at 1:1.13). The H share premium reflects that global investors favor Montage as a scarce China AI-exposed name without direct geopolitical risks, unlike many other Chinese semiconductor companies that face entity list restrictions or export control headwinds. At our target price, the implied P/E multiple on 2BF EPS for H shares is 59x.

RISKS**Montage Technology**

Key potential risks:

1. A decrease in memory and AIDC server demand;
2. Intensify competition that leads to share loss and lower margin;
3. Failure to introduce new products for the AIDC networking market.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

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Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

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Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

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Autonomous brand – preferred stocks

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- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

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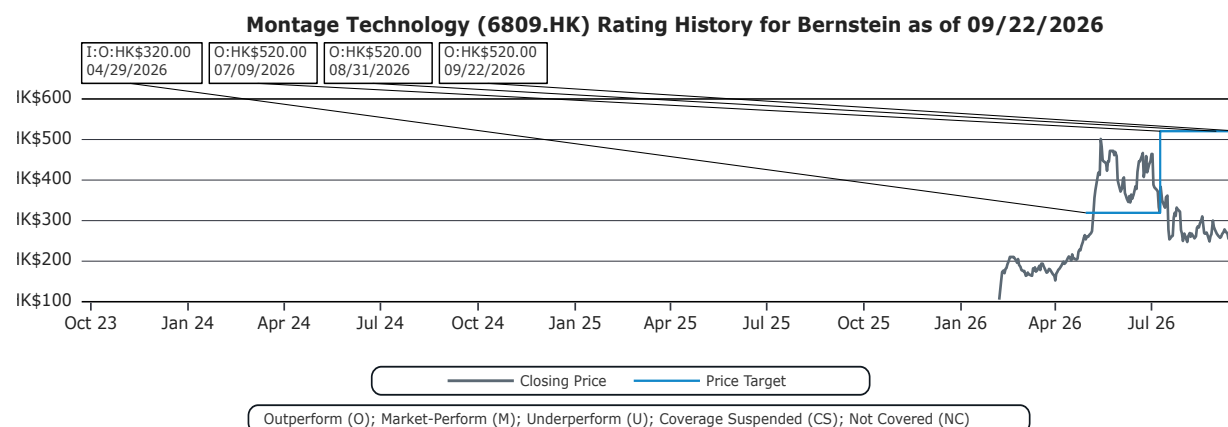
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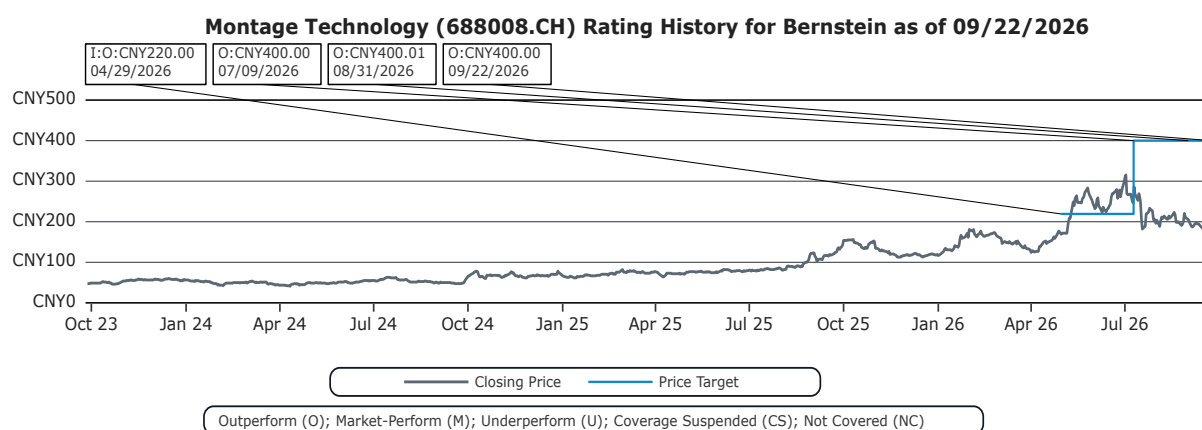
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